BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

# Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis



| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2020-040459 |
| Article Type: | Original research |
| Date Submitted by the Author: | 15-May-2020 |
| Complete List of Authors: | Li, Zhe; University of Western Ontario, Epidemiology and Biostatistics<br>Dawson , Emily ; Western University, Centre for Medical Evidence, Decision Integrity & Clinical Impact (MEDICI)<br>Moodie, Jessica; University of Western Ontario, Centre for Medical Evidence, Decision Integrity and Clinical Impact (MEDICI)<br>Martin, Janet; University of Western Ontario, High Impact Technology Evaluation Centre<br>Bagur, Rodrigo; Western University, Department of Epidemiology and Bio statistics<br>Cheng, Davy; London Health Sciences Centre University Hospital, Anesthesia & Perioperative Medicine<br>Kiaii, Bob; Western University, Department of Cardiology, Division of Cardiac Surgery<br>John-Baptiste, Ava; University of Western Ontario, Anesthesia & Perioperative Medicine |
| Keywords: | Valvular heart disease < CARDIOLOGY, EPIDEMIOLOGY, Cardiac surgery < SURGERY, VASCULAR SURGERY |
| | |



SCHOLARONE™
Manuscripts



*I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.*

*The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.*

*Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.*

For peer review only

# Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis

Zhe Li [a, b, c], Emily Dawson [a, b, c], Jessica Moodie [b, c], Janet Martin [a, b, c], Rodrigo Bagur [a], Davy Cheng [b, c, d], Bob Kiaii [e], Ava John-Baptiste [a, b, c, f]

**Affiliations**:

a. Department of Epidemiology & Biostatistics, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

b. Centre for Medical Evidence, Decision Integrity & Clinical Impact (MEDICI), Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

c. Department of Anesthesia & Perioperative Medicine, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

d. Department of Medicine, Division of Critical Care Medicine, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

For peer review only

e. Department of Surgery, Division of Cardiac Surgery, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

f. Schulich Interfaculty Program in Public Health, Western University, London, ON, Canada.

**Corresponding Author:** Dr. Ava John-Baptiste

The Western Centre for Public Health and Family Medicine

Rm 4111, 1465 Richmond St.

London, ON, Canada, N6G 2M1

e. ajohnbap@uwo.ca

t. 519.661.2111 (86265)

**Running head**: Prognosis of frail patients after TAVI

## Abstract

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Frail patients have poorer prognosis following transcatheter aortic valve implantation (TAVI), however no consensus exists on how to measure frailty. Our objectives were to review the literature to identify frailty instruments in use for TAVI recipients and synthesize prognostic data from these studies, in order to inform clinical management of frail patients undergoing TAVI. We systematically reviewed the literature published in 2006 or later. We included studies of patients with aortic stenosis, diagnosed as frail, who underwent a TAVI procedure that reported mortality or clinical outcomes. We categorized the frailty instruments and reported on the prevalence of frailty in each study. We summarized the frequency of clinical outcomes and pooled outcomes from multiple studies. We explored heterogeneity and performed sub-group analysis, where possible. We also used GRADE to assess the overall certainty of the estimates. Of 35 included studies, 15 used single-dimension measures to assess frailty, and 20 used multi-dimensional measures. Prevalence of frailty ranged from 7.21% to 90.07%. Albumin was the most commonly used single-dimension frailty measure and the Fried or modified Fried

For peer review only

phenotype were the most commonly used multi-dimensional measures. Meta-analyses of studies that used either the Fried or modified Fried phenotype showed a 30-day mortality of 7.86% (95% confidence interval, 5.20% to 11.70%) and a 1-year mortality of 26.91% (95% confidence interval, 21.50% to 33.11%). The GRADE system suggests very low and low certainty of the respective estimates. Frailty instruments varied across studies, leading to a wide range of frailty prevalence estimates for TAVI recipients and substantial heterogeneity. The results provide clinicians, patients, and health care administrators, with potentially useful information on the prognosis of frail patients undergoing TAVI, and highlight the need for standardization of frailty measurement to promote consistency.

## Strengths and limitations

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

- This study examines the heterogeneity across different frailty assessment tool and determines the frequency of adverse outcomes and pool the prognosis after transcatheter aortic valve implantation in frail patients.
- This study uses a comprehensive literature search strategy and includes frail patients from randomized controlled trials and observational studies.
- This study excluded studies in which dimensions of frailty were assessed without reference to the goal of frailty assessment.

**Keywords**: frailty; transcatheter aortic valve implantation; systematic review; meta-analysis

Transcatheter aortic valve implantation (TAVI) has become an alternative, less invasive treatment option for patients with severe symptomatic aortic stenosis.[1] The evidence continues to accumulate and synthesis of the evidence to better understand the prognosis of frail patients who undergo TAVI may be helpful.[2]



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Frailty is a biological syndrome characterized by an increased vulnerability to stressors.[3] When exposed to stressors, such as chronic illness and surgery, frail patients are susceptible to adverse events, procedural complications, prolonged recovery, functional decline, and reduced survival.[4] Clinical research has identified frailty as an important risk factor for mortality and morbidity following TAVI.[5] Health economics research has shown that compared to non-frail patients, frail older adults undergoing cardiac surgery incurred substantially higher hospitalization costs.[6] Given the clinical and economic implications of TAVI, searching for and synthesizing outcomes of frailt patients undergoing TAVI may provide information that can help optimize the selection of TAVI candidates and ultimately improve decision-making related to treatment of aortic stenosis. [2]

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

When considering valve procedures, clinical practice guidelines recommend assessing frailty as one component of risk.[7] We performed a systematic review of the literature to identify studies reporting the prognosis of frail patients undergoing TAVI. With no single standard method of measuring frailty and a diversity of frailty measurements, the optimal approach to assessing frailty in patients undergoing TAVI is unclear.[2, 5] We catalogued frailty measures used in identified studies, to perform sub-group analyses for studies using the most common measures.

## Methods

We searched PubMed, EMBASE, PsycINFO, Cochrane Library, Web of Science and ClinicalTrials.gov for articles published between January 2006 and October 10, 2018 (see supplementary appendix A for search strategy). Conference abstracts from relevant conferences held in the last 3 years were also searched. We included studies of patients who underwent a TAVI procedure that reported frailty and its impact on mortality, clinical outcomes, or health-related quality of life. The detailed inclusion and exclusion criteria were described in detail in the protocol.[8] This systematic review and



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

meta-analysis is reported according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) guidelines[9] and follows the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines.[10] The protocol was registered in PROSPERO (CRD42018090597) and published in *BJM Open*.[8]

Two authors independently assessed study eligibility and risk of bias and extracted data. Disagreements were resolved by consulting a third reviewer. The risk of bias in individual studies was appraised independently by two authors using the Quality in Prognosis Studies (QUIPS) tool.[11] We classified studies with four or five low risk domains as having a low risk of bias overall, studies with two or more high risk domains as high risk of bias overall, and the remaining studies as moderate risk of bias overall.

We summarised the method of measuring frailty used in each study including the frailty tool used, dimensions of frailty measured, the cut-off for frail status, and the prevalence of frailty in the study population as measured by the frailty tool. We only extracted data from the most commonly used frailty instruments if multiple frailty instruments were applied in the same patient group. We categorized clinical outcomes and reported the frequency at each time point. Heterogeneity across studies was assessed using the Cochrane Q-statistic ($I^2$).[12] Given the substantial heterogeneity, we pooled and estimated outcomes using random effects model. We pooled dichotomous clinical outcomes using the DerSimonian and Laird random effects model

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

and applying a logit transformation.[13] For the length of hospitalization, we pooled the values, estimating the mean and standard deviation using the random effects model for continuous variables.[14] For studies presenting Kaplan-Meier curves with time to death, we collected the information on numbers at risk and total number of events, and then created a single pooled Kaplan-Meier curve. We pooled time to death data from individual studies to obtain an overall estimate of survival, using an algorithm developed by Guyot et al.[15]

We conducted a sub-group analysis to see if the estimates of mortality rates differed for studies that used the Fried phenotype, the most common multi-dimensional measure, compared to studies that did not use the Fried phenotype. A two-sided p-value of 0.05 or less was considered statistically significant. All analyses were conducted using R software (version 3.5.0). Pre-specified statistical details were described in the protocol. [8]

We used the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) system to conduct an evaluation of the overall estimates based on considerations of risk of bias, consistency, precision, directness, and publication bias.[16] Given that cohort studies of prognosis exclude randomized controlled trial study designs, we did not downgrade the certainty of evidence due to observational study design.

## Patient and Public Involvement

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Due to the nature of the study, patients were not involved in research design or the outcome measures, nor were they involved in data analysis and write-up of the results.

## Results

Our search identified 2,370 records with 1,559 articles remaining after removing duplicates. After screening, 35 studies [17-51] were identified as eligible for inclusion in the review (Figure 1). The eligible studies enrolled a total of 11,323 frail patients with aortic stenosis undergoing TAVI.

The characteristics of the included studies are summarized in supplementary Appendix B. Three studies [38, 41, 51] enrolled patients from the Placement of Aortic Transcatheter Valves (PARTNER) trial reporting separately on outcomes of frail patients; the remaining studies reported on patients from a single cohort or registry. Most studies collected patient data prospectively; twelve studies [19, 21, 22, 24, 30, 33, 36, 39, 41, 43, 44, 46] were conducted retrospectively.

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary appendix C summarizes the risk of bias assessment of individual studies. Of the 35 studies, [12 23, 24, 26, 27, 29, 31, 35, 37, 38, 45, 47, 50] were rated at overall low risk of bias, [18 17-22, 30, 33, 34, 40-43, 46, 48, 49, 51] at moderate risk and 5 [25, 36, 39, 44, 48] at high risk of bias.

Table 1 summarizes frailty assessment in patients undergoing TAVI. Fifteen studies [17-31] used single-dimension measures, and 20 studies [32-51] used multi-dimensional measures. The prevalence of frailty varied widely among studies that assessed frailty with single dimension measures, ranging from 7.21% to 90.07%. Albumin, body mass index, and Katz Activity of Daily Living were the four most commonly used single-dimension measures when assessing frailty in TAVI patients. However, even with the same measure, different cut-points or definitions of frailty were used. For example, three studies [19, 21, 22] used albumin to assess frailty; two [19, 21] defined frailty as albumin level below 4g/dL, and one [22] as albumin level below 3.5g/dL.

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The prevalence of frailty reported by studies that assessed frailty using multi-dimensional measures ranged from 15.23% to 84.67%. Most of these studies assessed frailty based on the Fried frailty phenotype [3]; one study [32] assessed frailty based on the accumulated deficits frailty index. Of the 20 studies reporting multi-dimensional measures, four [44-46, 49] used the original Fried frailty phenotype and eight [33, 36-39, 41, 42, 51] modified the Fried frailty phenotype by examining fewer dimensions, altering cut-off values or measuring the same domains with different criteria. Among the eight studies [33, 36-39, 41, 42, 51] reporting the modified Fried frailty phenotype, measures used to assess mobility and disability were identical. Measures used to assess nutrition were different; seven studies [33, 36-38, 41, 42, 51] measured serum albumin, and 1 study [39] measured weight loss.

Supplementary appendix D summarized prognosis of frail TAVI recipients reported for each study. Fifteen studies [17, 19, 25-29, 32-34, 38, 39, 41, 48, 51] reported 30-day mortality, which ranged from 2.83% to 25%; the combined 30-day mortality estimate was 7.73% (95% confidence interval 5.20% to 11.33%, table 2, figure 2a, figure 2b.). Combining three studies [33, 38, 39] that measured frailty using the modified Fried frailty



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

phenotype, we observed a 30-day mortality of 7.86% (5.20% to 11.70%, table 2, figure 2c, figure 2d.).

Twelve studies [23, 28, 29, 32, 33, 35, 38, 41, 44, 47, 48, 51] reported 1-year mortality, ranging from 14.8% to 37.5%. The combined 1-year mortality estimate was 24.13% (20.91% to 27.68%, table 2, figure 3a, figure 3b.). When pooling two studies [33, 44] that used the Fried or modified Fried frailty phenotype to assess frailty, the estimated 1-year mortality was 26.91% (21.50% to 33.11%, table 2, figure 3c, figure 3d.). Subgroup analyses of studies reporting frailty measurement using the Fried phenotype compared to non-Fried phenotype did not find statistical differences in effect estimates on 30-day and 1-year mortality (see supplementary appendix E).

Fifteen studies [20, 21, 23-27, 29, 31, 32, 38, 39, 45, 46, 50] reported survival of frail patients after TAVI using a Kaplan-Meier curve. The combined survival estimates at 1, 2, and 3 years were 78.6% (77.1% to 80.1%, table 2), 67.8% (65.7% to 70%, table 2), and 50.8% (45.2% to 57.2%, table 2), respectively. Combining the studies that used the Fried or modified Fried phenotype, we found survival estimates at 1, 2, and 3 years were 73% (68.8% to 77.5%, table 2), 64.5% (56.4% to 73.9%, table 2), and 58.9% (49%



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

to 70.9%, table 2), respectively. Details of survival are provided in supplementary appendix F.

Two studies [40, 42] measured health-related quality of life. Kobe et al [40] assessed quality of life before and 30 days after TAVI using the Short Form-36 questionnaire; they found that at 30-day follow-up, the mean scores of all but role physical and social functioning were significantly lower for frail patients. Okoh et al [42] assessed quality of life preoperatively and at 30-day follow-up using the 12-item Kansas City Cardiomyopathy Questionnaire (KCCQ); they found that at 30 days, frail patients reported worsening in two domains, KCCQ-symptoms and KCCQ physical limitation, but quality of life improved slightly overall (supplementary appendix G).

Other commonly reported outcomes measuring the prognosis of frail TAVI recipients include procedural cardiac tamponade (ranging from 2% to 4%, reported by 3 studies [26, 29, 31]), convert to open heart surgery (ranging from 1.2% to 2.27%, reported by 2 studies [27, 29]), procedural life-threatening bleeding (ranging from 7.1% to 16.7%, reported by 5 studies [26, 28, 29, 31, 37]) and 2-valve implantation (ranging from 1.8% to



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2.55%, reported by 2 studies [29, 31]). Seven studies [30, 31, 36, 37, 39, 42 43] reported the mean length of hospitalization, ranging from 6 days to 12.1 days.

The GRADE certainty assessment per outcome, together with the pooled effects, is provided in table 2. Thirty-day mortality, procedural cardiac tamponade, procedural life threatening bleeding, and conversion to open heart surgery were rated as low certainty of evidence; the remaining outcomes were rated as very low certainty of evidence.

## Discussion

We found that multi-dimensional measures are more commonly used than single-dimension measures. Even with the same frailty measure, different definitions or cut-offs were used. The most frequently used frailty measure in TAVI the studies we identified was the modified Fried phenotype, in which disability, muscle strength, mobility, and nutrition were assessed. Approaches to modifying the Fried phenotype included measuring fewer domains, using different cut-offs, or using different tools to assess the same domain.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Greater heterogeneity of meta-analyses that included single measures suggests single measures did not measure the same frailty construct and did not reliably measure frailty. Single measures included a mix of biological variables (albumin and BMI) or single performance measures (gait speed or activities of daily living), which address only a single component of the frailty construct. Thus, our study confirms that frailty is a multi-dimensional phenomenon that cannot be captured by a single construct.

The variety of frailty definitions and the diversity of TAVI populations in the studies contribute to the wide range and substantial heterogeneity of patient outcomes after TAVI. We identified studies on a wide array of clinical outcomes for frail patients post-TAVI, but few studies reported on quality of life measures. Using GRADE to assess confidence in prognosis estimates from the meta-analyses, we found very low or low confidence in the overall estimates,

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

mainly due to inconsistency as influenced by heterogeneity of estimates and indirectness of frailty measures as influenced by lack of homogeneity across the TAVI populations identified in the studies.

Previous studies demonstrated that the assessment of frailty significantly enhances prediction of mortality after TAVI when combined with the European system for cardiac operative risk evaluation (Euro SCORE) or the Society of Thoracic Surgeons (STS) score.[47] There have been several studies reviewing frailty in cardiac surgical populations. Kim et al [5] conducted a systematic review of frailty instruments in older adults undergoing cardiac surgical procedures. Kim et al [5] found high quality evidence that used mobility assessment as a single frailty measure and found mobility to be the most frequently assessed domain. Sepehri et al [52] performed a systematic review to demonstrate the association of frailty with negative postoperative outcomes in patients undergoing cardiac surgery. Our study adds to the existing literature as we investigate the frequency of adverse outcomes and pool estimates of survival after TAVI in frail patients from multiple studies.

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The FRAILTY-AVR study [53] examined the validity of frailty measures in predicting mortality amongst TAVI recipients. The study added value to the literature by selecting frailty elements with the greatest predictive value, finding that the Essential Frailty Toolset (EFT) consisting of chair rise, cognition measured by the Mini-Mental State Examination, hemoglobin and serum albumin, performed best for predicting one-year mortality.[53] Due to the focus on predictive validity, the FRAILTY-AVR study [53] did not report outcomes separately for frail patients. As a result, the study did not meet the inclusion criteria for our systematic review, which was focused on prognostic information amongst frail patients only. The FRAILTY-AVR study [53] makes important efforts to define a standard frailty assessment tool. Although the Fried and modified Fried were the most commonly used instruments amongst studies included in our meta-analysis, the FRAILTY-AVR showed the Fried did not perform as well as the EFT in predicting mortality amongst TAVI patients. [53] We suggest use of a standard measure, such as the EFT, can enhance the quality of frailty research in the TAVI patient population. We also recognize that use of a standard frailty measure is unlikely as researchers and clinicians may value use of diverse measures which reflect different aspects of frailty. If the EFT

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

emerges as a standard, it may be used by clinicians to exclude frail patients from treatment, due to concerns about increased mortality. This would limit the opportunity to better understand the prognosis of frail patients undergoing TAVI, which was the primary goal of our study.

This review has several unique strengths. We performed a comprehensive literature search to identify both published and unpublished studies, in addition to searching citations from previous reviews. We included prognostic data from randomized controlled trials and observational studies. Using the QUIPS tool, two reviewers independently assessed the risk of bias, and the use of the GRADE system to assess the certainty of evidence offers a structured and transparent evaluation of our findings. We systematically reviewed the operationalization of frailty assessment in TAVI patients, and pooled clinical outcomes of frail TAVI recipients. We tested for heterogeneity and attempted to address heterogeneity by performing sensitivity analysis and sub-group analysis.

This review has some important limitations. Given the limited data reported by the



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

included studies, we were unable to perform meta-regression to further investigate the potential sources of heterogeneity and to determine the influence of mean age on outcomes. We therefore explored the causes and types of heterogeneity relying on the investigation of the $I^2$ statistic, which may be imprecise when the number of studies is small.[54] When extracting data, we encountered several studies that applied multiple frailty instruments in the same patient group, and in this situation, we only extracted data from the most commonly used frailty instrument, and this may introduce selection bias. Some studies defined an intermediate 'pre-frail' group, but we did not find sufficient data to synthesize outcomes for this important sub-group. Though less

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

vulnerable than the frail group, pre-frail patients may be at higher risk than robust patients for experiencing adverse outcomes.[55, 56] Individual-patient level data were not available, precluding adjustment for any study level differences in clinical or procedural variables that may have influenced prognosis across the cohorts. Therefore, clinical heterogeneity could not be ruled-out and along with high levels of heterogeneity, resulted in lower GRADE evaluations. Data on the quality of life was reported by few studies, so we were not able to assess the change in quality of life for frail patients after TAVI. The aim of this study was to characterize prognosis for frail patients undergoing TAVI, therefore, we did not directly compare prognosis to other groups of patients or to frail patients undergoing different therapies, nor were we able to determine which frailty measures perform best as prognostic tools for TAVI recipients.

When selecting candidates to undergo TAVI, several multivariate risk scores have been widely used to estimate operative mortality based on patient characteristics.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The STS score and the EuroSCORE are the most commonly used scoring systems.[57, 58] However, a disadvantage of both scores is that the main variables for scoring perioperative risk are medical diagnoses and comorbidities, which may not reflect the true 'biological status' of the patient.[57, 58] When considering valve procedures for patients, clinical practice guidelines recommend assessing frailty as one component of risk.[5, 7] Although a large number of frailty measures exist, there is currently little consensus on the optimal approach to assessing frailty in patients undergoing TAVI.[2] Frailty has consistently been shown to significantly predict mortality [52] and post-operative delirium [59], even after controlling for other risk factors, suggesting that use of any frailty assessment is better than none when selecting patients for TAVI. Systematically reviewing the operationalization of frailty assessment in TAVI patients and pooling clinical outcomes of frail TAVI recipients will help better understand how frailty is assessed among TAVI patients, provide information on the prognosis of frail patients after TAVI, and can ultimately improve decisions related to treatment of AS.

To help achieve consensus on frailty measures to be applied in TAVI recipients, future studies should evaluate the prognostic value of frailty measures in TAVI

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

recipients and determine the additional prognostic value of frailty measurement in addition to these established risk scores. Future studies should also compare prognosis of frail patients undergoing TAVI to frail patients undergoing surgical intervention or medical therapy. Few studies reported quality of life measures. In order to address the gaps in the literature future studies should measure quality of life before and after TAVI with use of standardized quality of life measurement tools such as the Short-Form 36.

In conclusion, frailty instruments for TAVI recipients varied across studies, leading to a range of frailty prevalence estimates and substantial heterogeneity. The results of this systematic review provide clinicians, patients, and health care administrators, with potentially useful evidence on the prognosis of frail patients.

**Acknowledgement:** We gratefully thank Dr. Farid Foroutan for his assistance with meta-analysis of survival data.

**Ethics committee approval:** Due to the nature of the study, there are no ethical concerns and informed consent was not required.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Contributorship statement:** Zhe Li, Emily Dawson, Jessica Moodie, Janet Martin, Rodrigo Bagur, Davy Cheng, Bob Kiaii, and Ava John-Baptiste proposed and designed the purpose, review questions, and methods of this study. Searching strategy was developed by Jessica Moodie, and revised by Zhe Li, Emily Dawson, and Ava John-Baptiste. Zhe Li, Emily Dawson and Ava John-Baptiste screened and critically appraised the articles. Data analysis was primarily conducted by Zhe Li. Ava John-Baptiste provided primary academic supervision. Zhe Li drafted the manuscript. All co-authors have contributed to the revision of the manuscript.

**Competing interests:** There is no competing interest to declare.

**Funding:** This work was supported by the Academic Medical Organization of Southwestern Ontario (grant number: INN 17-001).

**Data sharing statement**: There is no additional unpublished data from the study. All data are provided in full in the main manuscript document.

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

## References

[1] Siontis G, Overtchouk P, Cahill TJ, et al. Transcatheter aortic valve implantation vs. surgical aortic valve replacement for treatment of symptomatic severe aortic stenosis: an updated meta-analysis. European Heart Journal. 2019, 40: 3143-53

[2] Puri R, Iung B, Cohen DJ, Rodes-Cabau J. TAVI or No TAVI: identifying patients unlikely to benefit from transcatheter aortic valve implantation. European heart journal. 2016 Jan 26;37(28):2217-25.

[3] Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: evidence for a phenotype. The Journals of Gerontology Series A: Biological Sciences and Medical Sciences. 2001 Mar 1;56(3):M146-57.

[4] Shamliyan T, Talley KM, Ramakrishnan R, Kane RL. Association of frailty with survival: a systematic literature review. Ageing research reviews. 2013 Mar 1;12(2):719-36.

[5] Kim DH, Kim CA, Placide S, Lipsitz LA, Marcantonio ER. Preoperative frailty assessment and outcomes at 6 months or later in older adults undergoing cardiac



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

surgical procedures: a systematic review. Annals of internal medicine. 2016 Nov 1;165(9):650-60.

[6] Goldfarb M, Bendayan M, Rudski LG, et al. Cost of cardiac surgery in frail compared with nonfrail older adults. Canadian Journal of Cardiology. 2017 Aug 1;33(8):1020-6.

[7] Nishimura RA, Otto CM, Bonow RO, et al. 2017 AHA/ACC focused update of the 2014 AHA/ACC guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Journal of the American College of Cardiology. 2017 Jul 3;70(2):252-89.

[8] Li Z, Dawson E, Moodie J et al. Frailty in patients undergoing transcatheter aortic valve implantation: a protocol for a systematic review. *BMJ Open*. 2019; 9: e024163.

[9] Moher D, Liberati A, Tetzlaff J, Altman DG. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Annals of internal medicine. 2009 Aug 18;151(4):264-9.

[10] Stroup DF, Berlin JA, Morton SC, et al. Meta-analysis of observational studies in epidemiology: a proposal for reporting. Jama. 2000 Apr 19;283(15):2008-12.

[11] Hayden JA, van der Windt DA, Cartwright JL, Côté P, Bombardier C. Assessing bias in studies of prognostic factors. Annals of internal medicine. 2013 Feb 19;158(4):280-6.

[12] Jackson D, White IR, Thompson SG. Extending DerSimonian and Laird's methodology to perform multivariate random effects meta-analyses. Statistics in medicine. 2010 May 30;29(12):1282-97.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[13] Marín-Martínez F, Sánchez-Meca J. Weighting by inverse variance or by sample size in random-effects meta-analysis. Educational and Psychological Measurement. 2010 Feb;70(1):56-73.

[14] Guyot P, Ades AE, Ouwens MJ, Welton NJ. Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan-Meier survival curves. BMC medical research methodology. 2012 Dec;12(1):9.

[15] Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. Bmj. 2003 Sep 4;327(7414):557-60.

[16] Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. Bmj. 2008 Apr 24;336(7650):924-6.

[17] Alfredsson J, Stebbins A, Brennan JM, et al. Gait speed predicts 30-day mortality after transcatheter aortic valve replacement: results from the Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy Registry. Circulation. 2016 Apr 5;133(14):1351-9.

[18] Bagienski M, Kleczynski P, Dziewierz A, et al. Incidence of postoperative delirium and its impact on outcomes after transcatheter aortic valve implantation. The American journal of cardiology. 2017 Oct 1;120(7):1187-92.

[19] Bogdan A, Barbash IM, Segev A, et al. Albumin correlates with all-cause mortality in elderly patients undergoing transcatheter aortic valve implantation. EuroIntervention: journal of EuroPCR in collaboration with the Working Group on Interventional Cardiology of the European Society of Cardiology. 2016 Oct;12(8):e1057-64.

[20] Cockburn J, Singh MS, Rafi NH, Dooley M, Hutchinson N, Hill A, Trivedi U, de Belder A, Hildick-Smith D. Poor mobility predicts adverse outcome better than other



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

frailty indices in patients undergoing transcatheter aortic valve implantation. Catheterization and Cardiovascular Interventions. 2015 Dec 1;86(7):1271-7.

[21] Grossman Y, Barbash IM, Fefer P, et al. Addition of albumin to traditional risk score improved prediction of mortality in individuals undergoing transcatheter aortic valve replacement. Journal of the American Geriatrics Society. 2017 Nov;65(11):2413-7.

[22] Koifman E, Magalhaes MA, Ben-Dor I, et al. Impact of pre-procedural serum albumin levels on outcome of patients undergoing transcatheter aortic valve replacement. The American journal of cardiology. 2015 May 1;115(9):1260-4.

[23] Kleczynski P, Dziewierz A, Bagienski M, et al. Impact of frailty on mortality after transcatheter aortic valve implantation. American heart journal. 2017 Mar 1;185:52-8.

[24] Mok M, Allende R, Leipsic J, et al. Prognostic value of fat mass and skeletal muscle mass determined by computed tomography in patients who underwent transcatheter aortic valve implantation. The American journal of cardiology. 2016 Mar 1;117(5):828-33.

[25] Martin GP, Sperrin M, Ludman PF, et al. Do frailty measures improve prediction of mortality and morbidity following transcatheter aortic valve implantation? An analysis of the UK TAVI registry. BMJ open. 2018 Jun 1;8(6):e022543.

[26] Puls M, Sobisiak B, Bleckmann A, et al. Impact of frailty on short-and long-term morbidity and mortality after transcatheter aortic valve implantation: risk assessment by Katz Index of activities of daily living. EuroIntervention: journal of EuroPCR in collaboration with the Working Group on Interventional Cardiology of the European Society of Cardiology. 2014 Sep;10(5):609-19.

[27] Rodés-Cabau J, Webb JG, Cheung A, et al. Transcatheter aortic valve implantation for the treatment of severe symptomatic aortic stenosis in patients at very high or



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

prohibitive surgical risk: acute and late outcomes of the multicenter Canadian experience. Journal of the American College of Cardiology. 2010 Mar 16;55(11):1080-90.

[28] Stortecky S, Windecker S, Pilgrim T, et al. The association of body mass index with clinical outcome in patients with severe aortic stenosis undergoing transcatheter aortic valve implantation. EuroIntervention. 2012; 8 SUPPLEMENT N

[29] Shimura T, Yamamoto M, Kano S, et al. Impact of the clinical frailty scale on outcomes after transcatheter aortic valve replacement. Circulation. 2017 May 23;135(21):2013-24.

[30] Traynor MM, Greason KL, Nkomo VT, et al. Assisted care as a baseline patient risk characteristic affecting the outcome of transcatheter aortic valve insertion. Catheterization and Cardiovascular Interventions. 2017 Jun 1;89(7):1268-72.

[31] Yamamoto M, Hayashida K, Watanabe Y, et al. Effect of body mass index< 20 kg/m2 on events in patients who underwent transcatheter aortic valve replacement. The American journal of cardiology. 2015 Jan 15;115(2):227-33.

[32] Bureau ML, Liuu E, Christiaens L, et al. Using a multidimensional prognostic index (MPI) based on comprehensive geriatric assessment (CGA) to predict mortality in elderly undergoing transcatheter aortic valve implantation. International journal of cardiology. 2017 Jun 1;236:381-6.

[33] Chauhan D, Haik N, Merlo A, et al. Quantitative increase in frailty is associated with diminished survival after transcatheter aortic valve replacement. American heart journal. 2016 Dec 1;182:146-54.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[34] Capodanno D, Barbanti M, Tamburino C, et al. A simple risk tool (the OBSERVANT score) for prediction of 30-day mortality after transcatheter aortic valve replacement. The American journal of cardiology. 2014 Jun 1;113(11):1851-8.

[35] Eichler S, Salzwedel A, Harnath A, et al. Nutrition and mobility predict all-cause mortality in patients 12 months after transcatheter aortic valve implantation. Clinical Research in Cardiology. 2018 Apr 1;107(4):304-11.

[36] Ghatak A, O'Neill B, Patel R, et al. Frailty is a major determinant of length of stay after transcatheter aortic valve replacement. Journal of the American College of Cardiology. 2012, 60 (17 supplement) B253.

[37] Green P, Woglom AE, Genereux P, et al. The impact of frailty status on survival after transcatheter aortic valve replacement in older adults with severe aortic stenosis: a single-center experience. JACC: Cardiovascular Interventions. 2012 Sep 1;5(9):974-81.

[38] Green P, Arnold SV, Cohen DJ, et al. Relation of frailty to outcomes after transcatheter aortic valve replacement (from the PARTNER trial). The American journal of cardiology. 2015 Jul 15;116(2):264-9.

[39] Huded CP, Huded JM, Friedman JL, et al. Frailty status and outcomes after transcatheter aortic valve implantation. The American journal of cardiology. 2016 Jun 15;117(12):1966-71.

[40] Kobe AR, Meyer A, Elmubarak H, et al. Frailty assessed by the FORECAST is a valid tool to predict short-term outcome after transcatheter aortic valve replacement. Innovations. 2016 Nov;11(6):407-13.

[41] Maniar H, Zajarias A, Quader N, Arnold S, Cohen D, Mack M, Melby S, Alu M, Ayele G, Lindman B, Lasala J. THE ASSOCIATION OF FRAILTY WITH MORTALITY IN HIGH RISK PATIENTS UNDERGOING TRANSCATHETER AORTIC VALVE



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REPLACEMENT: AN ANALYSIS OF THE PARTNER II ALTERNATIVE ACCESS COHORTS. Journal of the American College of Cardiology. 2016 Apr 5;67(13 Supplement):241.

[42] Okoh AK, Chauhan D, Kang N, et al. The impact of frailty status on clinical and functional outcomes after transcatheter aortic valve replacement in nonagenarians with severe aortic stenosis. Catheterization and Cardiovascular Interventions. 2017 Nov 15;90(6):1000-6.

[43] Patel J, Banga S, Kim M, et al. Frailty profile is independently associated with higher cost for patients undergoing transcatheter aortic valve replacement for symptomatic severe aortic stenosis: a single center experience. Journal of the American College of Cardiology. 2016; 68(18 supplement) B277.

[44] Rabinovitz E, Finkelstein A, Assa EB, et al. Norton scale for predicting prognosis in elderly patients undergoing trans-catheter aortic valve implantation: A historical prospective study. Journal of cardiology. 2016 Jun 1;67(6):519-25.

[45] Rodríguez-Pascual C, Paredes-Galán E, Ferrero-Martínez AI, et al. The frailty syndrome and mortality among very old patients with symptomatic severe aortic stenosis under different treatments. International journal of cardiology. 2016 Dec 1;224:125-31.

[46] Rogers T, Alraies MC, Pacha HM, et al. Clinical frailty as an outcome predictor after transcatheter aortic valve implantation. The American journal of cardiology. 2018 Apr 1;121(7):850-5.

[47] Schoenenberger AW, Moser A, Bertschi D, et al. Improvement of risk prediction after transcatheter aortic valve replacement by combining frailty with conventional risk scores. JACC: Cardiovascular Interventions. 2018 Feb 19;11(4):395-403.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[48] Steinvil A, Buchanan KD, Kiramijyan S, et al. Utility of an additive frailty tests index score for mortality risk assessment following transcatheter aortic valve replacement. American heart journal. 2018 Jun 1;200:11-6.

[49] Shi S, Afilalo J, Lipsitz LA, et al. Frailty phenotype and deficit accumulation Frailty Index in predicting recovery after transcatheter and surgical aortic valve replacement. The Journals of Gerontology: Series A. 2018 Aug 25.

[50] Skaar E, Eide LS, Norekvål TM, et al. A novel geriatric assessment frailty score predicts 2-year mortality after transcatheter aortic valve implantation. European Heart Journal-Quality of Care and Clinical Outcomes. 2018 Oct 31;5(2):153-60.

[51] Zajarias A, Maniar H, Arnold S, et al. The association of frailty with mortality in high risk patients undergoing transcatheter aortic valve replacement: an analysis of the partner ii b transfemoral cohort. Journal of the American College of Cardiology. 2016 Apr 5;67(13 Supplement):249.

[52] Sepehri A, Beggs T, Hassan A, Rigatto C, Shaw-Daigle C, Tangri N, Arora RC. The impact of frailty on outcomes after cardiac surgery: a systematic review. The Journal of thoracic and cardiovascular surgery. 2014 Dec 1;148(6):3110-7.

[53] Afilalo J, Lauck S, Kim DH, Lefèvre T, Piazza N, Lachapelle K, Martucci G, Lamy A, Labinaz M, Peterson MD, Arora RC. Frailty in older adults undergoing aortic valve replacement: the FRAILTY-AVR study. Journal of the American College of Cardiology. 2017 Jul 31;70(6):689-700.

[54] von Hippel PT. The heterogeneity statistic I 2 can be biased in small meta-analyses. BMC Medical Research Methodology. 2015 Dec;15(1):35.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

[55] Bouillon K, Kivimaki M, Hamer M, Sabia S, Fransson EI, Singh-Manoux A, Gale CR, Batty GD. Measures of frailty in population-based studies: an overview. BMC geriatrics. 2013 Dec;13(1):64.

[56] Frost R, Belk C, Jovicic A, Ricciardi F, Kharicha K, Gardner B, Iliffe S, Goodman C, Manthorpe J, Drennan VM, Walters K. Health promotion interventions for community-dwelling older people with mild or pre-frailty: a systematic review and meta-analysis. BMC geriatrics. 2017 Dec;17(1):157.

[57] Sündermann S, Dademasch A, Praetorius J, Kempfert J, Dewey T, Falk V, Mohr FW, Walther T. Comprehensive assessment of frailty for elderly high-risk patients undergoing cardiac surgery. European Journal of Cardio-Thoracic Surgery. 2011 Jan 1;39(1):33-7.

[58] Vahanian A, Otto CM. Risk stratification of patients with aortic stenosis. European heart journal. 2010 Jan 4;31(4):416-23.

[59] Jung P, Pereira MA, Hiebert B, Song X, Rockwood K, Tangri N, Arora RC. The impact of frailty on postoperative delirium in cardiac surgery patients. The Journal of thoracic and cardiovascular surgery. 2015 Mar 1;149(3):869-75.

For peer review only

## Figure legends

**Figure 1. PRISMA flow diagram of included studies**

PRISMA, reporting in systematic reviews and meta-analyses. N, number of studies.

**Figure 2a. Meta-analysis of 30-day mortality in frail patients after TAVI.**

Frailty was measured using sing and multi-dimensional measures.

For peer review only

The squares indicate the 30-day mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 30-day mortality

**Figure 2b. Funnel plots, using data from all studies that reported 30-day mortality**

y-axis is the standard error of the 30-day mortality

x-axis is the 30-day mortality

**Figure 2c. Meta-analysis of 30-day mortality in frail patients after TAVI.**

Frailty was measured using modified Fried frailty phenotype.

The squares indicate the 30-day mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 30-day mortality

**Figure 2d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.**

For peer review only

y-axis is the standard error of the 30-day mortality

x-axis is the 30-day mortality

**Figure 3a. Meta-analysis of 1-year mortality in frail patients after TAVI.**

Frailty was measured using single and multi-dimensional measures.

The squares indicate the 1-year mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 1-year mortality

**Figure 3b. Funnel plots, using data from all studies that reported 30-day mortality**

y-axis is the standard error of the 1-year mortality

x-axis is the 1-year mortality

**Figure 3c. Meta-analysis of 1-year mortality in frail patients after TAVI.**

Frailty was measured using the Fried frailty phenotype.

The squares indicate the 1-year mortality reported by each study

For peer review only

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 1-year mortality

**Figure 3d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.**

y-axis is the standard error of the 1-year mortality

x-axis is the 1-year mortality

Table 1. Frailty assessment in patients undergoing TAVI

| Studies that used a single dimensionto assess frailty | | | | | |
|---|---|---|---|---|---|
| **Study, year** | **Measure** | **Dimensions** | **Definition** | **Total N** | **Frail n (%)** |
| Alfredsson, 2016 [17 a] | Gait speed | mobility | <0.83m/s or <6 s | 8039 | 6100 (75.88%) |
| Bagienski, 2017 [18 b] | Katz ADL | disability | <6 points | 141 | 127 (90.07%) |
| Bogdan, 2016 [19] | Albumin | nutrition | ≤4g/dl | 150 | 79 (52.67%) |
| Cockburn, 2015 [20] | Brighton mobility index | mobility | Poor mobility | 312 | 65 (20.83%) |
| Grossman, 2017 [21] | Albumin | nutrition | <4 g/dl | 426 | 192 (45.07%) |
| Koifman, 2015 [22 c] | Albumin | nutrition | <3.5g/dl | 476 | 238 (50%) |
| Kleczynski, 2017 [23] | ISAR | unclear | ≥2 points | 101 | 53 (52.48%) |
| Mok, 2016 [24] | Sarcopenia | nutrition | skeletal muscle mass index 2 SDs less than the mean SMM of young, healthy gender-specific reference ranges | 460 | 293 (63.70%) |
| Martin, 2018 [25] | CSHA score (1-7) | physical function | Scores 5-7 | 2624 | 1043 (39.75%) |
| Puls, 2014 [26] | Katz ADL | disability | <6 points | 300 | 144 (48%) |
| Rodes-Cabau, 2010 [27] | Clinical judgment | subjective | Unclear | 339 | 85 (25.07%) |
| Stortecky, 2012 [28] | BMI | nutrition | <20kg/m$^2$ | 256 | 24 (9.38%) |
| Shimura, 2017 [29] | CFS | subjective | ≥5 points (score ranges 0-9) | 1215 | 353 (29.05%) |
| Traynor, 2017 [30] | Assisted care | unclear | Need assisted care | 597 | 60 (10.05%) |
| Yamamoto, 2015 [31] | BMI | nutrition | <20kg/m$^2$ | 777 | 56 (7.21%) |

| Studies that used multiple dimensions to assess frailty | | | | | | |
|---|---|---|---|---|---|---|
| **Study, year** | **Name** | **Measures** | **Dimensions** | **Definition** | **Total N** | **Frail n (%)** |



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | | |
|---|---|---|---|---|---|---|
| Bureau, 2017 [32] | MPI | ADL | disability | MPI≥0.34 (the sum of all domain values is divided by 8 to obtain the MPI score between 0 and 1) | 116 | 71 (61.21%) |
| | | IADL | disability | | | |
| | | SPMSQ | cognition | | | |
| | | CIRS-CI | medical | | | |
| | | MNA-SF | nutrition | | | |
| | | ESS | medical | | | |
| | | Number of medications | medical | | | |
| | | Social support network | living status | | | |
| Chauhan, 2016 [33 d] | Modified Fried phenotype | ADL | disability | Presence of 2 or more criteria | 343 | 233 (67.93%) |
| | | Hand strength | muscle strength | | | |
| | | Gait speed | mobility | | | |
| | | Albumin | nutrition | | | |
| Capodanno, 2014 [34] | GSS | Not reported | not reported | Value of 2 or 3 | 1256 | 306 (24.36%) |
| Eichler, 2017 [35] | FI | MMSE | cognition | ≥3 points (score ranges 0-7) | 333 | 152 (45.65%) |
| | | MNA | nutrition | | | |
| | | ADL | disability | | | |
| | | IADL | disability | | | |
| | | Time up and go test | mobility | | | |
| | | Subjective mobility disability | mobility | | | |
| Ghatak, 2012 [36] | Modified Fried phenotype | Albumin | nutrition | Presence of 3 or more criteria | 45 | 22 (48.89%) |
| | | Katz ADL | disability | | | |
| | | 5MWT | mobility | | | |
| | | Grip strength | muscle strength | | | |
| Green, 2015 [37] | Modified Fried phenotype | Gait speed | mobility | Frailty score ≥6 | 244 | 110 (45.08%) |
| | | Grip strength | muscle strength | | | |
| | | Albumin | nutrition | | | |
| | | ADL | disability | | | |
| Green, 2012 [38] | Modified Fried phenotype | Gait speed | mobility | Frailty score ≥5 points | 159 | 76 (47.80%) |
| | | Grip strength | muscle strength | | | |
| | | Albumin | nutrition | | | |
| | | ADL | disability | | | |
| Huded, 2016 [39] | Modified Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 191 | 64 (33.51%) |
| | | Grip strength | muscle strength | | | |
| | | 5MWT | mobility | | | |
| | | Katz ADL | disability | | | |
| Kobe, 2016 [40] | FORCAST | Chair rise | muscle strength | ≥ 4 points (score ranges 0-12) | 130 | 71 (54.62%) |
| | | Weakness | muscle strength | | | |
| | | Stair | mobility | | | |
| | | CFS | subjective | | | |
| | | Creatinine level | medical | | | |
| Maniar, 2016 [41] | Modified Fried phenotype | Serum albumin | nutrition | ≥6 points (score ranges 0-12) | 219 | 73 (33.3%) |
| | | Gait speed | mobility | | | |
| | | Grip strength | muscle strength | | | |
| | | Katz ADL | disability | | | |
| Okoh, 2017 [42] | Modified Fried phenotype | Hand grip strength | muscle strength | FI ≥3/4 | 75 | 30 (40%) |
| | | Gait speed | mobility | | | |
| | | Serum albumin | nutrition | | | |
| | | ADL | disability | | | |
| Patel, 2016 [43] | NA | Gait speed | mobility | Gait speed≥6s or/and albumin<3.5g/dl | 117 | 31 (26.50%) |
| | | Albumin | nutrition | | | |
| Rabinovitz, 2016 [44] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 302 | 46 (15.23%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | | |
|---|---|---|---|---|---|---|
| | | Walk speed | mobility | | | |
| | | Low physical activity | physical activity | | | |
| Rodriguez-Pascual, 2016 [45] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 109 | 68 (62.39%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | physical activity | | | |
| Rogers, 2018 [46] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 544 | 242 (44.49%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | disability | | | |
| Schoenenberger, 2018 [47] | NA | MMSE | cognition | ≥3 points (score ranges 0-7) | 330 | 169 (51.21%) |
| | | Time up and go | mobility | | | |
| | | MNA | nutrition | | | |
| | | Basic ADL | disability | | | |
| | | Incremental ADL | disability | | | |
| Steinvil, 2018 [48] | NA | BMI | nutrition | Presence of 3 or more criteria | 498 | 232 (46.59%) |
| | | Albumin | nutrition | | | |
| | | Katz ADL | disability | | | |
| | | Grip strength | muscle strength | | | |
| | | Walk test | mobility | | | |
| Shi, 2018 [49] | Fried phenotype | Weight loss | nutrition | Presence of 3 or more criteria | 137 | 116 (84.67%) |
| | | Exhaustion | Exhaustion | | | |
| | | Minnesota leisure time activity | Physical activity | | | |
| | | 5m walk test | mobility | | | |
| | | Grip strength | muscle strength | | | |
| Skaar, 2018 [50] | Geriatric assessment tool (0-9) | MMSE | cognition | Scores≥4 | 142 | 34 (23.94%) |
| | | Nottingham extended ADL | disability | | | |
| | | BMI<20.5 | nutrition | | | |
| | | Low energy | exhaustion | | | |
| | | Weight loss | nutrition | | | |
| | | Chair stand | muscle strength | | | |
| | | Charlson comorbidity index | comorbidity | | | |
| | | Hospital anxiety and depression scale | psychological | | | |
| Zajarias, 2016 [51] | Modified Fried phenotype | Albumin | nutrition | ≥6 points (score ranges 0-12) | 553 | 265 (47.92%) |
| | | Gait speed | mobility | | | |
| | | Grip strength | muscle strength | | | |
| | | Katz ADL | disability | | | |

a. Alfredsson (2016) enrolled patient populations from the STS/ACC registry. Chauhan (2016), Green (2012), Green (2015), Huded (2016), Okoh (2017), Rogers (2018), Steinvil (2018), Traynor (2017), and Bagienski (2017) enrolled patients from the participating centres of STS/ACC registry.

b. Bagienski (2017) and Kleczynski (2017) enrolled patients from the same medical centre but used different frailty definitions.

c. Koifman (2015), Rogers (2018), and Steinvil (2018) enrolled patients from the same medical centre but used different frailty definitions.

d. Chauhan (2016) and Okoh (2017) enrolled patients from the same medical centre but used different frailty definitions.

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

N, the number of patients. SMM, skeletal muscle mass. 5MWT, 5-meter walk test. ADL, activities of daily living. MPI, multidimensional prognostic index. SPMSQ, short portable mental status questionnaire. CIRS-CI, cumulative illness rating scale comorbidity index. MNA-SF, mini-nutritional assessment short form. ESS, Exton Smith scale. FI, frailty index. GSS, geriatric status scale. MMSE, mini-mental state examination. BMI, body mass index. ISAR, identification of seniors at risk. FORCAST, frailty predicts death one year after elective cardiac surgery test. CFS, clinical frailty scale. NA, not applicable.

Table 2. Results of meta-analysis and GRADE assessment**

| Effects | | | | | GRADE assessment | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| # included study | Frailty measures* | # individuals | # events | Estimate (95% CI) | Study design | Risk of bias | Inconsistency | Indirectness | Imprecision | Other considerations | Certainty |
| Procedural death | | | | | | | | | | | |
| 4 | Single | 6682 | 350 | 6.49% (2.46%-16.4%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 30-day mortality | | | | | | | | | | | |
| 9 | All | 8743 | 545 | 7.73% (5.20%-11.33%) | Observational | Strongly serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 6 | Multi | 1016 | 82 | 8.31% (6.68%-10.28%) | Observational | Serious | Not serious | Strongly serious | Not serious | None | Low |
| 3 | Modified Fried | 407 | 31 | 7.86% (5.20%-11.70%) | Observational | Serious | Not serious | Strongly serious | Serious | None | Very low |
| Cardiovascular death at 30 days | | | | | | | | | | | |
| 2 | Single | 6453 | 259 | 3.37% (1.93%-5.81%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| 6-month mortality | | | | | | | | | | | |
| 2 | Multi | 187 | 30 | 16.12% (11.50%-22.13%) | Observational | Serious | Not serious | Strongly serious | Strongly serious | None | Very low |
| 1-year mortality | | | | | | | | | | | |
| 8 | All | 922 | 217 | 24.13% (20.91%-27.68%) | Observational | Serious | Not serious | Strongly serious | Serious | None | Very low |
| 6 | Multi | 845 | 191 | 22.75% (20.03%-25.71%) | Observational | Serious | Not serious | Strongly serious | Serious | None | Very low |
| 2 | Fried and modified Fried | 223 | 60 | 26.91% (21.50%-33.11%) | Observational | Serious | Not serious | Strongly serious | Strongly serious | None | Very low |
| Survival | | | | | | | | | | | |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 15 | All | 3296 | NA | 1-year survival: 78.6% (77.1%-80.1%) 2-year survival: 67.8% (65.7%-70%) 3-year survival: 50.8% (45.2%-57.2%) | Observational | Strongly serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 4 | Fried and modified Fried | 484 | NA | 1-year survival: 73% (68.8%-77.5%) 2-year survival: 64.5% (56.4%-73.9%) 3-year survival: 58.9% (49%-70.9%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Procedural acute kidney injury | | | | | | | | | | | |
| 3 | Single | 6509 | 450 | 9.57% (5.28%-16.73%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural cardiac tamponade | | | | | | | | | | | |
| 3 | Single | 553 | 17 | 3.19% (1.99%-5.07%) | Observational | Not serious | Not serious | Strongly serious | Not serious | None | Low |
| Convert to open heart surgery | | | | | | | | | | | |
| 2 | Single | 438 | 9 | 2.11% (1.10%-4.00%) | Observational | Not serious | Not serious | Strongly serious | Not serious | None | Low |
| Procedural life-threatening bleeding | | | | | | | | | | | |
| 5 | All | 653 | 63 | 9.75% (7.69%-12.29%) | Observational | Not serious | Not serious | Strongly serious | Not serious | None | Low |
| Procedural major bleeding | | | | | | | | | | | |
| 4 | Single | 791 | 104 | 9.93% (3.99%-22.63%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural minor bleeding | | | | | | | | | | | |
| 3 | Single | 735 | 143 | 20.46% (11.35%-34.08%) | Observational | Serious | Strongly serious | Strongly serious | Serious | None | Very low |
| Procedural major vascular complications | | | | | | | | | | | |
| 3 | Single | 647 | 63 | 10.49% (4.76%-21.54%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 30-day major vascular complications | | | | | | | | | | | |
| 2 | All | 189 | 7 | 2.97% (0.34%-21.67%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Procedural minor vascular complications | | | | | | | | | | | |
| 2 | Single | 591 | 43 | 7.37% (3.24%-15.93%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural major access-site complications | | | | | | | | | | | |
| 2 | Single | 109 | 14 | 13.07% (7.89%-20.88%) | Observational | Serious | Not serious | Strongly serious | Strongly serious | None | Very low |
| Procedural permanent pacemaker | | | | | | | | | | | |
| 4 | All | 468 | 22 | 5.16% (3.01%-8.71%) | Observational | Serious | Not serious | Strongly serious | Not serious | None | Very low |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Readmission within 30 days | | | | | | | | | | | |
| 2 | Modified Fried | 174 | 11 | 6.63% (2.19%-18.37%) | Observational | Strongly serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Procedural stroke | | | | | | | | | | | |
| 6 | All | 761 | 24 | 3.66% (2.25%-5.91%) | Observational | Serious | Not serious | Strongly serious | Not serious | None | Very low |
| Stroke within 30 days | | | | | | | | | | | |
| 2 | Single | 6185 | 132 | 2.14% (1.81%-2.53%) | Observational | Serious | Not serious | Strongly serious | Not serious | None | Very low |
| Transfusion | | | | | | | | | | | |
| 3 | All | 458 | 191 | 41.01% (34.02%-48.39%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| 2-valve implantation | | | | | | | | | | | |
| 2 | Single | 409 | 10 | 2.46% (1.33%-4.51%) | Observational | Not serious | Not serious | Strongly serious | Not serious | None | Low |
| Length of hospitalization | | | | | | | | | | | |
| 6 | All | 308 | NA | 8.25 (6.62-10.27) | Observational | Strongly serious | Strongly serious | Strongly serious | Strongly serious | None | Very low |

**Meta-analyses conducted using random effects model
*Frailty measures are categorized as single, multi-measures, Fried, modified Fried and all.
Single indicates single measures
Multi indicates multi-measures
Fried indicates the Fried phenotype
Modified Fried indicates the modified Fried phenotype
Fried and modified Fried includes the Fried phenotype and modified Fried phenotype
All includes all single- and multi-measures
CI indicates confidence interval.
GRADE indicates Grading of Recommendations Assessment, Development and Evaluation

For peer review only


For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

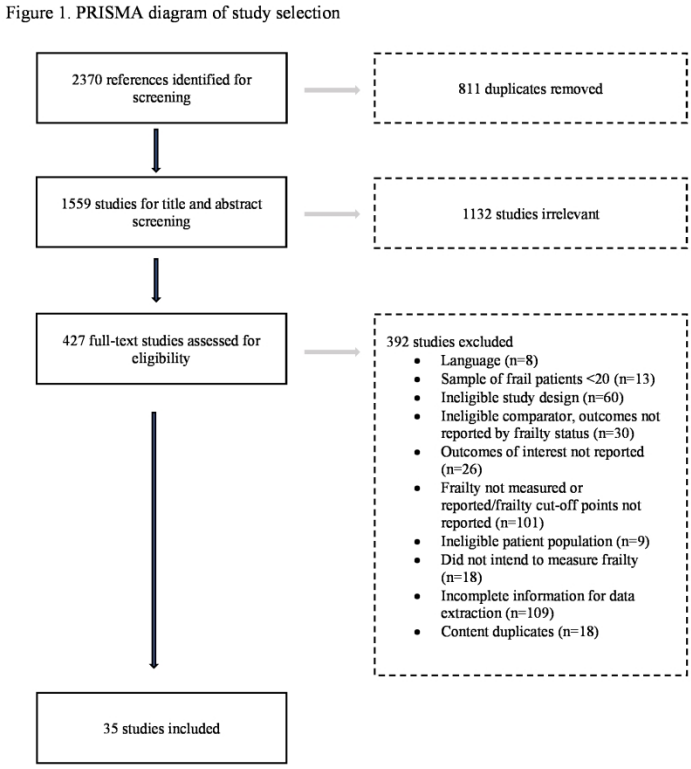



Figure 1. PRISMA flow diagram of included studies
PRISMA, reporting in systematic reviews and meta-analyses. N, number of studies.

215x279mm (300 x 300 DPI)

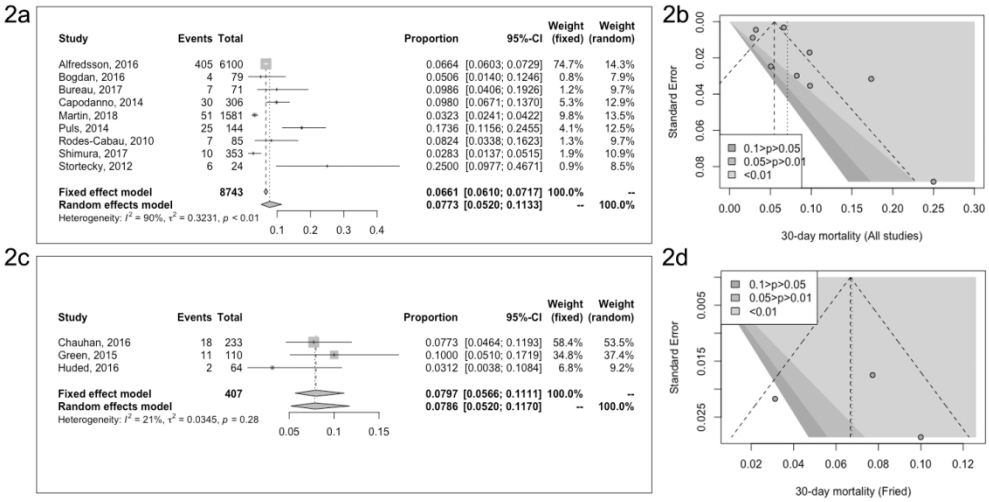



Figure 2a. Meta-analysis of 30-day mortality in frail patients after TAVI.
Frailty was measured using sing and multi-dimensional measures.
The squares indicate the 30-day mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 30-day mortality

Figure 2b. Funnel plots, using data from all studies that reported 30-day mortality
y-axis is the standard error of the 30-day mortality
x-axis is the 30-day mortality

Figure 2c. Meta-analysis of 30-day mortality in frail patients after TAVI.
Frailty was measured using modified Fried frailty phenotype.
The squares indicate the 30-day mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 30-day mortality

Figure 2d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.
y-axis is the standard error of the 30-day mortality
x-axis is the 30-day mortality

529x275mm (72 x 72 DPI)

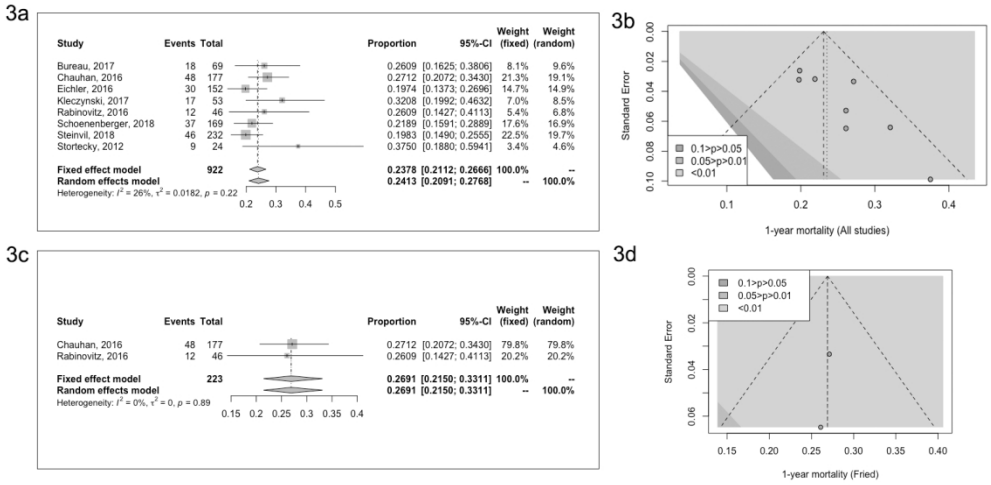



Figure 3a. Meta-analysis of 1-year mortality in frail patients after TAVI.
Frailty was measured using single and multi-dimensional measures.
The squares indicate the 1-year mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 1-year mortality

Figure 3b. Funnel plots, using data from all studies that reported 30-day mortality
y-axis is the standard error of the 1-year mortality
x-axis is the 1-year mortality

Figure 3c. Meta-analysis of 1-year mortality in frail patients after TAVI.
Frailty was measured using the Fried frailty phenotype.
The squares indicate the 1-year mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 1-year mortality

Figure 3d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.
y-axis is the standard error of the 1-year mortality
x-axis is the 1-year mortality

**Supplementary Appendix A Literature Search**

**Search: TAVI in Frail Patients with Aortic Stenosis**

Database: Embase Classic+Embase 1947 to 2018 October 08, 2018

Date run: Oct 09, 2018

| # | Searches | Results |
|---|---|---|
| 1 | frailty/ | 4,474 |
| 2 | frail elderly/ | 9,200 |
| 3 | geriatric assessment/ | 14,090 |
| 4 | frail*.mp. | 29,693 |
| 5 | fragile.mp. | 25,961 |
| 6 | or/1-5 | 66,792 |
| 7 | geriatric patient/ | 22,004 |
| 8 | very elderly/ | 148,862 |
| 9 | aged/ | 2,835,855 |
| 10 | aged hospital patient/ | 752 |
| 11 | geriatrician/ | 1,701 |
| 12 | *geriatrics/ | 25,458 |
| 13 | (aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*).mp. | 4,829,113 |
| 14 | or/7-13 | 4,829,113 |
| 15 | exhaustion/ | 6,833 |
| 16 | limited mobility/ | 1,507 |
| 17 | "timed up and go test"/ | 1,775 |
| 18 | exp walk test/ | 5,786 |
| 19 | walking speed/ | 11,203 |
| 20 | gait/ | 48,761 |
| 21 | physical activity/ | 132,030 |
| 22 | Performance Oriented Mobility Assessment/ | 283 |
| 23 | grip strength/ | 17,571 |
| 24 | hand strength/ | 3,100 |
| 25 | muscle strength/ | 54,850 |
| 26 | daily life activity/ | 78,985 |
| 27 | exp "activity of daily living assessment"/ | 21,992 |
| 28 | exp ADL disability/ | 13,335 |

For peer review only

| | | |
|---|---|---|
| 29 | exp disability/ | 180,851 |
| 30 | exp functional status assessment/ | 121,162 |
| 31 | exp neuropsychological test/ | 913,342 |
| 32 | mental function assessment/ | 384 |
| 33 | cognition assessment/ | 2,695 |
| 34 | exp cognition/ | 2,059,036 |
| 35 | exp cognitive defect/ | 433,929 |
| 36 | memory assessment/ | 714 |
| 37 | nutritional assessment/ | 26,256 |
| 38 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) adj7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)).mp. | 854,212 |
| 39 | (chair adj2 (rise or stand)).mp. | 1,696 |
| 40 | ((grip* or grasp* or hand* or musc*) adj2 strength).mp. | 82,777 |
| 41 | (tug adj3 test).mp. | 1,953 |
| 42 | (timed* adj6 (up* or go)).mp. | 6,854 |
| 43 | timedupandgo.mp. | 2 |
| 44 | weight loss.mp. | 129,598 |
| 45 | (activit* adj3 (living or life)).mp. | 100,405 |
| 46 | (adl or iadl or badl).mp. | 18,280 |
| 47 | or/15-46 | 3,442,519 |
| 48 | 14 and 47 | 781,563 |
| 49 | 6 or 48 | 824,674 |
| 50 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) adj3 aortic valv*).mp. | 19,014 |
| 51 | transcatheter aortic valve implantation/ | 16,180 |
| 52 | or/50-51 | 19,014 |
| 53 | 49 and 52 | 1,414 |
| 54 | limit 53 to yr="2006 -Current" | 1,409 |

For peer review only

Database: Ovid MEDLINE(R) ALL 1946 to October 08, 2018

Date run: October 09, 2018

| # | Searches | Results |
|---|---|---|
| 1 | Frailty/ | 662 |
| 2 | Frail Elderly/ | 9,739 |
| 3 | Geriatric Assessment/ | 24,413 |
| 4 | frail*.mp. | 21,060 |
| 5 | fragile.mp. | 19,323 |
| 6 | or/1-5 | 61,189 |
| 7 | "Aged, 80 and over"/ | 816,726 |
| 8 | Aged/ | 2,823,799 |
| 9 | Geriatricians/ | 42 |
| 10 | *Geriatrics/ | 25,486 |
| 11 | (aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*).mp. | 5,286,097 |
| 12 | or/7-11 | 5,286,097 |
| 13 | Physical Exertion/ | 55,505 |
| 14 | Physical Endurance/ | 18,188 |
| 15 | Walk Test/ | 677 |
| 16 | Exercise Test/ | 59,444 |
| 17 | Exercise Tolerance/ | 11,201 |
| 18 | Walking Speed/ | 557 |
| 19 | Gait/ | 24,764 |
| 20 | Exercise/ | 94,223 |
| 21 | mobility limitation/ | 3,922 |
| 22 | Hand Strength/ | 12,733 |
| 23 | Muscle Strength/ | 16,326 |
| 24 | exp "Activities of Daily Living"/ | 64,195 |
| 25 | "Quality of Life"/ | 167,141 |
| 26 | exp Neuropsychological Tests/ | 164,970 |
| 27 | exp Cognition/ | 142,641 |
| 28 | Cognitive Dysfunction/ | 9,643 |
| 29 | Cognition Disorders/ | 61,522 |
| 30 | Nutritional Assessment/ | 13,529 |
| 31 | Nutritional Status/ | 39,381 |

For peer review only

| | | |
|---|---|---|
| 32 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) adj7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)).mp. | 556,389 |
| 33 | (chair adj2 (rise or stand)).mp. | 1,106 |
| 34 | ((grip* or grasp* or hand* or musc*) adj2 strength).mp. | 49,580 |
| 35 | (tug adj3 test).mp. | 1,150 |
| 36 | (timed* adj6 (up* or go)).mp. | 4,104 |
| 37 | timedupandgo.mp. | 1 |
| 38 | weight loss.mp. | 87,560 |
| 39 | (activit* adj3 (living or life)).mp. | 80,777 |
| 40 | (adl or iadl or badl).mp. | 10,012 |
| 41 | or/13-40 | 1,308,674 |
| 42 | 12 and 41 | 555,051 |
| 43 | 6 or 42 | 592,801 |
| 44 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) adj3 aortic valv*).mp. | 8,618 |
| 45 | Transcatheter Aortic Valve Replacement/ | 2,879 |
| 46 | or/44-45 | 8,618 |
| 47 | 43 and 46 | 479 |
| 48 | limit 47 to yr="2006 -Current" | 477 |

For peer review only

Database: Cochrane Central Register of Controlled Trials : Issue 4 of 12, April 2018

Date run: May 15, 2018

| # | Searches | Results |
|---|---|---|
| 1 | MeSH descriptor: [Frailty] this term only | 14 |
| 2 | MeSH descriptor: [Frail Elderly] this term only | 632 |
| 3 | MeSH descriptor: [Geriatric Assessment] this term only | 1357 |
| 4 | frail*:ti,ab,kw (Word variations have been searched) | 2,289 |
| 5 | fragile:ti,ab,kw (Word variations have been searched) | 1322 |
| 6 | {OR #1-#5} | 4,707 |
| 7 | MeSH descriptor: [Aged, 80 and over] this term only | 275 |
| 8 | MeSH descriptor: [Aged] this term only | 1,057 |
| 9 | MeSH descriptor: [Geriatrics] this term only | 194 |
| 10 | aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*:ti,ab,kw (Word variations have been searched) | 543,310 |
| 11 | {or #7-#10} | 543,310 |
| 12 | MeSH descriptor: [Mobility Limitation] this term only | 383 |
| 13 | MeSH descriptor: [Physical Exertion] this term only | 3,745 |
| 14 | MeSH descriptor: [Physical Endurance] this term only | 3,066 |
| 15 | MeSH descriptor: [Walk Test] this term only | 133 |
| 16 | MeSH descriptor: [Exercise Test] this term only | 7,895 |
| 17 | MeSH descriptor: [Exercise Tolerance] this term only | 2,321 |
| 18 | MeSH descriptor: [Walking Speed] this term only | 61 |
| 19 | MeSH descriptor: [Gait] this term only | 1,711 |
| 20 | MeSH descriptor: [Exercise] this term only | 13,622 |
| 21 | MeSH descriptor: [Hand Strength] this term only | 1,237 |
| 22 | MeSH descriptor: [Muscle Strength] this term only | 3,554 |
| 23 | MeSH descriptor: [Activities of Daily Living] explode all trees | 4,849 |
| 24 | MeSH descriptor: [Quality of Life] this term only | 20,519 |
| 25 | MeSH descriptor: [Neuropsychological Tests] explode all trees | 14,997 |
| 26 | MeSH descriptor: [Cognition] explode all trees | 9,262 |
| 27 | MeSH descriptor: [Cognition Disorders] this term only | 3,300 |
| 28 | MeSH descriptor: [Cognitive Dysfunction] this term only | 682 |
| 29 | MeSH descriptor: [Nutrition Assessment] this term only | 643 |
| 30 | MeSH descriptor: [Nutritional Status] this term only | 2,164 |

For peer review only

| | | |
|---|---|---|
| 31 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) next/7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)):ti,ab,kw (Word variations have been searched) | 71,128 |
| 32 | (chair next/2 (rise or stand)):ti,ab,kw (Word variations have been searched) | 577 |
| 33 | ((grip* or grasp* or hand* or musc*) next/2 strength):ti,ab,kw (Word variations have been searched) | 12,819 |
| 34 | (tug next/3 test):ti,ab,kw (Word variations have been searched) | 277 |
| 35 | (timed* next/6 (up* or go)):ti,ab,kw (Word variations have been searched) | 1,747 |
| 36 | "weight loss":ti,ab,kw (Word variations have been searched) | 14,222 |
| 37 | (activit* next/3 (living or life)):ti,ab,kw (Word variations have been searched) | 8,991 |
| 38 | adl or iadl or badl:ti,ab,kw (Word variations have been searched) | 2,433 |
| 39 | {or #12-#38} | 147,840 |
| 40 | #11 and #39 | 94,059 |
| 41 | #6 or #40 | 96,452 |
| 42 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) next/3 aortic valv*):ti,ab,kw (Word variations have been searched) | 730 |
| 43 | MeSH descriptor: [Transcatheter Aortic Valve Replacement] this term only | 111 |
| 44 | #42 or #43 | 730 |
| 45 | #41 and #44 | 72 |

For peer review only

Database: PsycINFO (ProQuest) - updated weekly

Date run: October 09, 2018

| # | Searches | Results |
|---|---|---|
| 1 | MAINSUBJECT.EXACT("Health Impairments" OR "Geriatric Assessment") | 9,138 |
| 2 | AB,TI,SU(frail* OR fragile) | 10,660 |
| 3 | 1 OR 2 | 18,152 |
| 4 | MAINSUBJECT.EXACT("Geriatric Patients" OR "Geriatrics" OR "Gerontology") | 31,680 |
| 5 | AB,TI,SU(aged OR aging OR older OR elderly OR senior* OR geriatric* OR centenarian* OR nonagenarian* OR octogenarian* OR septuagenarian* OR sexagenarian*) | 1,926,992 |
| 6 | 4 OR 5 | 1,927,152 |
| 7 | MAINSUBJECT.EXACT("Fatigue" OR "Physical Mobility" OR "Locomotion" OR "Physical Agility" OR "Physical Activity" OR "Physical Endurance" OR "Physical Fitness" OR "Physical Strength" OR "Exercise" OR "Ability Level" OR "Activity Level" OR "Walking" OR "Gait" OR "Activities of Daily Living" OR "Daily Activities" OR "Quality of Life" OR "Neuropsychological Assessment" OR "Cognition" OR "Cognitive Impairment" OR "Memory Disorders" OR "Nutritional Deficiencies") | 257,149 |
| 8 | AB,TI,SU(exhaustion OR fatigue* OR tired* OR mobility OR gait OR walk* OR stand OR balance OR bath* OR dress* OR toilet* OR continence OR feeding OR cognition OR memory OR mental OR disability OR NYHA OR Karnofsky OR CSHA OR functional* OR Katz OR Fried OR Rockwood OR frailty OR nutrition*) AND AB,TI,SU(assess* OR phenotype* OR eval* OR test* OR exam* OR instrument* OR index OR indices OR scale* OR score* OR tool* OR declin* OR dependenc* OR impair*) | 882,574 |
| 9 | chair NEAR/2 (rise OR stand) | 356 |
| 10 | strength NEAR/2 (grip* OR grasp* OR hand* OR musc*) | 6,781 |
| 11 | tug NEAR/3 test | 218 |
| 12 | timed* NEAR/6 (up* OR go) | 1,197 |
| 13 | activit* NEAR/3 (living OR life) | 32,094 |
| 14 | AB,TI,SU(timedupandgo OR weight loss OR adl OR iadl OR badl) | 17,547 |
| 15 | 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 | 1,008,472 |
| 16 | 6 AND 15 | 555,046 |
| 17 | 3 OR 16 | 563,316 |
| 18 | (transfemoral* OR trans-femoral* OR transapical* OR trans-apical* OR transaxillary OR trans-axillary OR transarterial* OR trans-arterial* OR subclavian* OR sub-clavian* OR transcatheter* OR trans-catheter* OR transcutaneous* OR trans-cutaneous* OR per-cutaneous* OR percutaneous* OR transcaval* OR trans-caval* OR "direct aortic" OR tavi OR tavr OR pavi OR pavr OR sapien OR cribier OR revalv* OR lotus OR "direct flow" OR jenavalve OR portico OR engager OR evolut) AND (aortic valv*) | 18 |
| 19 | TAVI | 27 |
| 20 | 18 OR 19 | 34 |
| 21 | 17 AND 20 | 9 |

For peer review only

Database: Web of Science Core Collection (Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI Timespan=All years) updated daily

Date run: October 09, 2018

| # | Searches | Results |
|---|---|---|
| 1 | TS=(frail*) | 22,247 |
| 2 | TS=(fragile) | 37,652 |
| 3 | #1 OR #2 | 59,834 |
| 4 | TS=(aged OR aging OR older OR elderly OR senior* OR geriatric* OR centenarian* OR nonagenarian* OR octogenarian* OR septuagenarian* OR sexagenarian*) | 3,797,370 |
| 5 | TS=((exhaustion OR fatigue* OR tired* OR mobility OR gait OR walk* OR stand OR balance OR bath* OR dress* OR toilet* OR continence OR feeding OR cognition OR memory OR mental OR disability OR NYHA OR Karnofsky OR CSHA OR functional* OR Katz OR Fried OR Rockwood OR frailty OR nutrition*) NEAR/7 (assess* OR phenotype* OR eval* OR test* OR exam* OR instrument* OR index OR indices OR scale* OR score* OR tool* OR declin* OR dependenc* OR impair*)) | 773,498 |
| 6 | TS=(chair NEAR/2 (rise OR stand)) | 2,002 |
| 7 | TS=((grip* OR grasp* OR hand* OR musc*) NEAR/2 strength) | 39,945 |
| 8 | TS=(tug NEAR/3 test) | 1,187 |
| 9 | TS=(timed* NEAR/6 (up OR go)) | 3,859 |
| 10 | TS=(weight loss) | 174,213 |
| 11 | TS=(activit* NEAR/3 (living OR life)) | 43,381 |
| 12 | TS=(ADL OR IADL OR BADL) | 11,123 |
| 13 | #12 OR #11 OR #10 OR #9 OR #8 OR #7 OR #6 OR #5 | 1,000,842 |
| 14 | #13 AND #4 | 263,645 |
| 15 | #14 OR #3 | 315,519 |
| 16 | TS=((transfemoral* OR trans-femoral* OR transapical* OR trans-apical* OR transaxillary OR trans-axillary OR transarterial* OR trans-arterial* OR subclavian* OR sub-clavian* OR transcatheter* OR trans-catheter* OR transcutaneous* OR trans-cutaneous* OR per-cutaneous* OR percutaneous* OR transcaval* OR trans-caval* OR "direct aortic" OR tavi OR tavr OR pavi OR pavr OR sapien OR cribier OR revalv* OR lotus OR "direct flow" OR jenavalve OR portico OR engager OR evolut) NEAR/3 aortic valv*) | 12,250 |
| 17 | #16 AND #15 | 403 |

For peer review only

**Search numbers for PRISMA**

| Database | Search results | Internal and cross-database duplicates | De-duped results for screening |
|---|---|---|---|
| Ovid Embase | 1,409 | 158 | 1,251 |
| Ovid MEDLINE | 477 | 373 | 104 |
| Cochrane Library | 72 | 68 | 4 |
| PsycINFO | 9 | 7 | 2 |
| Web of Science | 403 | 347 | 56 |
| TOTALS: | 2,370 | 953 | 1,417 |
| | | Search updates (to October 09 2018): | 142 |
| | | **Total screened (from databases):** | **1,559** |

For peer review only

## Supplementary appendix B. Baseline demographics and population characteristics

| Study | Recruitment period | N (frail) | Age (mean±SD) | Female (n,%) | NYHA III/IV (n,%) | STS (mean±SD) | Logistic Euro score (mean±SD) | TF-approach (n, %) | TA-approach (n,%) | AVA, $cm^2$ (mean±SD) | Creatinine (mg/dL) (mean±SD) | Haemoglobin, g/dL (mean±SD) | LVEF (mean±SD) | Aortic gradient, mmHg (mean±SD) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | 2011-2014 | 6100 | NR | 3421 (56.08%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bureau (2017) | 2013-2015 | 71 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | 127 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bogdan (2016) | 2009-2012 | 79 | 82±5 | 43 (55%) | 46 (58.2%) | 5.8±3.12 | 19.44±14.98 | NR | NR | 0.66±0.14 | 1.35±0.93 | 11.27±1.58 | 53.22±12.64 | 43.54±15.19 |
| Capodanno (2014) | 2010-2012 | 306 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | 2012-2015 | 233 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2016) | 2008-2014 | 65 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Eichler (2017) | 2013-2015 | 152 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Ghatak (2012) | NR | 22 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Green (2015) | PARTNER[1] | 110 | 87.1** (82.7-90.3) | 58 (53%) | NR | 11.3** (9.6-13.8) | NR | 57 (52%) | NR | 0.62** (0.51-0.72) | NR | NR | 55** (35-60) | 45.2** (34.9-59.7) |
| Green (2012) | PARTNER[1] | 76 | 87.1±6.6 | 40 (53%) | NR | 11.9±4 | NR | NR | 30 (39%) | 0.6±0.2 | NR | 11.1±1.6 | 47±15 | 45±15 |
| Grossman (2017) | 2008-2014 | 192 | 84.3±7.5 | 109 (56.8) | 122 (66%) | 6.2±4.5 | 20.8±16 | 132 (73%) | 42 (23%) | NR | NR | NR | NR | NR |
| Huded (2016) | 2012-2015 | 64 | 83.1±7.5 | 42 (66%) | 53 (90%) | 6.8±2.8 | NR | 57 (89%) | NR | NR | 1.19±0.45 | 11.8±1.8 | NR | NR |
| Koifman (2015) | 2007-2014 | 238 | 84±8 | 81 (21%) | 210 (94%) | 10±4.7 | NR | 181 (76%) | NR | 0.64±0.13 | 1.3±0.9 | 11±1.4 | NR | NR |
| Kleczynski (2017) | NR | 53 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kobe (2016) | 2011-2014 | 71 | NR | 37 (52.11%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | 2013-2014 | 1466 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | NR | 293 | 81.3±7.5 | 105 (36%) | 240 (82%) | 7.1±4 | NR | 174 (59.4%) | 119 (40.6%) | 0.68±0.21 | NR | NR | 54±14 | NR |
| Maniar (2016) | PARTNER II[2] | 73 | 84±7 | 49 (67.6%) | NR | 12.3±5.7 | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | NR | 30 | 92±2 | 24 (80%) | 24 (80%) | 8.4±5.4 | NR | 20 (67%) | NR | NR | NR | NR | NR | NR |
| Patel (2016) | 2012-2015 | 31 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | 2008-2012 | 144 | 83.1±5.2 | 106 (74%) | 136 (94.4%) | 8.8±6.7 | 28.8±16.3 | NR | NR | NR | NR | NR | NR | NR |
| Rabinovitz (2016) | 2009-2013 | 46 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | 2005-2009 | 85 | 83±7 | 58 (68.2%) | NR | 11.6±8.3 | NR | 42 (49.4%) | 43 (50.6%) | NR | 112±73 (umol/L) | NR | NR | NR |

For peer review only

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Rodriguez-Pascual (2016) | 2010-2015 | 68 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | Since 2007 | 242 | 83±7.6 | 140 (58%) | NR | 8.8±6.1 | NR | NR | NR | NR | NR | NR | NR | NR |
| Schoenenberger (2018) | 2009-2013 | 169 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2017) | 2013-2016 | 353 | NR | 276 (78.2%) | 208 (58.92%) | NR | NR | 285 (80.74%) | 58 (16.43%) | NR | NR | NR | NR | NR |
| Steinvil (2018) | 2011-2016 | 232 | 83±7 | 132 (57%) | 193 (83%) | 8.8±6.1 | NR | NR | NR | 0.7 ±0.2 | NR | NR | NR | 43±13 |
| Shi (2018) | 2014-2016 | 116 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | 2011-2015 | 34 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | 2007-2010 | 24 | 84±4 | NR | NR | NR | 24.8±16 | NR | NR | 0.5±0.2 | NR | NR | 52±19 | NR |
| Traynor (2017) | 2008-2015 | 60 | 85.4±7.4 | 38 (63.3%) | 49 (81.7%) | 12.7±11.2 | NR | 34 (56.67%) | NR | NR | 1.4±1.1 | NR | 56±14 | NR |
| Yamamoto (2015) | 2010-2011 | 56 | 85±6.7 | 38 (67.90%) | 47 (83.9%) | 10.9±7.3 | 23.6±11.2 | 41 (73.2%) | NR | 0.58±0.19 | NR | NR | NR | 53.3±21.5 |
| Zajarias (2016) | PARTNER II[2] | 265 | 85±6 | 157 (59.7%) | NR | 11.8±5.8 | NR | NR | NR | NR | NR | NR | NR | NR |

## Baseline demographics and population characteristics (continued)

| Study | AF (n, %) | Active cancer (n, %) | BMI (mean±SD) | COPD (n,%) | Cerebrovascular Disease (n,%) | CAD (n, %) | Chronic Lung disease (n,%) | Chronic Dialysis (n,%) | Chronic Kidney disease (n, %) | dyslipidemia (n,%) | Diabetes Mellitus (n,%) | Hyperlipidemia (n,%) | Hypertension (n,%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson, (2016) | NR | NR | NR | NR | NR | NR | 1611 (26.41%) | 186 (3.05%) | NR | NR | 2137 (35.03%) | NR | 5510 (90.33%) |
| Bureau (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bogdan (2016) | 14 (17%) | NR | NR | 23 (29%) | NR | NR | NR | NR | NR | 39 (49%) | 27 (34%) | NR | 56 (70%) |
| Capodanno (2014) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2015) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Eichler (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Ghatak (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Green (2015) | NR | NR | NR | 46 (42%) | 26 (26%) | 91 (83%) | NR | NR | NR | NR | 28 (26%) | NR | 95 (86%) |
| Green (2012) | NR | NR | 25±6.4 | NR | NR | NR | NR | NR | NR | NR | 19 (25%) | 38 (50%) | 58 (76%) |
| Grossman | 57 | NR | NR | NR | NR | 100 | NR | NR | NR | NR | 71 | 136 | 160 |

For peer review only

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (2017) | (32%) | | | | | (56%) | | | | | (39%) | (74%) | (86%) |
| Huded (2016) | 26 (41%) | NR | 29.1±6.9 | 5 (8%) | NR | 54 (84%) | NR | NR | 14 (22%) | NR | 24 (38%) | 37 (58%) | 50 (78%) |
| Koifman (2015) | 101 (44%) | NR | NR | 77 (34%) | 32 (16%) | NR | NR | NR | NR | NR | 79 (35%) | NR | 211 (92%) |
| Kleczynski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kobe (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | 107 (37%) | NR | 25.4±4.9 | NR | NR | 206 (71%) | 94 (32%) | NR | NR | 208 (71%) | 85 (29%) | NR | 246 (84%) |
| Maniar (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | 11 (37%) | NR | 25.4±3.7 | NR | NR | 22 (73%) | 7 (23%) | 1 (3%) | NR | 17 (57%) | NR | NR | 28 (93%) |
| Patel (2016) | 19 (61.3%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | NR | NR | 26.9±5.1 | NR | NR | 101 (70%) | NR | NR | NR | NR | 50 (34%) | NR | NR |
| Rabinovitz (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | NR | NR | 25±5 | 23 (27.1%) | 15 (17.7%) | 57 (67.1%) | NR | 3 (3.5%) | NR | 55 (65.5%) | 21 (26.6%) | NR | 58 (68.2%) |
| Rodriguez-Pascual (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | 109 (45%) | NR | 29±11 | 116 (48%) | NR | NR | NR | 15 (6%) | 90 (37%) | NR | 82 (34%) | 198 (82%) | 223 (92%) |
| Schoenenberger (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2017) | NR | 8 (2.3%) | NR | NR | NR | NR | NR | NR | 203 (57.5%) | NR | 94 (26.6%) | NR | 258 (73.1%) |
| Steinvil (2018) | 102 (44%) | NR | 29±11 | 114 (49%) | NR | NR | NR | 14 (6%) | 86 (37%) | NR | 79 (34%) | 190 (82%) | 216 (93%) |
| Shi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Traynor (2017) | NR | NR | NR | NR | 18 (30%) | NR | 32 (55.2%) | 2 (3.3%) | NR | NR | 28 (46.7%) | NR | 57 (95%) |
| Yamamoto (2015) | NR | NR | NR | 17 (30.40%) | NR | NR | NR | 1 (1.80%) | 42 (75%) | 19 (33.90%) | 5 (8.90%) | NR | 37 (66.10%) |
| Zajarias (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |

Baseline demographics and population characteristics (continued)

For peer review only

| Study | Liver disease (n,%) | Previous stroke (n, %) | Previous MI (n, %) | Previous BVA (n, %) | Previous PCI (n, %) | Previous CABG (n,%) | Pulmonary disease (n,%) | Peripheral arterial disease (n,%) | Peripheral Vascular disease (n, %) | Permanent pacemaker (n,%) | Valve type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | NR | 753 (12.34%) | 1422 (23.31%) | 774 (12.69%) | 2103 (34.48%) | 1800 (29.51%) | NR | 1930 (31.64%) | NR | NR | NR |
| Bureau (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve<br>Edwards Sapien<br>Jena<br>Lotus<br>NVT |
| Bogdan (2016) | 1 (1%) | NR | NR | NR | NR | NR | NR | NR | 9 (11%) | NR | CoreValve: 60 (75.94%)<br>Edwards Sapien: 19 (24.06%) |
| Capodanno (2014) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2015) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve: 277 (89%)<br>Edwards Sapien: 24 (7.7%)<br>Lotus: 10 (3.2%) |
| Eichler (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve (classic, Evolut R)<br>Edwards Sapien<br>Edwards Sapien XT |
| Ghatak (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien |
| Green (2015) | 9 (8%) | NR | NR | 35 (32%) | 45 (41%) | 47 (43%) | NR | NR | 46 (42%) | 27 (25%) | Edwards Sapien |
| Green (2012) | NR | 9 (12%) | NR | NR | NR | 27 (36%) | 22 (29%) | NR | 16 (21%) | 24 (32%) | Edwards Sapien |
| Grossman (2017) | NR | 6 (3%) | NR | NR | NR | NR | NR | NR | 19 (11%) | 13 (7%) | CoreValve<br>Edwards Sapien XT |
| Huded (2016) | NR | 13 (20%) | NR | NR | NR | 17 (27%) | NR | 6 (9%) | NR | NR | Edwards Sapien<br>Edwards Sapien XT<br>Edwards Sapien 3<br>CoreValve<br>Direct Flow Medical |
| Koifman (2015) | NR | NR | NR | NR | 60 (26%) | 67 (29%) | NR | NR | 76 (34%) | 37 (21%) | |
| Kleczynski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien<br>Edwards Sapien XT<br>Edwards Sapien 3<br>CoreValve<br>JenaValve |
| Kobe (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | NR | 56 (19%) | 84 (29%) | NR | NR | 57 (20%) | NR | NR | 87 (30%) | NR | NR |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Maniar (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | NR | NR | NR | 1 (3%) | 4 (13%) | 4 (13%) | NR | NR | 7 (23%) | 7 (23%) | |
| Patel (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | NR | NR | NR | NR | 40 (28%) | 16 (11%) | 43 (30%) | NR | 42 (29%) | NR | CoreValve<br>Edwards Sapien |
| Rabinovitz (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | NR | NR | 39 (45.9%) | NR | 26 (30.6%) | 20 (23.5%) | NR | NR | 29 (34.1%) | NR | Cribier-Edwards<br>Edwards Sapien<br>Edwards Sapien XT |
| Rodriguez-Pascual (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | NR | 17 (7%) | 41 (17%) | 53 (22%) | 80 (33%) | 58 (24%) | NR | NR | 58 (24%) | 65 (27%) | Balloon expandable (67%)<br>Self-expanding (33%) |
| Schoenenberger (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve<br>Edwards Sapien XT |
| Shimura (2017) | 17 (4.8%) | 65 (18.4%) | 28 (7.9%) | NR | 94 (26.6%) | 21 (5.9%) | 108 (30.6%) | 67 (19%) | NR | NR | NR |
| Steinvil (2018) | NR | 16 (7%) | 42 (18%) | 49 (21%) | 79 (34%) | 58 (25%) | NR | NR | 53 (23%) | 67 (29%) | NR |
| Shi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | NR | NR | NR | NR | NR | 1 (4.2%) | NR | NR | NR | NR | NR |
| Traynor (2017) | NR | NR | 18 (30%) | NR | NR | 15 (25%) | NR | NR | 38 (63.3%) | 12 (20%) | NR |
| Yamamoto (2015) | NR | 6 (10.70%) | 5 (8.90%) | NR | NR | NR | NR | 8 (14.30%) | NR | 10 (17.90%) | CoreValve<br>Edwards Sapien |
| Zajarias (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien XT |

N, sample size of frail TAVI recipients. SD, standard deviation. NYHA, New York Heart Association. STS, Society of Thoracic Surgeons. TF, transfemoral approach. TA, transapical approach. AVA, aortic valve area. LVEF, left ventricular ejection fraction. AF, atrial fibrillation. BMI, body mass index. COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. MI, myocardial infarction. BVA, balloon aortic valvuloplasty. PCI, percutaneous coronary intervention. CABG, coronary artery bypass grafting.

NR, not reported

§ = Euro score II

For peer review only

**= median and IQR

1, recruitment period, 2007-2010

2, recruitment period, 2011-2016

For peer review only

| Study | Study Participation | Study Attrition | Prognostic Factor Measurement | Outcome Measurement | Statistical Analysis Reporting | Overall Risk of Bias | Study Design | Abstract |
|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | high | moderate | low | low | low | moderate | prospective | no |
| Bureau (2017) | high | moderate | low | low | low | moderate | prospective | no |
| Bagienski (2017) | moderate | moderate | low | low | low | moderate | prospective | no |
| Bogdan (2016) | moderate | high | low | low | moderate | moderate | retrospective | no |
| Capodanno (2014) | moderate | moderate | low | low | low | moderate | prospective | no |
| Chauhan (2016) | moderate | high | low | low | low | moderate | retrospective | no |
| Cockburn (2015) | moderate | low | moderate | low | moderate | moderate | prospective | no |
| Eichler (2017) | high | low | low | low | low | low | prospective | no |
| Ghatak (2012) | high | high | moderate | low | moderate | high | retrospective | yes |
| Grossman (2017) | moderate | high | moderate | low | low | moderate | retrospective | no |
| Green (2012) | moderate | low | low | low | low | low | prospective | no |
| Green (2015) | moderate | low | low | low | low | low | prospective | no |
| Huded (2016) | high | high | low | low | low | high | retrospective | no |
| Kleczynski (2017) | low | low | low | low | low | low | prospective | no |
| Kobe (2016) | moderate | moderate | low | low | low | moderate | prospective | no |
| Koifman (2015) | moderate | high | low | low | low | moderate | retrospective | no |
| Martin (2018) | high | high | low | low | low | high | prospective | no |
| Maniar (2016) | high | moderate | moderate | low | low | moderate | retrospective | yes |
| Mok (2016) | high | low | low | low | low | low | retrospective | no |
| Okoh (2017) | Low | high | moderate | low | low | moderate | prospective | no |
| Patel (2016) | moderate | high | low | low | low | moderate | retrospective | no |
| Puls (2014) | moderate | low | low | low | low | low | prospective | no |
| Rabinovitz (2016) | high | high | moderate | low | low | high | retrospective | no |
| Rodes-Cabau (2010) | low | low | high | low | low | low | prospective | no |
| Rodriguez-Pascual (2016) | low | low | low | low | low | low | prospective | no |
| Rogers (2018) | moderate | moderate | low | low | low | moderate | retrospective | no |
| Schoenenberger (2018) | high | low | low | low | low | low | prospective | no |
| Shimura (2017) | low | moderate | low | low | low | low | prospective | no |
| Steinvil (2018) | moderate | moderate | low | low | low | moderate | prospective | no |
| Shi (2018) | moderate | high | low | low | low | moderate | prospective | no |
| Skaar (2018) | low | low | low | low | low | low | prospective | no |
| Stortecky (2012) | high | moderate | moderate | moderate | high | high | prospective | yes |
| Traynor (2017) | low | low | moderate | moderate | low | moderate | retrospective | no |
| Yamamoto (2015) | moderate | low | low | low | low | low | prospective | no |
| Zajarias (2016) | high | moderate | low | low | low | moderate | prospective | yes |

For peer review only

# Supplementary appendix C. Assessment of quality in individual studies

*Risk of bias was assessed using the Quality in Prognostic Studies tool (QUIPS)
Low risk= studies with four or five low risk domains
High risk=studies with two or more high risk domains
Moderate risk=remaining studies

Reference. Hayden JA, van der Windt DA, Cartwright JL, Côté P, Bombardier C. Assessing bias in studies of prognostic factors. Annals of internal medicine. 2013 Feb 19;158(4):280-6.

For peer review only

# Supplementary appendix D. Prognosis of frail TAVI recipients

| Study | Procedural/in-hospital | 30 days | 6 months | 8 months | 1 year | 2 year |
|---|---|---|---|---|---|---|
| ***Primary outcomes*** | | | | | | |
| ***Death*** | | | | | | |
| Alfredsson, 2016 | 300/6100 (4.9%) | 405/6100 (6.6%) | | | | |
| Bureau, 2017 | | 7/71 (9.9%) | 13/71 (18.3%) | | 18/69 (26.1%) | |
| Bogdan, 2016 | | 4/79 (5.1%) | | | | 35.4% (2.1 years) |
| Capodanno, 2014 | | 30/306 (9.8%) | | | | |
| Chauhan, 2016 | | 18/233 (7.7%) | | | 48/177 (27.1%) | |
| Eichler, 2017 | | | | | 30/152 (19.7%) | |
| Green, 2015 | | 11/110 (10%) | | | 32.7% | |
| Huded, 2016 | | 2/64 (3.13%) | | | | |
| Kleczynski, 2017 | | | | | 17/53 (32.1%) | |
| Martin, 2018 | | 51/1581(3.23%) | | | | |
| Maniar, 2016 | | 12.3% | | | 36.5% | |
| Okoh, 2017 | | | | | | 30% |
| Puls, 2014 | 33/144 (23%) | 25/144 (17%) | | | | |
| Rabinovitz, 2016 | | | | | 12/46 (26.1%) | |
| Rodriguez-Pascual, 2016 | | | | | | 39.7%*(98weeks) |
| Rodes-Cabau, 2010 | 2/85 (2.4%) | 7/85 (8.2%) | | 19/85 (22.4%) | | |
| Schoenenberger, 2018 | | | | | 37/169(21.9%) | |
| Shimura, 2017 | 15/353 (4.25%) | 10/353 (2.83%) | | | | |
| Steinvil, 2018 | | 14/232 (6.03%) | | | 46/232 (19.8%) | |
| Shi, 2018 | | | 17/116 (15%) | | | |
| Skaar, 2018 | | | | | | 9/34 (26.47%) |
| Stortecky, 2012 | | 6/24 (25%) | | | 9/24 (37.5%) | |
| Traynor, 2017 | 2/60 (3.3%) | | | | 14.8% | |
| Zajarias, 2016 | | 17/265 (6.4%) | | | 88/265 (33.2%) | |
| ***Death (cardiovascular cause)*** | | | | | | |
| Alfredsson, 2016 | | 251/6100 (4.1%) | | | | |
| Green, 2015 | | 8/110 (7%) | | | | |
| Shimura, 2017 | 4/353(1.13%) | 8/353 (2.27%) | | | | |
| ***Secondary outcomes*** | | | | | | |
| ***Acute kidney injury*** | | | | | | |
| Alfredsson, 2016 | 397/6100 (6.5%) | | | | | |
| Bogdan, 2016 | | 25/79 (31.6%) | | | | |
| Green, 2012 | 3/76 (3.95%) | | | | | |
| Okoh, 2017 | 2/30 (6.67%) | | | | | |
| Shimura, 2017 | 47/353 (13.31%) | | | | | |
| Yamamoto, 2015 | 6/56 (10.7%) | | | | | |
| ***Arrhythmia*** | | | | | | |
| Rodes-Cabau, 2010 | 9/85 (10.6%) | | | | | |
| ***Annulus rupture*** | | | | | | |
| Yamamoto, 2015 | 1/56 (1.8%) | | | | | |
| ***Bleeding*** | | | | | | |
| Zajarias, 2016 | 15/265 (43.4%) | | | | | |
| ***Cardiac tamponade*** | | | | | | |
| Puls, 2014 | 3/144 (2%) | | | | | |
| Shimura, 2017 | 11/353 (3.11%) | | | | | |
| Yamamoto, 2015 | 3/56 (5.4%) | | | | | |
| ***Conversion to open surgery*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| Shimura, 2017 | 8/353 (2.27%) | | | | | |
| ***Coronary obstruction*** | | | | | | |
| Shimura, 2017 | 3/353 (0.85%) | | | | | |
| ***Delirium*** | | | | | | |
| Bagienski, 2017 | 105/127 (82.68%) | | | | | |
| ***Life-threatening bleeding*** | | | | | | |
| Alfredsson, 2016 | | 391/6100 (6.4%) | | | | |
| Green, 2012 | 7/76 (9.21%) | | | | | |
| Puls, 2014 | 15/144 (10%) | | | | | |
| Stortecky, 2012 | 4/24 (16.7%) | | | | | |
| Shimura, 2017 | 33/353 (9.35%) | | | | | |
| Yamamoto, 2015 | 4/56 (7.1%) | | | | | |
| ***Major bleeding*** | | | | | | |
| Green, 2015 | | 10/110 (9%) | | | | |
| Koifman, 2015 | 6/238 (3%) | | | | | |
| Puls, 2014 | 12/144 (8%) | | | | | |
| Shimura, 2017 | 77/353 (21.81%) | | | | | |
| Yamamoto, 2015 | 9/56 (16.1%) | | | | | |
| ***Minor bleeding*** | | | | | | |
| Koifman, 2015 | 31/238 (13%) | | | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Puls, 2014 | 50/144 (35%) | | | | | |
| Shimura, 2017 | 62/353 (17.56%) | | | | | |
| ***Major vascular complications*** | | | | | | |
| Bogdan, 2016 | | 0/79 (0%) | | | | |
| Green, 2015 | | 7/110 (6%) | | | | |
| Koifman, 2015 | 37/238 (16%) | | | | | |
| Shimura, 2017 | 18/353 (5.1%) | | | | | |
| Yamamoto, 2015 | 8/56 (14.3%) | | | | | |
| ***Minor vascular complications*** | | | | | | |
| Bogdan, 2016 | | 2/79 (2.5%) | | | | |
| Koifman, 2015 | 26/238 (11%) | | | | | |
| Shimura, 2017 | 17/353 (4.82%) | | | | | |
| ***Major access site complications*** | | | | | | |
| Rodes-Cabau, 2010 | 12/85 (14.1%) | | | | | |
| Stortecky, 2012 | 2/24 (8.3%) | | | | | |
| ***Major vascular injury*** | | | | | | |
| Huded, 2016 | | 4/64 (6%) | | | | |
| ***Myocardial infarction*** | | | | | | |
| Rodes-Cabau, 2010 | | 2/85 (2.4%) | | | | |
| Yamamoto, 2015 | 0/56 (0%) | | | | | |
| ***Need for second valve*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| ***Permanent pacemaker*** | | | | | | |
| Green, 2015 | | 10/110 (9%) | | | | |
| Koifman, 2015 | 12/238 (5%) | | | | | |
| Okoh, 2017 | 2/30 (6.67%) | | | | | |
| Puls, 2014 | 3/144 (2%) | | | | | |
| Rodes-Cabau, 2010 | | 7/85 (8.2%) | | | | |
| Yamamoto, 2015 | 5/56 (8.9%) | | | | | |
| ***Percutaneous closure device failure*** | | | | | | |
| Bogdan, 2016 | | 10/79 (12.7%) | | | | |
| ***Renal failure*** | | | | | | |
| Green, 2015 | | 9/110 (8%) | | | | |
| ***Respiratory failure*** | | | | | | |
| Huded, 2016 | | 1/64 (2%) | | | | |
| ***Readmission*** | | | | | | |
| Green, 2015 | | 4/110(4%) | | | | |
| Huded, 2016 | | 7/64 (11%) | | | | |
| ***Stroke*** | | | | | | |
| Alfredsson, 2016 | | 129/6100 (2.1%) | | | | |
| Green, 2015 | | 1/110 (1%) | | | | |
| Huded, 2016 | | 4/64 (6%) | | | | |
| Koifman, 2015 | 11/238 (5%) | | | | | |
| Rodes-Cabau, 2010 | | 3/85 (3.5%) | | | | |
| Okoh, 2017 | 0/30 (0%) | | | | | |
| Shimura, 2017 | 6/353 (1.7%) | | | | | |
| Stortecky, 2012 | 2/24 (8.3%) | | | | | |
| Traynor, 2017 | 2/60 (3.3%) | | | | | |
| Yamamoto, 2015 | 3/56 (5.4%) | | | | | |
| ***Sepsis*** | | | | | | |
| Rodes-Cabau, 2010 | | 4/85 (4.7%) | | | | |
| ***Transfusion*** | | | | | | |
| Green, 2012 | 24/76 (32.9%) | | | | | |
| Koifman, 2015 | 100/238 (43%) | | | | | |
| Puls, 2014 | 67/144 (47%) | | | | | |
| ***Valve embolization*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| ***Vascular complications*** | | | | | | |
| Bogdan, 2016 | | 12/79 (15.18%) | | | | |
| ***2-valve implantation*** | | | | | | |
| Shimura, 2017 | 9/353 (2.55%) | | | | | |
| Yamamoto, 2015 | 1/56 (1.8%) | | | | | |
| ***Length of hospital stay for frail TAVI recipients, days*** | | | | | | |
| Ghatak, 2012 | 12.1±8.4 | | | | | |
| Green, 2012 | 9±6 | | | | | |
| Huded, 2016 | 6±3.5 | | | | | |
| Okoh, 2017 | 7±8 | | | | | |
| Patel, 2016 | 9** | | | | | |
| Traynor, 2017 | 6.8±5.2 | | | | | |
| Yamamoto, 2015 | 10.3±7.5 | | | | | |
| ***Intensive care unit (ICU) stay, days*** | | | | | | |
| Patel, 2016 | 2.7** | | | | | |
| Yamamoto, 2015 | 4.9±4 | | | | | |

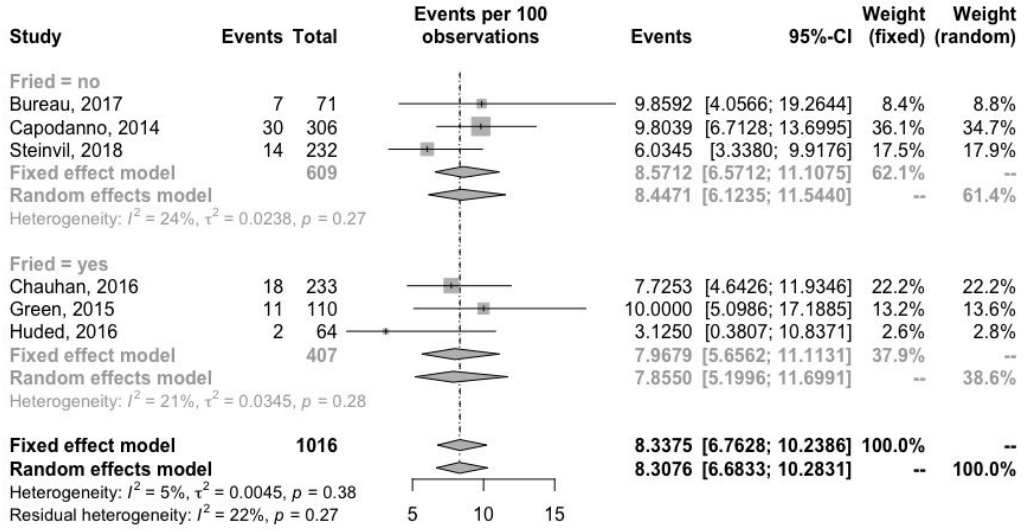
*= median + IQR
**=mean without standard deviation reported

For peer review only

For peer review only

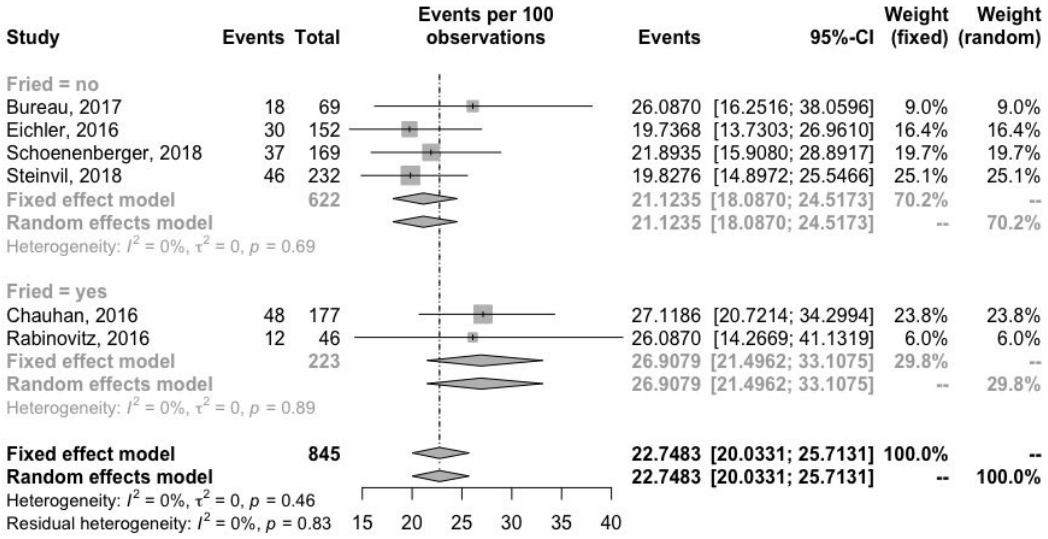
**Supplementary appendix G. Subgroup analyses of studies reporting frailty measurement using the Fried phenotype compared to non-Fried phenotype. The squares indicate the mortality reported by each study. The diamonds indicate the pooled estimates for mortality.**

**Subgroup analysis of 30-day mortality**




Test for subgroup differences: Q=0.08 (p=0.78)

**Subgroup analysis of 1-year mortality**

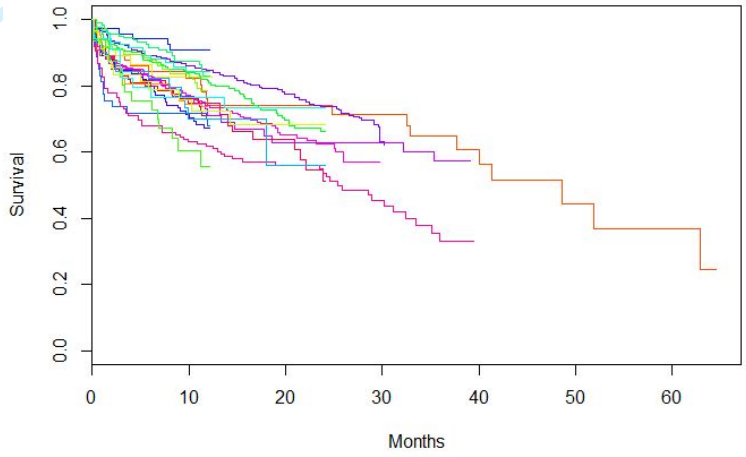


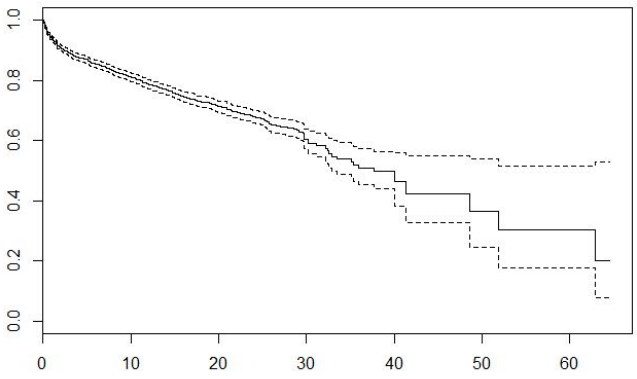

Test for subgroup differences: Q=3.12 (p=0.08)

Appendix F. Survival of frail patients after TAVI

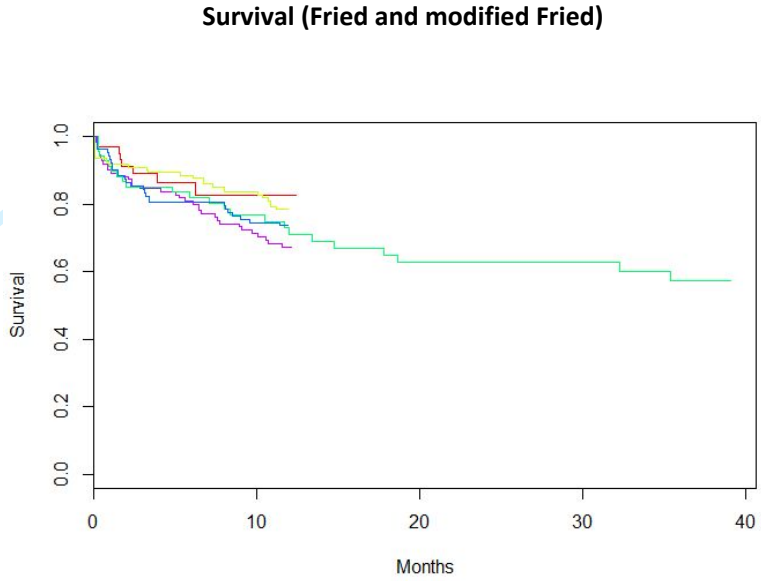
Survival after TAVI of frail patients.
Frailty was measured using single- and multi-dimensional measures.
Individual patient data estimated using algorithm developed by Guyot et al 2012.

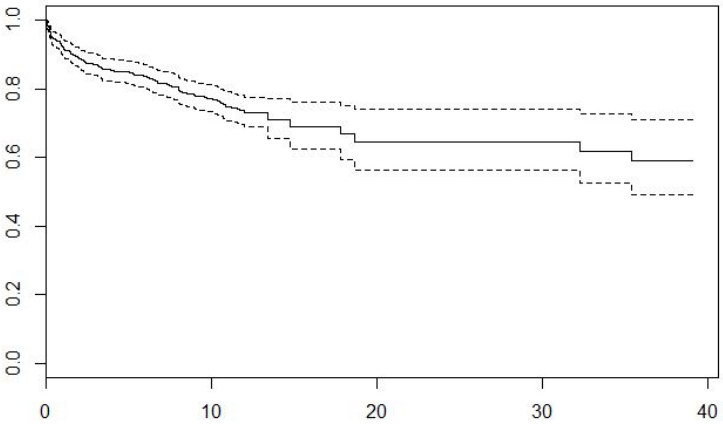
**Survival (All)**



Survival after TAVI of frail patients.
Frailty was measured using the Fried and modified Fried frailty phenotype.
Individual patient data estimated using algorithm developed by Guyot et al 2012.



**Survival rate of frail patients after TAVI (combined all studies)**

| Time (Months) | n.risk | n.event | survival | Std.err | Lower 95%CI | Upper 95% CI |
|---|---|---|---|---|---|---|
| 1 | 3088 | 198 | 0.94 | 0.00411 | 0.932 | 0.949 |
| 2 | 2915 | 307 | 0.907 | 0.00506 | 0.897 | 0.917 |
| 3 | 2790 | 359 | 0.89 | 0.00546 | 0.88 | 0.901 |
| 4 | 2668 | 399 | 0.878 | 0.00575 | 0.866 | 0.889 |
| 5 | 2573 | 422 | 0.87 | 0.00592 | 0.858 | 0.882 |
| 6 | 2458 | 463 | 0.856 | 0.00621 | 0.844 | 0.868 |
| 7 | 2371 | 491 | 0.846 | 0.00641 | 0.834 | 0.859 |
| 8 | 2279 | 525 | 0.834 | 0.00666 | 0.821 | 0.847 |
| 9 | 2183 | 556 | 0.822 | 0.00688 | 0.809 | 0.836 |
| 10 | 2103 | 587 | 0.811 | 0.0071 | 0.797 | 0.825 |
| 11 | 2005 | 613 | 0.8 | 0.00728 | 0.786 | 0.815 |
| 12 | 1756 | 648 | 0.786 | 0.00754 | 0.771 | 0.801 |
| 13 | 1274 | 660 | 0.779 | 0.00774 | 0.764 | 0.794 |
| 14 | 1194 | 676 | 0.769 | 0.00804 | 0.753 | 0.785 |
| 15 | 1116 | 696 | 0.756 | 0.00843 | 0.739 | 0.773 |
| 16 | 1043 | 713 | 0.744 | 0.00877 | 0.727 | 0.761 |
| 17 | 974 | 721 | 0.738 | 0.00895 | 0.721 | 0.756 |
| 18 | 915 | 734 | 0.728 | 0.00926 | 0.71 | 0.746 |
| 19 | 835 | 737 | 0.725 | 0.00934 | 0.707 | 0.744 |
| 20 | 775 | 753 | 0.711 | 0.00982 | 0.692 | 0.731 |
| 21 | 720 | 762 | 0.702 | 0.0101 | 0.683 | 0.722 |
| 22 | 662 | 768 | 0.696 | 0.0103 | 0.676 | 0.717 |
| 23 | 606 | 777 | 0.687 | 0.0107 | 0.666 | 0.708 |
| 24 | 554 | 784 | 0.678 | 0.011 | 0.657 | 0.7 |
| 25 | 363 | 787 | 0.673 | 0.0113 | 0.651 | 0.696 |
| 26 | 327 | 799 | 0.65 | 0.0128 | 0.625 | 0.675 |
| 27 | 284 | 801 | 0.645 | 0.0131 | 0.62 | 0.671 |
| 28 | 244 | 803 | 0.64 | 0.0134 | 0.614 | 0.667 |
| 29 | 198 | 805 | 0.634 | 0.014 | 0.607 | 0.662 |
| 30 | 134 | 813 | 0.604 | 0.017 | 0.571 | 0.638 |
| 31 | 71 | 815 | 0.591 | 0.019 | 0.555 | 0.63 |
| 32 | 65 | 816 | 0.583 | 0.0205 | 0.544 | 0.624 |
| 33 | 61 | 820 | 0.547 | 0.0259 | 0.498 | 0.6 |
| 34 | 57 | 821 | 0.538 | 0.0271 | 0.487 | 0.593 |
| 35 | 56 | 821 | 0.538 | 0.0271 | 0.487 | 0.593 |
| 36 | 51 | 824 | 0.508 | 0.0305 | 0.452 | 0.572 |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| 37 | 49 | 824 | 0.508 | 0.0305 | 0.452 | 0.572 |
| 38 | 46 | 825 | 0.497 | 0.0317 | 0.439 | 0.564 |
| 39 | 45 | 825 | 0.497 | 0.0317 | 0.439 | 0.564 |

**Survival rate of frail patients after TAVI**
**(combined studies used Fried/modified Fried phenotype)**

| Time (Months) | n.risk | n.event | survival | Std.err | Lower 95%CI | Upper 95% CI |
|---|---|---|---|---|---|---|
| **1** | 422 | 36 | 0.922 | 0.0125 | 0.898 | 0.947 |
| **2** | 398 | 52 | 0.887 | 0.0148 | 0.858 | 0.916 |
| **3** | 382 | 59 | 0.871 | 0.0157 | 0.841 | 0.902 |
| **4** | 367 | 67 | 0.853 | 0.0166 | 0.821 | 0.886 |
| **5** | 359 | 69 | 0.848 | 0.0169 | 0.816 | 0.882 |
| **6** | 347 | 74 | 0.836 | 0.0175 | 0.803 | 0.871 |
| **7** | 333 | 82 | 0.817 | 0.0184 | 0.781 | 0.853 |
| **8** | 319 | 88 | 0.802 | 0.019 | 0.765 | 0.84 |
| **9** | 305 | 96 | 0.782 | 0.0198 | 0.744 | 0.821 |
| **10** | 297 | 100 | 0.771 | 0.0202 | 0.733 | 0.812 |
| **11** | 284 | 109 | 0.748 | 0.0211 | 0.708 | 0.79 |
| **12** | 108 | 114 | 0.73 | 0.0223 | 0.688 | 0.775 |
| **13** | 36 | 114 | 0.73 | 0.0223 | 0.688 | 0.775 |
| **14** | 35 | 115 | 0.71 | 0.0295 | 0.655 | 0.77 |
| **15** | 34 | 116 | 0.69 | 0.0349 | 0.625 | 0.762 |
| **16** | 32 | 116 | 0.69 | 0.0349 | 0.625 | 0.762 |
| **17** | 32 | 116 | 0.69 | 0.0349 | 0.625 | 0.762 |
| **18** | 30 | 117 | 0.668 | 0.0403 | 0.593 | 0.751 |
| **19** | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **20** | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **21** | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **22** | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **23** | 27 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **24** | 27 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **25** | 26 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **26** | 26 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **27** | 26 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **28** | 25 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **29** | 25 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **30** | 24 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **31** | 24 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **32** | 23 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| **33** | 22 | 119 | 0.617 | 0.0508 | 0.525 | 0.725 |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| **34** | 22 | 119 | 0.617 | 0.0508 | 0.525 | 0.725 |
| **35** | 22 | 119 | 0.617 | 0.0508 | 0.525 | 0.725 |
| **36** | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |
| **37** | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |
| **38** | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |
| **39** | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |

For peer review only

## Supplementary appendix G. Quality of life among frail TAVI recipients

| Study | QoL measures | Baseline (SD) | 30-day (SD) | Change in QoL at 30 days |
|---|---|---|---|---|
| Okoh, 2017 | KCCQ overall score | | | -5.86 (p=.62) |
| | KCCQ physical limitation | | | -8.34 (p=0.46) |
| | KCCQ symptoms | | | -5.68 (p=0.65) |
| | KCCQ quality of life | | | 6.36 (p=0.63) |
| | KCCQ social limitation | | | -7.57 (p=0.58) |
| Kobe, 2016 Subgroup of frail patients with a frailty score 4-6 | SF-36 physical functioning | 55.8 (15.3) | 49.6 (22.2) | |
| | SF-36 role physical | 37.9 (32.8) | 33.8 (43.1) | |
| | SF-36 bodily pain | 88.1 (23.2) | 74.4 (23.7) | |
| | SF-36 role emotional | 88.2 (23.6) | 38.3 (42.3) | |
| | SF-36 general health | 73.4 (18.7) | 62 (15.5) | |
| | SF-36 vitality | 45.5 (24.6) | 43.4 (15.5) | |
| | SF-36 social functioning | 91.5 (17.8) | 68.8 (26.4) | |
| | SF-36 mental health | 80.4 (16.8) | 67.2 (23.2) | |
| | SF-36 physical component summary | 42.9 (6.54) | 42.3 (8.02) | |
| | SF-36 mental component summary | 55 (8.79) | 42.8 (9.86) | |
| Kobe, 2016 Subgroup of frail patients with a frailty score 6-14 | SF-36 physical functioning | 40.9 (20.5) | 46.5 (30.2) | |
| | SF-36 role physical | 25.6 (33.2) | 27.5 (39.7) | |
| | SF-36 bodily pain | 69.1 (35.6) | 60.4 (34.8) | |
| | SF-36 role emotional | 82.5 (32.9) | 45 (46.2) | |
| | SF-36 general health | 63.1 (22.7) | 50.6 (20.9) | |
| | SF-36 vitality | 35.4 (25.8) | 44.1 (24.9) | |
| | SF-36 social functioning | 79.7 (28.7) | 67.9 (23.9) | |
| | SF-36 mental health | 75 (16.3) | 66 (23.3) | |
| | SF-36 physical component summary | 36.1 (9.57) | 38.6 (8.68) | |
| | SF-36 mental component summary | 53 (9.23) | 45.2 (11.6) | |

KCCQ, Kansas City Cardiomyopathy Questionnaire. SF-36, Short Form-36 questionnaire.

For peer review only

# MOOSE Checklist for Meta-analyses of Observational Studies

| Item No | Recommendation | Reported on Page No |
|---|---|---|
| Reporting of background should include | | |
| 1 | Problem definition | 4 |
| 2 | Hypothesis statement | - |
| 3 | Description of study outcome(s) | 5 |
| 4 | Type of exposure or intervention used | 5 |
| 5 | Type of study designs used | 5 |
| 6 | Study population | 5 |
| Reporting of search strategy should include | | |
| 7 | Qualifications of searchers (eg, librarians and investigators) | 1, 5 |
| 8 | Search strategy, including time period included in the synthesis and key words | 5, Supplementary appendix A |
| 9 | Effort to include all available studies, including contact with authors | 5 |
| 10 | Databases and registries searched | 5 |
| 11 | Search software used, name and version, including special features used (eg, explosion) | Protocol |
| 12 | Use of hand searching (eg, reference lists of obtained articles) | Protocol |
| 13 | List of citations located and those excluded, including justification | Fig 1 |
| 14 | Method of addressing articles published in languages other than English | - |
| 15 | Method of handling abstracts and unpublished studies | Protocol |
| 16 | Description of any contact with authors | - |
| Reporting of methods should include | | |
| 17 | Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested | 5-6, protocol |
| 18 | Rationale for the selection and coding of data (eg, sound clinical principles or convenience) | 5-6, protocol |
| 19 | Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability) | 5-6, protocol |
| 20 | Assessment of confounding (eg, comparability of cases and controls in studies where appropriate) | - |
| 21 | Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results | 6, protocol |
| 22 | Assessment of heterogeneity | 7 |
| 23 | Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated | 6-7, protocol |
| 24 | Provision of appropriate tables and graphics | Tables 1-2, Figs 1 Supplementary appendix A-G |
| Reporting of results should include | | |
| 25 | Graphic summarizing individual study estimates and overall estimate | Table 1-2, Supplementary appendix D-G |
| 26 | Table giving descriptive information for each study included | Table 1, Supplementary |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | |
|---|---|---|
| | | appendix B |
| 27 | Results of sensitivity testing (eg, subgroup analysis) | 9-10 Table 2, Supplementary appendix E, F |
| 28 | Indication of statistical uncertainty of findings | 9-12 |

| Item No | Recommendation | Reported on Page No |
|---|---|---|
| Reporting of discussion should include | | |
| 29 | Quantitative assessment of bias (eg, publication bias) | 8, Supplementary appendix C |
| 30 | Justification for exclusion (eg, exclusion of non-English language citations) | 5, protocol |
| 31 | Assessment of quality of included studies | 6 |
| Reporting of conclusions should include | | |
| 32 | Consideration of alternative explanations for observed results | 12-16 |
| 33 | Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review) | 17 |
| 34 | Guidelines for future research | 16 |
| 35 | Disclosure of funding source | 17 |

*From*: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

For peer review only

BMJ Open

# Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis



| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2020-040459.R1 |
| Article Type: | Original research |
| Date Submitted by the Author: | 27-Nov-2020 |
| Complete List of Authors: | Li, Zhe; University of Western Ontario, Epidemiology and Biostatistics<br>Dawson , Emily ; Western University, Centre for Medical Evidence, Decision Integrity & Clinical Impact (MEDICI)<br>Moodie, Jessica; University of Western Ontario, Centre for Medical Evidence, Decision Integrity and Clinical Impact (MEDICI)<br>Martin, Janet; University of Western Ontario, High Impact Technology Evaluation Centre<br>Bagur, Rodrigo; Western University, Department of Epidemiology and Bio statistics<br>Cheng, Davy; London Health Sciences Centre University Hospital, Anesthesia & Perioperative Medicine<br>Kiaii, Bob; Western University, Department of Cardiology, Division of Cardiac Surgery<br>Hashi, Adam; Faculty of Sciences<br>Bi, Ran; Schulich Interfaculty Program in Public Health<br>Yeschin, Michelle ; Department of Epidemiology and Biostatistics, Western University<br>John-Baptiste, Ava; University of Western Ontario, Anesthesia & Perioperative Medicine |
| Primary Subject Heading: | Cardiovascular medicine |
| Secondary Subject Heading: | Cardiovascular medicine, Epidemiology |
| Keywords: | Valvular heart disease < CARDIOLOGY, EPIDEMIOLOGY, Cardiac surgery < SURGERY, VASCULAR SURGERY |
| | |
| Note: The following files were submitted by the author for peer review, but cannot be converted to PDF. You must view these files (e.g. movies) online. | |
| Supplementary appendix F Survival of frail patients 11.20.docx | |

SCHOLARONE™
Manuscripts



*I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.*

*The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.*

*Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.*

For peer review only

# Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis

Zhe Li [a, b, c], Emily Dawson [a, b, c], Jessica Moodie [b, c], Janet Martin [a, b, c], Rodrigo Bagur [a], Davy Cheng [b, c, d], Bob Kiaii [e], Adam Hashi [f], Ran Bi [g], Michelle Yeschin [a], Ava John-Baptiste [a, b, c, g]

**Affiliations**:

a. Department of Epidemiology & Biostatistics, Schulich School of Medicine and Dentistry,

Western University, London, ON, Canada.

b. Centre for Medical Evidence, Decision Integrity & Clinical Impact (MEDICI), Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

c. Department of Anesthesia & Perioperative Medicine, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

d. Department of Medicine, Division of Critical Care Medicine, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

For peer review only

e. Department of Surgery, Division of Cardiac Surgery, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

f. Faculty of Science, Western University, London, ON, Canada.

g. Schulich Interfaculty Program in Public Health, Western University, London, ON, Canada.

**Corresponding Author:** Dr. Ava John-Baptiste

The Western Centre for Public Health and Family Medicine

Rm 4111, 1465 Richmond St.

London, ON, Canada, N6G 2M1

e. ajohnbap@uwo.ca

t. 519.661.2111 (86265)

**Running head**: Prognosis of frail patients after TAVI

## Abstract

For peer review only

**Objectives**: Our objectives were to review the literature to identify frailty instruments in use for TAVI recipients and synthesize prognostic data from these studies, in order to inform clinical management of frail patients undergoing TAVI. **Methods**: We systematically reviewed the literature published in 2006 or later. We included studies of patients with aortic stenosis, diagnosed as frail, who underwent a TAVI procedure that reported mortality or clinical outcomes. We categorized the frailty instruments and reported on the prevalence of frailty in each study. We summarized the frequency of clinical outcomes and pooled outcomes from multiple studies. We explored heterogeneity and performed sub-group analysis, where possible. We also used GRADE to assess the overall certainty of the estimates. **Results**: Of 49 included studies, 21 used single-dimension measures to assess frailty, 3 used administrative data-based measures, and 25 used multi-dimensional measures. Prevalence of frailty ranged from 5.67% to 90.07%. Albumin was the most commonly used single-dimension frailty measure and the Fried or modified Fried phenotype were the most commonly used multi-dimensional measures. Meta-analyses of studies that used either the Fried or modified Fried phenotype showed a 30-day mortality of 7.86% (95% confidence interval,

For peer review only

5.20% to 11.70%) and a 1-year mortality of 26.91% (95% confidence interval, 21.50% to 33.11%). The GRADE system suggests very low certainty of the respective estimates.

**Conclusions**: Frailty instruments varied across studies, leading to a wide range of frailty prevalence estimates for TAVI recipients and substantial heterogeneity. The results provide clinicians, patients, and health care administrators, with potentially useful information on the prognosis of frail patients undergoing TAVI. This review highlights the need for standardization of frailty measurement to promote consistency.

## Strengths and limitations

- This study examines the heterogeneity across different frailty assessment tool and determines the frequency of adverse outcomes and pool the prognosis after transcatheter aortic valve implantation in frail patients.
- This study uses a comprehensive literature search strategy and includes frail patients from randomized controlled trials and observational studies.
- This study excluded studies in which dimensions of frailty were assessed without reference to the goal of frailty assessment.

For peer review only

**Keywords**: frailty; transcatheter aortic valve implantation; systematic review; meta-analysis

Transcatheter aortic valve implantation (TAVI) has become an alternative, less invasive treatment option for patients with severe symptomatic aortic stenosis.[1] The evidence continues to accumulate and synthesis of the evidence to better understand the prognosis of frail patients who undergo TAVI may be helpful.[2]

Frailty is a biological syndrome characterized by an increased vulnerability to stressors.[3] When exposed to stressors, such as chronic illness and surgery, frail patients are susceptible to adverse events, procedural complications, prolonged recovery, functional decline, and reduced survival.[4] Clinical research has identified frailty as an important risk factor for mortality and morbidity following TAVI.[5] Health economics research has shown that compared to non-frail patients, frail older adults undergoing cardiac surgery incurred substantially higher hospitalization costs.[6] Given the clinical and economic implications of TAVI, searching for and synthesizing outcomes of frail patients undergoing TAVI may provide information that can help optimize the selection of TAVI candidates and ultimately improve decision-making related to treatment of aortic stenosis.[2]

When considering valve procedures, clinical practice guidelines recommend assessing frailty as one component of risk.[7] We performed a systematic review of the literature to identify studies reporting the prognosis of frail patients undergoing TAVI. With no single standard method of measuring frailty and a diversity of frailty measurements, the optimal approach to assessing frailty in patients undergoing TAVI is unclear.[2,5] We catalogued frailty measures used in identified studies, to perform sub-group analyses for studies using the most common measures.

For peer review only

## Methods

This systematic review and meta-analysis is reported according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) guidelines[8] and follows the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines.[9] The protocol was registered in PROSPERO (CRD42018090597) and published in *BJM Open*.[10]

### Literature search and eligibility criteria

We searched PubMed, EMBASE, PsycINFO, Cochrane Library, Web of Science and ClinicalTrials.gov for articles published between January 2006 and September 23, 2020 (Supplementary appendix A). Conference abstracts from relevant conferences held in the last 3 years were also searched. We included studies of patients who underwent a TAVI procedure that reported frailty and its impact on mortality, clinical outcomes, or health-related quality of life. The detailed inclusion and exclusion criteria were described in detail in the protocol.[10] Authors (ZL, ED, AH, BR and MY) independently assessed study eligibility. Disagreements were resolved by consulting a third reviewer.

### Risk of bias assessment

The risk of bias in individual studies was appraised independently by two authors (ZL and ED) using the Quality in Prognosis Studies (QUIPS) tool.[11] We classified studies with four or five low risk domains as having a low risk of bias overall, studies with two or more high risk domains as high risk of bias overall, and the remaining studies as moderate risk of bias overall.

### Data synthesis and meta-analysis

Pre-specified statistical details were described in the protocol.[10] We summarized the method of measuring frailty used in each study including the frailty tool used, dimensions of frailty measured, the cut-off for frail status, and the prevalence of frailty in the study population as measured by the frailty tool. We only extracted data from the most commonly used frailty instruments if multiple frailty instruments were applied in the same patient group. We

For peer review only

categorized clinical outcomes and reported the frequency at each time point. Heterogeneity across studies was assessed using the Cochrane Q-statistic ($I^2$).[12]

For adverse clinical outcomes, we pooled proportions using the inverse-variance weighted DerSimonian and Laird model and incorporated the Freeman-Tukey double arcsine transformation.[13,14] A funnel plot was used to plot the effect estimates from individual studies against the standard error of each study. In the absence of bias and heterogeneity, the funnel plot will be symmetrical.[15] For the length of hospitalization, we pooled the values, estimating the mean and standard deviation using the random effects model for continuous variables.[16] For studies presenting Kaplan-Meier curves with time to death, we collected the information on numbers at risk and total number of events, and then created a single pooled Kaplan-Meier curve. We pooled time to death data from individual studies to obtain an overall estimate of survival, based on an algorithm developed by Guyot et al.[17] All analyses were conducted using R software (version 3.5.0). A two-sided p-value of 0.05 or less was considered statistically significant.

**Subgroup analysis**

We conducted a sub-group analysis to see if the estimates of mortality rates differed for studies that used the Fried phenotype, the most common multi-dimensional measure, compared to studies that did not use the Fried phenotype.

**Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) assessment**

We used the GRADE system to conduct an evaluation of the overall estimates based on considerations of risk of bias, consistency, precision, directness, and publication bias.[18] Given that cohort studies of prognosis exclude randomized controlled trial study designs, we did not downgrade the certainty of evidence due to observational study design.

**Results**

**Characteristics of included studies**

Our search identified 4,944 records with 2,635 articles remaining after removing duplicates. After screening, 49

For peer review only

studies[19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67] were identified as eligible for inclusion in the review (figure 1).

The characteristics of the included studies are summarized in supplementary appendix B. Three studies[40,43,53] enrolled patients from the Placement of Aortic Transcatheter Valves (PARTNER) trial reporting separately on outcomes of frail patients; the remaining studies reported on patients from a single cohort or registry. Most studies collected patient data prospectively; seventeen studies[21,23,24,26,32,35,38,41,43,45,46,48,57,60,61,62,65] were conducted retrospectively.

Supplementary appendix C summarizes the risk of bias assessment of individual studies. Of the 49 studies, 21 [25,26,28,29,31,33,37,39,40,47,49,52,55,58,59,60,62,63,64,65,66] were rated at overall low risk of bias, 21 [19-24, 32,35,36, 42-45, 48, 50,51, 53,54,56,61,] at moderate risk and 7 [27, 38, 41, 46, 50, 57,67] at high risk of bias.

**Measurement of frailty in patients undergoing TAVI**

Table 1 summarizes frailty assessment in patients undergoing TAVI. Twenty one studies [19-33, 54-59] used single-dimension measures, 3 studies [60-62] used administrative data-based measures, and 25 studies [34-53, 63-67] used multi-dimensional measures. The prevalence of frailty varied widely among studies that assessed frailty with single dimension measures, ranging from 5.67% to 90.07%. Albumin, body mass index, and Katz Activity of Daily Living were the three most commonly used single-dimension measures when assessing frailty in TAVI patients. However, even with the same measure, different cut-points or definitions of frailty were used. For example, four studies [21,23,24,59] used albumin to assess frailty; two [21,23] defined frailty as albumin level below 4g/dL, and two [24, 59] as albumin level below 3.5g/dL.

Among studies that used frailty indices based on administrative data, the prevalence of frailty ranged from 5.54% to 47.64%. Two studies [60, 62] used the Hospital Frailty Risk Score, a frailty algorithm calculated based on a list of predefined International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10) diagnostic codes. Frailty prevalence reported among the two studies was 41.06% and 47.64%, respectively. One study [61] used Johns Hopkins Adjusted Clinical Groups frailty-defining diagnosis indicator that was based on 10 clusters of frailty-defining diagnoses.

For peer review only

The prevalence of frailty reported by studies that assessed frailty using multi-dimensional measures ranged from 15.23% to 84.67%. Most of these studies assessed frailty based on the Fried frailty phenotype; one study[34] assessed frailty based on the accumulated deficits frailty index. Of the 25 studies reporting multi-dimensional measures, four [46-48, 51] used the original Fried frailty phenotype and eight [35,38-41,43,44,53] modified the Fried frailty phenotype by examining fewer dimensions, altering cut-off values or measuring the same domains with different criteria. Among the eight studies [35,38-41,43,44,53] reporting the modified Fried frailty phenotype, measures used to assess mobility and disability were identical. Measures used to assess nutrition were different; seven studies [35,38-40,43,44,53] measured serum albumin, and 1 study [41] measured weight loss.

**Prognosis of frail TAVI recipients**

Supplementary appendix D summarized prognosis of frail TAVI recipients reported for each study. Twenty studies [19,21,27-31,34-36,40,41,43,50,53,56,58,62,65,66] reported 30-day mortality, which ranged from 2.83% to 25%; the combined 30-day mortality estimate was 7.32% (95% confidence interval 5.66% to 9.42%, table 2, figure 2). Combining three studies [35,40,41] that measured frailty using the modified Fried frailty phenotype, we estimated a 30-day mortality of 7.86% (5.20% to 11.70%, table 2, figure 2).

Fifteen studies[25,30,31,34,35,37,40,43,46,49,50,53,56,58,62] reported 1-year mortality, ranging from 14.8% to 37.5%. The combined 1-year mortality estimate was 23.98% (20.71% to 27.58%, table 2, figure 3).When pooling two studies[35,46] that used the Fried or modified Fried frailty phenotype to assess frailty, the estimated 1-year mortality was 26.91% (21.50% to 33.11%, table 2, figure 3). Subgroup analyses of studies reporting frailty measurement using the Fried phenotype compared to non-Fried phenotype did not find statistical differences in effect estimates on 30-day and 1-year mortality (supplementary appendix E).

Seventeen studies [22,23,25-29,31,33,34,40,41,47,48,52,56,58,60,62]reported survival of frail patients after TAVI using a Kaplan-Meier curve. The combined survival estimates at 1, 2, and 3 years were 75.6% (75.2% to 76.0%, table 2), 65.0% (63.3% to 66.7%, table 2), and 48.7% (43.3% to 54.7%, table 2), respectively. Combining the studies that used the Fried or modified Fried phenotype, we found survival estimates at 1, 2, and 3 years were 73% (68.8% to 77.5%, table 2), 64.5% (56.4%

For peer review only

to 73.9%, table 2), and 58.9% (49% to 70.9%, table 2), respectively. Details of survival are provided in supplementary appendix F.

Five studies [42,44,63,65,67] measured health-related quality of life (supplementary appendix G). Three studies [44,65,67] assessed quality of life preoperatively using the 12-item Kansas City Cardiomyopathy Questionnaire (KCCQ). Two studies [65,67] assessed quality of life post-TAVI; both studies found improved quality of life overall. Okoh et al [44] assessed quality of life at 30 days following TAVI, and found that at 30 days, frail patients reported worsening in two domains, KCCQ-symptoms and KCCQ physical limitation, but quality of life improved slightly overall. Kobe et al [42] assessed quality of life before and 30 days after TAVI using the Short Form-36 questionnaire; they found that at 30-day follow-up, the mean scores of all but role physical and social functioning were significantly lower for frail patients. Goudzwaard et al [63] assessed quality of life using the Euro-QoL-5-dimension (EQ-5D) scale; they found that at 12 months follow-up, the mean EQ-5D decreased while the mean EQ-visual analogue scale increased.

Other commonly reported outcomes measuring the prognosis of frail TAVI recipients include procedural acute kidney injury (ranging from 3.95% to 20.51%), conversion to open heart surgery (ranging from 0% to 9.9%), life-threatening bleeding (ranging from 4.86% to 16.7%), major bleeding (ranging from 2.56% to 21.81%), permanent pacemaker implantation (ranging from 2% to 12.82%), and stroke (ranging from 0% to 8.3%). Eight studies[32,33,38,39,41,44,45,56] reported the mean length of hospitalization, ranging from 5 days to 12.1 days.

### GRADE assessment

The GRADE certainty assessment per outcome, together with the pooled effects, is provided in table 2. Due to inconsistency as influenced by heterogeneity of estimates and indirectness of frailty measures as influenced by lack of homogeneity across the TAVI populations, confidence in the overall estimates was very low.

## Discussion

### Principal findings

For Peer Review Only

We found that multi-dimensional measures are more commonly used than single-dimension measures. Even with the same frailty measure, different definitions or cut-offs were used. The most frequently used frailty measure in TAVI the studies we identified was the modified Fried phenotype, in which disability, muscle strength, mobility, and nutrition were assessed. Approaches to modifying the Fried phenotype included measuring fewer domains, using different cut-offs, or using different tools to assess the same domain.

Greater heterogeneity of meta-analyses that included single measures suggests single measures did not measure the same frailty construct and did not reliably measure frailty. Single measures included a mix of biological variables (albumin and BMI) or single performance measures (gait speed or activities of daily living), which address only a single component of the frailty construct. Thus, our study suggests that frailty is a multi-dimensional phenomenon that cannot be captured by a single construct.

The variety of frailty definitions and the diversity of TAVI populations in the studies contribute to the wide range and substantial heterogeneity of patient outcomes after TAVI. Using GRADE to assess confidence in prognosis estimates from the meta-analyses, we found very low confidence in the overall estimates, mainly due to inconsistency as influenced by heterogeneity of estimates and indirectness of frailty measures as influenced by lack of homogeneity across the TAVI populations identified in the studies.

**Comparison with other studies**

Previous studies demonstrated that the assessment of frailty significantly enhances prediction of mortality after TAVI when combined with the European system for cardiac operative risk evaluation (EuroSCORE) or the Society of Thoracic Surgeons (STS) score.[49] There have been several studies reviewing frailty in cardiac surgical populations. Kim et al[5] conducted a systematic review of frailty instruments in older adults undergoing cardiac surgical procedures. Kim et al[5] found high quality evidence that used mobility assessment as a single frailty measure and found mobility to be the most frequently assessed domain. Sepehri et al[68] performed a systematic review to demonstrate the association of frailty with negative postoperative outcomes in patients undergoing cardiac surgery. Our study adds to the existing literature as we investigate the frequency of adverse outcomes and pool estimates of survival after TAVI in frail patients from multiple studies.

For peer review only

The FRAILTY-AVR study[69] examined the validity of frailty measures in predicting mortality amongst TAVI recipients. The study added value to the literature by selecting frailty elements with the greatest predictive value, finding that the Essential Frailty Toolset (EFT) consisting of chair rise, cognition measured by the Mini-Mental State Examination, hemoglobin and serum albumin, performed best for predicting one-year mortality.[69] Due to the focus on predictive validity, the FRAILTY-AVR study[69] did not report outcomes separately for frail patients. As a result, the study[69] did not meet the inclusion criteria for our systematic review, which was focused on prognostic information amongst frail patients only. The FRAILTY-AVR study[69] makes important efforts to define a standard frailty assessment tool. Although the Fried and modified Fried were the most commonly used instruments amongst studies included in our meta-analysis, the FRAILTY-AVR showed the Fried did not perform as well as the EFT in predicting mortality amongst TAVI patients.[69] We suggest use of a standard measure, such as the EFT, can enhance the quality of frailty research in the TAVI patient population. We also recognize that use of a standard frailty measure is unlikely as researchers and clinicians may value use of diverse measures which reflect different aspects of frailty. If the EFT emerges as a standard, it may be used by clinicians to exclude frail patients from treatment, due to concerns about increased mortality. This would limit the opportunity to better understand the prognosis of frail patients undergoing TAVI, which was the primary goal of our study.

**Strengths and limitations**

This review has several unique strengths. We performed a comprehensive literature search to identify both published and unpublished studies, in addition to searching citations from previous reviews. We included prognostic data from randomized controlled trials and observational studies. Using the QUIPS tool, two reviewers independently assessed the risk of bias, and the use of the GRADE system to assess the certainty of evidence offers a structured and transparent evaluation of our findings. We systematically reviewed the operationalization of frailty assessment in TAVI patients, and pooled clinical outcomes of frail TAVI recipients. We tested for heterogeneity and attempted to address heterogeneity by performing sensitivity analysis and sub-group analysis.

This review has some important limitations. Given the limited data reported by the

For peer review only

included studies, we were unable to perform meta-regression to further investigate the potential sources of heterogeneity and to determine the influence of mean age on outcomes. We therefore explored the causes and types of heterogeneity relying on the investigation of the $I^2$ statistic, which may be imprecise when the number of studies is small.[70] When extracting data, we encountered several studies that applied multiple frailty instruments in the same patient group, and in this situation, we only extracted data from the most commonly used frailty instrument, and this may introduce selection bias. Some studies defined an intermediate 'pre-frail' group, but we did not find sufficient data to synthesize outcomes for this important sub-group. Though less vulnerable than the frail group, pre-frail patients may be at higher risk than robust patients for experiencing adverse outcomes.[71,72] Individual-patient level data were not available, precluding adjustment for any study level differences in clinical or procedural variables that may have influenced prognosis across the cohorts. Therefore, clinical heterogeneity could not be ruled-out and along with high levels of heterogeneity, resulted in lower GRADE evaluations. The aim of this study was to characterize prognosis for frail patients undergoing TAVI, therefore, we did not directly compare prognosis to other groups of patients or to frail patients undergoing different therapies, nor were we able to determine which frailty measures perform best as prognostic tools for TAVI recipients.

**Implications**

When selecting candidates to undergo TAVI, several multivariate risk scores have been widely used to estimate operative mortality based on patient characteristics. The STS score and the EuroSCORE are the most commonly used scoring systems.[73,74] However, a disadvantage of both scores is that the main variables for scoring perioperative risk are medical diagnoses and comorbidities, which may not reflect the true 'biological status' of the patient.[73,74] When considering valve procedures for patients, clinical practice guidelines recommend assessing frailty as one component of risk.[5,7] Although a large number of frailty measures exist, there is currently little consensus on the optimal approach to assessing frailty in patients undergoing TAVI.[2] Frailty has consistently been shown to significantly predict mortality[68] and post-operative delirium,[75] even after controlling for other risk factors, suggesting that use of any frailty assessment is better than none when selecting patients for TAVI. Systematically reviewing the operationalization of frailty assessment in TAVI patients and pooling clinical outcomes of frail

For peer review only

TAVI recipients will help better understand how frailty is assessed among TAVI patients, provide information on the prognosis of frail patients after TAVI, and can ultimately improve decisions related to treatment of AS.

To help achieve consensus on frailty measures to be applied in TAVI recipients, future studies should evaluate the prognostic value of frailty measures in TAVI recipients and determine the additional prognostic value of frailty measurement in addition to these established risk scores. Future studies should also compare prognosis of frail patients undergoing TAVI to frail patients undergoing surgical intervention or medical therapy. Few studies reported quality of life measures. In order to address the gaps in the literature future studies should measure quality of life before and after TAVI with use of standardized quality of life measurement tools such as the Short-Form 36.

**Conclusion**

In conclusion, frailty instruments for TAVI recipients varied across studies, leading to a range of frailty prevalence estimates and substantial heterogeneity. The results of this systematic review provide clinicians, patients, and health care administrators, with potentially useful evidence on the prognosis of frail patients.

**Acknowledgement:** We gratefully thank Dr. Farid Foroutan for his assistance with meta-analysis of survival data.

**Ethics committee approval:** Due to the nature of the study, there are no ethical concerns and informed consent was not required.

**Contributorship statement:** Zhe Li, Emily Dawson, Jessica Moodie, Janet Martin, Rodrigo Bagur, Davy Cheng, Bob Kiaii, and Ava John-Baptiste proposed and designed

For peer review only

the purpose, review questions, and methods of this study. Searching strategy was developed by Jessica Moodie, and revised by Zhe Li, Emily Dawson, and Ava John-Baptiste. Zhe Li, Emily Dawson, Ava John-Baptiste, Adam Hashi, Ran Bi and Michelle Yeschin screened articles and verified data abstraction. Zhe Li and Emily Dawson abstracted data from articles. Zhe Li, Emily Dawson and Ava John-Baptiste critically appraised the articles. Data analysis was primarily conducted by Zhe Li. Ava John-Baptiste provided primary academic supervision. Zhe Li drafted the manuscript. All co-authors have contributed to the revision of the manuscript.

**Competing interests:** There is no competing interest to declare.

**Funding:** This work was supported by the Academic Medical Organization of Southwestern Ontario (grant number: INN 17-001).

**Data sharing statement**: There is no additional unpublished data from the study. All data are provided in full in the main manuscript document.

## References

1. Siontis GCM, Overtchouk P, Cahill TJ, et al. Transcatheter aortic valve

For peer review only

implantation vs. surgical aortic valve replacement for treatment of symptomatic severe aortic stenosis: An updated meta-analysis. *Eur Heart J*. 2019;40(38):3143-3153. doi:10.1093/eurheartj/ehz275

2. Puri R, Iung B, Cohen DJ, Rodés-Cabau J. TAVI or No TAVI: Identifying patients unlikely to benefit from transcatheter aortic valve implantation. *Eur Heart J*. 2016;37(28):2217-2225. doi:10.1093/eurheartj/ehv756
3. Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: Evidence for a phenotype. *Journals Gerontol - Ser A Biol Sci Med Sci*. 2001;56(3):146-157. doi:10.1093/gerona/56.3.m146
4. Shamliyan T, Talley KMC, Ramakrishnan R, Kane RL. Association of frailty with survival: A systematic literature review. *Ageing Res Rev*. 2013;12(2):719-736. doi:10.1016/j.arr.2012.03.001
5. Kim DH, Kim CA, Placide S, Lipsitz LA, Marcantonio E. Preoperative frailty assessment and outcomes at 6 months or later in older adults undergoing cardiac surgical procedures: A systematic review. *Ann Intern Med*. 2016;165(9):650-660. doi:10.7326/M16-0652.Preoperative
6. Goldfarb M, Bendayan M, Rudski LG, et al. Cost of Cardiac Surgery in Frail Compared With Nonfrail Older Adults. *Can J Cardiol*. 2017;33:1020-1026. doi:10.1016/j.cjca.2017.03.019
7. Nishimura RA, Otto CM, Bonow RO, et al. 2017 AHA/ACC focused update of the 2014 AHA/ACC guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2017;70(2):252-289. doi:10.1016/j.jacc.2017.03.011
8. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M SP, LA. S. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev*. 2015;4(1):1-19.
9. Stroup DF, Berlin JA, Morton SC, Olkin I, Williamson GD, Rennie D, Moher D, Becker BJ, Sipe TA TS. Meta-analysis of observational studies in Epidemiology. *Mod Methods Epidemiol*. Published online 2012:173-189. doi:10.1007/978-94-007-3024-3_10

For peer review only

10. Li Z, Dawson E, Moodie J, et al. Frailty in patients undergoing transcatheter aortic valve implantation : a protocol for a systematic review. 2019;(1):1-7. doi:10.1136/bmjopen-2018-024163
11. Hayden J a, Windt D a Van Der, Cartwright JL, Co P. Assessing Bias in Studies of Prognostic Factors. *Ann Intern Med*. 2013;144:427-437. doi:10.7326/0003-4819-158-4-201302190-00009
12. Higgins JPT, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *Br Med J*. 2003;327(7414):557-560. doi:10.1136/bmj.327.7414.557
13. Jackson D, White IR, Thompson SG. Extending DerSimonian and Laird's methodology to perform multivariate random effects meta-analyses. *Stat Med*. 2010;29(12):1282-1297. doi:10.1002/sim.3602
14. Nyaga VN, Arbyn M, Aerts M. Metaprop: A Stata command to perform meta-analysis of binomial data. *Arch Public Heal*. 2014;72(1):1-10. doi:10.1186/2049-3258-72-39
15. Sterne JAC, Sutton AJ, Ioannidis JPA, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ*. 2011;343(7818):1-8. doi:10.1136/bmj.d4002
16. Marín-Martínez F, Sánchez-Meca J. Weighting by Inverse Variance or by Sample Size in Random-Effects Meta-Analysis. *Educ Psychol Meas*. 2010;70(1):56-73. doi:10.1177/0013164409344534
17. Guyot P, Ades AE, Ouwens MJNM, Welton NJ. Enhanced secondary analysis of survival data: Reconstructing the data from published Kaplan-Meier survival curves. *BMC Med Res Methodol*. 2012;12. doi:10.1186/1471-2288-12-9
18. Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P SH. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *Rev Colomb Obstet Ginecol*. 2016;67(1):26-35. doi:10.18597/rcog.371
19. Alfredsson J, Stebbins A, Brennan JM, et al. Gait Speed Predicts 30-Day Mortality After Transcatheter Aortic Valve Replacement Cardiology Transcatheter Valve Therapy Registry. *Ciculation*. 2016;133:1351-1359.

For peer review only

doi:10.1161/CIRCULATIONAHA.115.020279

20. Bagienski M, Kleczynski P, Dziewierz A, et al. Incidence of Postoperative Delirium and Its Impact on Outcomes After Transcatheter Aortic Valve Implantation. *Am J Cardiol*. 2017;120(7):1187-1192. doi:10.1016/j.amjcard.2017.06.068
21. Bogdan A, Barbash IM, Segev A, Fefer P, Bogdan SN, Asher E, Fink N, Hamdan A, Spiegelstein D, Raanani E G V. Albumin correlates with all-cause mortality in elderly patients undergoing transcatheter aortic valve implantation. *EuroIntervention*. 2016;12(e1057-64).
22. Cockburn J, Singh MS, Rafi NHM, et al. Poor mobility predicts adverse outcome better than other frailty indices in patients undergoing transcatheter aortic valve implantation. *Catheter Cardiovasc Interv*. 2015;86(7):1271-1277. doi:10.1002/ccd.25991
23. Grossman Y, Barbash IM, Fefer P, et al. Addition of albumin to Traditional Risk Score Improved Prediction of Mortality in Individuals Undergoing Transcatheter Aortic Valve Replacement. *J Am Geriatr Soc*. 2017;65(11):2413-2417. doi:10.1111/jgs.15070
24. Koifman E, Magalhaes MA, Ben-Dor I, et al. Impact of pre-procedural serum albumin levels on outcome of patients undergoing transcatheter aortic valve replacement. *Am J Cardiol*. 2015;115(9):1260-1264. doi:10.1016/j.amjcard.2015.02.009
25. Kleczynski P, Dziewierz A, Bagienski M, et al. Impact of frailty on mortality after transcatheter aortic valve implantation. *Am Heart J*. 2017;185:52-58. doi:10.1016/j.ahj.2016.12.005
26. Mok M, Allende R, Leipsic J, et al. Prognostic value of fat mass and skeletal muscle mass determined by computed tomography in patients who underwent transcatheter aortic valve implantation. *Am J Cardiol*. 2016;117(5):828-833. doi:10.1016/j.amjcard.2015.12.015
27. Martin GP, Sperrin M, Ludman PF, et al. Do frailty measures improve prediction of mortality and morbidity following transcatheter aortic valve implantation? An analysis of the UK TAVI registry. *BMJ Open*. 2018;8(6):1-9. doi:10.1136/bmjopen-2018-022543

For peer review only

28. Puls M, Sobisiak B, Bleckmann A, Jacobshagen C, Danner BC, Huenlich M, Beissbarth T, Schoendube F, Hasenfuss G, Seipelt R SW. Impact of frailty on short-and long-term morbidity and mortality after transcatheter aortic valve implantation: risk assessment by Katz Index of activities of daily living. *EuroIntervention J Eur Collab with Work Gr Interv Cardiol Eur Soc Cardiol*. 2014;10(609-19).

29. Rodés-cabau J, Webb JG, Cheung A, et al. Transcatheter Aortic Valve Implantation for the Treatment of Severe Symptomatic Aortic Stenosis in Patients at Very High or Prohibitive Surgical Risk Acute and Late Outcomes of the Multicenter Canadian Experience. *JAC*. 2010;55(11):1080-1090. doi:10.1016/j.jacc.2009.12.014

30. Stortecky S, Windecker S, Pilgrim T et al. The association of body mass index with clinical outcome in patients with severe aortic stenosis undergoing transcatheter aortic valve implantation. *EuroIntervention*. 2012;8.

31. Shimura T, Yamamoto M, Kano S, et al. Impact of the clinical frailty scale on outcomes after transcatheter aortic valve replacement. *Circulation*. 2017;135(21):2013-2024. doi:10.1161/CIRCULATIONAHA.116.025630

32. Traynor MM, Greason KL, Nkomo VT, et al. Assisted care as a baseline patient risk characteristic affecting the outcome of transcatheter aortic valve insertion. *Catheter Cardiovasc Interv*. 2017;89(7):1268-1272. doi:10.1002/ccd.26580

33. Yamamoto M, Hayashida K, Watanabe Y, et al. Effect of body mass index <20 kg/m2 on events in patients who underwent transcatheter aortic valve replacement. *Am J Cardiol*. 2015;115(2):227-233. doi:10.1016/j.amjcard.2014.10.026

34. Bureau ML, Liuu E, Christiaens L, et al. Using a multidimensional prognostic index (MPI) based on comprehensive geriatric assessment (CGA) to predict mortality in elderly undergoing transcatheter aortic valve implantation. *Int J Cardiol*. 2017;236:381-386. doi:10.1016/j.ijcard.2017.02.048

35. Chauhan D, Haik N, Merlo A, Haik BJ, Chen C. Quantitative increase in frailty is associated with diminished survival after transcatheter aortic valve replacement. Published online 2015. doi:10.1016/j.ahj.2016.06.028

For peer review only

36. Capodanno D, Barbanti M, Tamburino C, et al. A simple risk tool (the OBSERVANT score) for prediction of 30-day mortality after transcatheter aortic valve replacement. *Am J Cardiol*. 2014;113(11):1851-1858. doi:10.1016/j.amjcard.2014.03.014
37. Eichler S, Salzwedel A, Harnath A, et al. Nutrition and mobility predict all-cause mortality in patients 12 months after transcatheter aortic valve implantation. *Clin Res Cardiol*. 2018;107(4):304-311. doi:10.1007/s00392-017-1183-1
38. Abhijit Ghatak, Brian O'Neill, Ruchi Patel, Sachil Shah, Claudia Martinez, Mauro Moscucci, Pedro Clark, Mauricio Cohen, Alan Heldman WO and CA. Frailty is a Major Determinant of Length of Stay After Transcatheter Aortic Valve Replacement. *J Am Coll Cardiol*. 2012;60(17). doi:10.1016/j.jacc.2012.08.921
39. Green P, Woglom AE, Genereux P, et al. The impact of frailty status on survival after transcatheter aortic valve replacement in older adults with severe aortic stenosis: A single-center experience. *JACC Cardiovasc Interv*. 2012;5(9):974-981. doi:10.1016/j.jcin.2012.06.011
40. Green P, Arnold S V., Cohen DJ, et al. Relation of Frailty to Outcomes After Transcatheter Aortic Valve Replacement (from the PARTNER Trial). *Am J Cardiol*. 2015;116(2):264-269. doi:10.1016/j.amjcard.2015.03.061
41. Huded CP, Huded JM, Friedman JL, et al. Frailty Status and Outcomes After Transcatheter Aortic Valve Implantation. Published online 2016. doi:10.1016/j.amjcard.2016.03.044
42. Kobe AR, Meyer A, Elmubarak H, et al. Frailty assessed by the FORECAST is a valid tool to predict short-term outcome after transcatheter aortic valve replacement. *Innov Technol Tech Cardiothorac Vasc Surg*. 2016;11(6):407-413. doi:10.1097/IMI.0000000000000321
43. Maniar H, Zajarias A, Quader N, Arnold S, Cohen D, Mack M, Melby S, Alu M, Ayele G, Lindman B LJ. THE ASSOCIATION OF FRAILTY WITH MORTALITY IN HIGH RISK PATIENTS UNDERGOING TRANSCATHETER AORTIC VALVE REPLACEMENT: AN ANALYSIS OF THE PARTNER II ALTERNATIVE ACCESS COHORTS. *J Am Coll Cardiol*. 2016;67 (13 Sup:241.
44. Okoh AK, Chauhan D, Kang N, et al. The impact of frailty status on clinical and

For peer review only

functional outcomes after transcatheter aortic valve replacement in nonagenarians with severe aortic stenosis. *Catheter Cardiovasc Interv*. 2017;90(6):1000-1006. doi:10.1002/ccd.27083

45. Patel J, Banga S, Kim MC, Wattanakit K, Barzallo MA MS. TCT-684 Frailty Profile is Independently Associated with Higher Cost for Patients undergoing Transcatheter Aortic Valve Replacement for Symptomatic Severe Aortic Stenosis: a Single Center Experience. *J Am Coll Cardiol*. 2016;68(18 Supp:B277.
46. Rabinovitz E, Finkelstein A, Ben E, et al. Norton scale for predicting prognosis in elderly patients undergoing trans-catheter aortic valve implantation : A historical prospective study. *J Cardiol*. 2016;67(6):519-525. doi:10.1016/j.jjcc.2016.01.017
47. Rodríguez-Pascual C, Paredes-Galán E, Ferrero-Martínez AI, et al. The frailty syndrome and mortality among very old patients with symptomatic severe aortic stenosis under different treatments. *Int J Cardiol*. 2016;224:125-131. doi:10.1016/j.ijcard.2016.09.020
48. Rogers T, Alraies MC, Moussa Pacha H, et al. Clinical Frailty as an Outcome Predictor After Transcatheter Aortic Valve Implantation. *Am J Cardiol*. 2018;121(7):850-855. doi:10.1016/j.amjcard.2017.12.035
49. Schoenenberger AW, Bertschi D, Wenaweser P, et al. Improvement of Risk Prediction After Transcatheter Aortic Valve Replacement by Combining Frailty With Conventional Risk Scores. 2018;11(4). doi:10.1016/j.jcin.2017.11.012
50. Steinvil A, Buchanan KD, Kiramijyan S, et al. Utility of an additive frailty tests index score for mortality risk assessment following transcatheter aortic valve replacement. *Am Heart J*. 2018;200:11-16. doi:10.1016/j.ahj.2018.01.007
51. Shi S, Afilalo J, Lipsitz LA, et al. Frailty phenotype and deficit accumulation frailty index in predicting recovery after transcatheter and surgical aortic valve replacement. *Journals Gerontol - Ser A Biol Sci Med Sci*. 2019;74(8):1249-1256. doi:10.1093/gerona/gly196
52. Skaar E, Eide LSP, Norekvål TM, et al. A novel geriatric assessment frailty score predicts 2-year mortality after transcatheter aortic valve implantation. *Eur Hear J - Qual Care Clin Outcomes*. 2019;5(2):153-160. doi:10.1093/ehjqcco/qcy044
53. Zajarias A, Maniar H, Arnold S, Quader N, Cohen D, Melby S, Alu M, Ayele G,

For peer review only

Lindman B, Lasala J MR. The association of frailty with mortality in high risk patients undergoing transcatheter aortic valve replacement: an analysis of the partner ii b transfemoral cohort. *J Am Coll Cardiol*. 2016;67(13 Supp:249.

54. Welle GA, Mahowald M, Alqahtani F, Fatema K, Alkhouli M. Five Meter Walk Test Is an Independent Predictor of Mortality in Transcutaneous Aortic Valve Replacement. *J Am Coll Cardiol*. 2020;75(11):1439. doi:10.1016/s0735-1097(20)32066-0
55. Mach M, Watzal V, Hasan W, et al. Fitness-Tracker Assisted Frailty-Assessment Before Transcatheter Aortic Valve Implantation: Proof-of-Concept Study. *JMIR mHealth uHealth*. 2020;8(10):e19227. doi:10.2196/19227
56. Kiani S, Stebbins A, Thourani VH, et al. The Effect and Relationship of Frailty Indices on Survival After Transcatheter Aortic Valve Replacement. *JACC Cardiovasc Interv*. 2020;13(2):219-231. doi:10.1016/j.jcin.2019.08.015
57. Gharibeh S, Zweiker D, Binder J, et al. A routine nursing questionnaire adds predictive value to conventional risk scores for TAVI ( Transcatheter Aortic Valve Implantation ) outcome. *https://academic.oup.com/eurheartj/article/40/Supplement_1/ehz7450596/5594816*. 2019;(September):3743.
58. Voigtländer L, Twerenbold R, Schäfer U, et al. Prognostic Impact of Underweight (Body Mass Index <20 kg/m2) in Patients With Severe Aortic Valve Stenosis Undergoing Transcatheter Aortic Valve Implantation or Surgical Aortic Valve Replacement (from the German Aortic Valve Registry [GARY]). *Am J Cardiol*. 2020;129(1):79-86. doi:10.1016/j.amjcard.2020.05.002
59. Shimura T, Yamamoto M, Kano S, et al. Predictors and prognostic impact of nutritional changes after transcatheter aortic valve replacement. *Cardiovasc Revascularization Med*. 2020;(xxxx). doi:10.1016/j.carrev.2020.08.031
60. Malik AH, Yandrapalli S, Zaid S, et al. Impact of Frailty on Mortality, Readmissions, and Resource Utilization After TAVI. *Am J Cardiol*. 2020;127:120-127. doi:10.1016/j.amjcard.2020.03.047
61. Sami F, Ranka S, Shah A, Torres C, Villablanca P. Impact of Frailty on Outcomes in Patients Undergoing Transcatheter Aortic Valve Replacement: a Report From

For peer review only

National Inpatient Sample. *J Am Coll Cardiol*. 2020;75(11):1487. doi:10.1016/s0735-1097(20)32114-8

62. Kundi H, Popma JJ, Reynolds MR, et al. Frailty and related outcomes in patients undergoing transcatheter valve therapies in a nationwide cohort. *Eur Heart J*. 2019;40(27):2231-2239. doi:10.1093/eurheartj/ehz187
63. Goudzwaard JA, de Ronde-Tillmans MJAG, van Hoorn FED, et al. Impact of frailty on health-related quality of life 1 year after transcatheter aortic valve implantation. *Age Ageing*. Published online 2020:1-6. doi:10.1093/ageing/afaa071
64. Goudzwaard JA, de Ronde-Tillmans MJAG, de Jager TAJ, et al. Incidence, determinants and consequences of delirium in older patients after transcatheter aortic valve implantation. *Age Ageing*. 2020;49(3):389-394. doi:10.1093/ageing/afaa001
65. Patel JN, Ahmad M, Kim M, et al. Relation of Frailty to Cost for Patients Undergoing Transcatheter Aortic Valve Implantation. *Am J Cardiol*. 2020;125(3):469-474. doi:10.1016/j.amjcard.2019.10.021
66. Drudi LM, Ades M, Asgar A, et al. Interaction Between Frailty and Access Site in Older Adults Undergoing Transcatheter Aortic Valve Replacement. *JACC Cardiovasc Interv*. 2018;11(21):2185-2192. doi:10.1016/j.jcin.2018.06.037
67. Morris J, Bruckel J, Welch T, et al. Subjective Versus Objective Frailty Assessment As a Predictor of Outcomes With Transcatheter Aortic Valve Replacement. *J Am Coll Cardiol*. 2020;75(11):1176. doi:10.1016/s0735-1097(20)31803-9
68. Sepehri A, Beggs T, Hassan A. The impact of frailty on outcomes after cardiac surgery : A systematic review. *J Thorac Cardiovasc Surg*. 2014;148(6):3110-3117. doi:10.1016/j.jtcvs.2014.07.087
69. Afilalo J, Lauck S, Kim DH, et al. Frailty in older adults undergoing aortic valve replacement: the FRAILTY-AVR study. *J Am Coll Cardiol*. 2017;70(6):689-700. doi:10.1016/j.jacc.2017.06.024
70. Hippel PT Von. The heterogeneity statistic I 2 can be biased in small meta-analyses. Published online 2015:1-8. doi:10.1186/s12874-015-0024-z
71. Bouillon K, Kivimaki M, Hamer M, et al. Measures of frailty in population-based

For peer review only

studies: An overview. *BMC Geriatr*. 2013;13(1). doi:10.1186/1471-2318-13-64

72. Frost R, Belk C, Jovicic A, et al. Health promotion interventions for community-dwelling older people with mild or pre-frailty : a systematic review and meta-analysis. *BMC Geriatr*. 2017;1:157. doi:10.1186/s12877-017-0547-8
73. Sündermann S, Dademasch A, Praetorius J, Kempfert J, Dewey T, Falk V, Mohr FW WT. Comprehensive assessment of frailty for elderly high-risk patients undergoing cardiac surgery. *Eur J Cardio-thoracic Surg*. 2011;39:33-37. doi:10.1016/j.ejcts.2010.04.013
74. Vahanian A, Otto CM. Risk stratification of patients with aortic stenosis. *Eur Heart J*. 2010;31(4):416-423. doi:10.1093/eurheartj/ehp575
75. Jung P, Pereira MA, Hiebert B, et al. The impact of frailty on postoperative delirium in cardiac surgery patients. *J Thorac Cardiovasc Surg*. 2015;149(3):869-875.e2. doi:10.1016/j.jtcvs.2014.10.118

## Figure legends

**Figure 1. PRISMA flow diagram of included studies**

PRISMA, reporting in systematic reviews and meta-analyses. N, number of studies.

**Figure 2a. Meta-analysis of 30-day mortality in frail patients after TAVI.**

Frailty was measured using sing and multi-dimensional measures.

The squares indicate the 30-day mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 30-day mortality

For peer review only

**Figure 2b. Funnel plots, using data from all studies that reported 30-day mortality**

y-axis is the standard error of the 30-day mortality

x-axis is the logit of 30-day mortality

**Figure 2c. Meta-analysis of 30-day mortality in frail patients after TAVI.**

Frailty was measured using modified Fried frailty phenotype.

The squares indicate the 30-day mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 30-day mortality

**Figure 2d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.**

y-axis is the standard error of the 30-day mortality

x-axis is the logit of 30-day mortality

For peer review only

**Figure 3a. Meta-analysis of 1-year mortality in frail patients after TAVI.**

Frailty was measured using single and multi-dimensional measures.

The squares indicate the 1-year mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 1-year mortality

**Figure 3b. Funnel plots, using data from all studies that reported 30-day mortality**

y-axis is the standard error of the 1-year mortality

x-axis is the logit of 1-year mortality

**Figure 3c. Meta-analysis of 1-year mortality in frail patients after TAVI.**

Frailty was measured using the Fried frailty phenotype.

The squares indicate the 1-year mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 1-year mortality

For peer review only

**Figure 3d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.**

y-axis is the standard error of the 1-year mortality

x-axis is the logit of 1-year mortality

Table 1. Frailty assessment in patients undergoing TAVI

| Studies that used a single dimension to assess frailty | | | | | |
|---|---|---|---|---|---|
| **Study, year** | **Measure** | **Dimensions** | **Definition** | **Total N** | **Frail n (%)** |
| Alfredsson, 2016 [19 a] | Gait speed (5 meters) | mobility | <0.83m/s or >6 s | 8,039 | 6,100 (75.88%) |
| Bagienski, 2017 [20 b] | Katz ADL | disability | <6 points | 141 | 127 (90.07%) |
| Bogdan, 2016 [21] | Albumin | nutrition | ≤4g/dl | 150 | 79 (52.67%) |
| Cockburn, 2015 [22] | Brighton mobility index | mobility | Poor mobility | 312 | 65 (20.83%) |
| Grossman, 2017 [23] | Albumin | nutrition | <4 g/dl | 426 | 192 (45.07%) |
| Koifman, 2015 [24 c] | Albumin | nutrition | <3.5g/dl | 476 | 238 (50%) |
| Kleczynski, 2017 [25] | ISAR | unclear | ≥2 points | 101 | 53 (52.48%) |
| Mok, 2016 [26] | Sarcopenia | nutrition | skeletal muscle mass index 2 SDs less than the mean SMM of young, healthy gender-specific reference ranges | 460 | 293 (63.70%) |
| Martin, 2018 [27] | CSHA score (1-7) | physical function | Scores 5-7 | 2624 | 1043 (39.75%) |
| Puls, 2014 [28] | Katz ADL | disability | <6 points | 300 | 144 (48%) |
| Rodes-Cabau, 2010 [29] | Clinical judgment | subjective | Unclear | 339 | 85 (25.07%) |
| Stortecky, 2012 [30] | BMI | nutrition | <20kg/m$^2$ | 256 | 24 (9.38%) |
| Shimura, 2017 [31 d] | CFS | subjective | ≥5 points (score ranges 0-9) | 1215 | 353 (29.05%) |
| Traynor, 2017 [32] | Assisted care | unclear | Need assisted care | 597 | 60 (10.05%) |
| Yamamoto, 2015 [33] | BMI | nutrition | <20kg/m$^2$ | 777 | 56 (7.21%) |
| Welle, 2020 [54] | Gait speed (5 meters) | mobility | ≥6 s | 723 | 483 (66.8%) |
| Mach, 2020 [55] | Fitness-tracker assisted frailty score | unclear | ≥1 point | 50 | 39 (78%) |
| Kiani, 2020 [56 a] | Gait speed (5 meters) | mobility | <0.83m/s or >6 s (including unable to perform the test) | 56,500 | 11,316 (20.03%) |
| Gharibeh, 2019 [57] | Clinical judgment | subjective | Indicators for limited self-dependence | 461 | 186 (40.35%) |
| Voigtlander, 2020 [58] | BMI | nutrition | <20kg/m$^2$ | 16,865 | 956 (5.67%) |
| Shimura, 2020 [59 d] | Albumin | nutrition | <3.5 g/dl | 1,524 | 284 (18.64%) |
| **Studies that used administrative database algorithms to assess frailty** | | | | | |
| **Study, year** | **Measure** | **Definition/ cut-off points** | | **Total N** | **Frail n (%)** |
| Malik, 2020 [60] | Hospital Frailty Risk Score | Hospital Frailty Risk Score ≥ 5 points | | 20,504 | 8,419 (41.06%) |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Sami, 2020 [61] | Johns-Hopkins ACG frailty indicator | | A dichotomous indicator defined based on 10 clusters of frailty-defining diagnoses | | 51,685 | 2,865 (5.54%) |
| Kundi, 2019 [62] | Hospital Frailty Risk Score | | Hospital Frailty Risk Score ≥ 5 points | | 28,531 | 13,593 (47.64%) |
| **Studies that used multiple dimensions to assess frailty** | | | | | | |
| **Study, year** | **Name** | **Measures** | **Dimensions** | **Definition** | **Total N** | **Frail n (%)** |
| Bureau, 2017 [34] | MPI | ADL | disability | MPI≥0.34 (the sum of all domain values is divided by 8 to obtain the MPI score between 0 and 1) | 116 | 71 (61.21%) |
| | | IADL | disability | | | |
| | | SPMSQ | cognition | | | |
| | | CIRS-CI | medical | | | |
| | | MNA-SF | nutrition | | | |
| | | ESS | medical | | | |
| | | Number of medications | medical | | | |
| | | Social support network | living status | | | |
| Chauhan, 2016 [35 e] | Modified Fried phenotype | ADL | disability | Presence of 2 or more criteria | 343 | 233 (67.93%) |
| | | Hand strength | muscle strength | | | |
| | | Gait speed | mobility | | | |
| | | Albumin | nutrition | | | |
| Capodanno, 2014 [36] | GSS | Not reported | not reported | Value of 2 or 3 | 1256 | 306 (24.36%) |
| Eichler, 2017 [37] | FI | MMSE | cognition | ≥3 points (score ranges 0-7) | 333 | 152 (45.65%) |
| | | MNA | nutrition | | | |
| | | ADL | disability | | | |
| | | IADL | disability | | | |
| | | Time up and go test | mobility | | | |
| | | Subjective mobility disability | mobility | | | |
| Ghatak, 2012 [38] | Modified Fried phenotype | Albumin | nutrition | Presence of 3 or more criteria | 45 | 22 (48.89%) |
| | | Katz ADL | disability | | | |
| | | 5MWT | mobility | | | |
| | | Grip strength | muscle strength | | | |
| Green, 2015 [39] | Modified Fried phenotype | Gait speed | mobility | Frailty score ≥6 | 244 | 110 (45.08%) |
| | | Grip strength | muscle strength | | | |
| | | Albumin | nutrition | | | |
| | | ADL | disability | | | |
| Green, 2012 [40] | Modified Fried phenotype | Gait speed | mobility | Frailty score ≥5 points | 159 | 76 (47.80%) |
| | | Grip strength | muscle strength | | | |
| | | Albumin | nutrition | | | |
| | | ADL | disability | | | |
| Huded, 2016 [41] | Modified Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 191 | 64 (33.51%) |
| | | Grip strength | muscle strength | | | |
| | | 5MWT | mobility | | | |
| | | Katz ADL | disability | | | |
| Kobe, 2016 [42] | FORCAST | Chair rise | muscle strength | ≥ 4 points (score ranges 0-12) | 130 | 71 (54.62%) |
| | | Weakness | muscle strength | | | |
| | | Stair | mobility | | | |
| | | CFS | subjective | | | |
| | | Creatinine level | medical | | | |
| Maniar, 2016 [43] | Modified Fried phenotype | Serum albumin | nutrition | ≥6 points (score ranges 0-12) | 219 | 73 (33.3%) |
| | | Gait speed | mobility | | | |
| | | Grip strength | muscle strength | | | |
| | | Katz ADL | disability | | | |
| Okoh, 2017 [44 e] | Modified Fried phenotype | Hand grip strength | muscle strength | FI ≥3/4 | 75 | 30 (40%) |
| | | Gait speed | mobility | | | |
| | | Serum albumin | nutrition | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| | | ADL | disability | | | |
| Patel, 2016 [45] | NA | Gait speed | mobility | Gait speed≥6s or/and albumin<3.5g/dl | 117 | 31 (26.50%) |
| | | Albumin | nutrition | | | |
| Rabinovitz, 2016 [46] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 302 | 46 (15.23%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | physical activity | | | |
| Rodriguez-Pascual, 2016 [47] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 109 | 68 (62.39%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | physical activity | | | |
| Rogers, 2018 [48] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 544 | 242 (44.49%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | disability | | | |
| Schoenenberger, 2018 [49] | NA | MMSE | cognition | ≥3 points (score ranges 0-7) | 330 | 169 (51.21%) |
| | | Time up and go | mobility | | | |
| | | MNA | nutrition | | | |
| | | Basic ADL | disability | | | |
| | | Incremental ADL | disability | | | |
| Steinvil, 2018 [50] | NA | BMI | nutrition | Presence of 3 or more criteria | 498 | 232 (46.59%) |
| | | Albumin | nutrition | | | |
| | | Katz ADL | disability | | | |
| | | Grip strength | muscle strength | | | |
| | | Walk test | mobility | | | |
| Shi, 2018 [51] | Fried phenotype | Weight loss | nutrition | Presence of 3 or more criteria | 137 | 116 (84.67%) |
| | | Exhaustion | Exhaustion | | | |
| | | Minnesota leisure time activity | Physical activity | | | |
| | | 5m walk test | mobility | | | |
| | | Grip strength | muscle strength | | | |
| Skaar, 2018 [52] | Geriatric assessment tool (0-9) | MMSE | cognition | Scores≥4 | 142 | 34 (23.94%) |
| | | Nottingham extended ADL | disability | | | |
| | | BMI<20.5 | nutrition | | | |
| | | Low energy | exhaustion | | | |
| | | Weight loss | nutrition | | | |
| | | Chair stand | muscle strength | | | |
| | | Charlson comorbidity index | comorbidity | | | |
| | | Hospital anxiety and depression scale | psychological | | | |
| Zajarias, 2016 [53] | Modified Fried phenotype | Albumin | nutrition | ≥6 points (score ranges 0-12) | 553 | 265 (47.92%) |
| | | Gait speed | mobility | | | |
| | | Grip strength | muscle strength | | | |
| | | Katz ADL | disability | | | |
| Goudzwaard, 2020 [63] | Erasmus Frailty Score | MMSE | cognition | Presence of 3 or more criteria | 330 | 97 (29.50%) |
| | | Hand grip test | Muscle strength | | | |
| | | Malnutrition universal screening tool | Nutrition | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| | | Katz ADL | Inactivity in basic activities of daily living | | | |
| | | Lawton and Brody index | Inactivity in instrumental activities of daily living | | | |
| Goudzwaard, 2020 [64] | Erasmus Frailty Score | MMSE | cognition | Presence of 3 or more criteria | 239 | 70 (29.3%) |
| | | Hand grip test | Muscle strength | | | |
| | | Malnutrition universal screening tool | Nutrition | | | |
| | | Katz ADL | Inactivity in basic activities of daily living | | | |
| | | Lawton and Brody index | Inactivity in instrumental activities of daily living | | | |
| Patel, 2020 [65] | A composite of two frailty markers | Gait speed | mobility | Presence of both criteria | 407 | 74 (18.18%) |
| | | Serum albumin | nutrition | | | |
| Drudi, 2018 [66] | Essential frailty toolset | Muscle weakness | Muscle strength | ≥3 scores (out of 5) | 723 | 254 (35.13%) |
| | | Cognitive impairment | Cognition | | | |
| | | Anemia | Nutrition | | | |
| | | Hypoalbuminemia | Nutrition | | | |
| Morris, 2020 [67] | Essential frailty toolset | Muscle weakness | Muscle strength | ≥3 scores (out of 5) | 559 | 234 (41.86%) |
| | | Cognitive impairment | Cognition | | | |
| | | Anemia | Nutrition | | | |
| | | Hypoalbuminemia | Nutrition | | | |

a. Alfredsson (2016) and Kiani (2020) enrolled patient populations from the STS/ACC registry. Chauhan (2016), Green (2012), Green (2015), Huded (2016), Okoh (2017), Rogers (2018), Steinvil (2018), Traynor (2017), and Bagienski (2017) enrolled patients from the participating centres of STS/ACC registry.
b. Bagienski (2017) and Kleczynski (2017) enrolled patients from the same medical centre but used different frailty definitions.
c. Koifman (2015), Rogers (2018), and Steinvil (2018) enrolled patients from the same medical centre but used different frailty definitions.
d. Shimura (2020) and Shimura (2017) enrolled patients from the same registry but used different frailty definitions.
e. Chauhan (2016) and Okoh (2017) enrolled patients from the same medical centre but used different frailty definitions.

N, the number of patients. SMM, skeletal muscle mass. 5MWT, 5-meter walk test. ADL, activities of daily living. MPI, multidimensional prognostic index. SPMSQ, short portable mental status questionnaire. CIRS-CI, cumulative illness rating scale comorbidity index. MNA-SF, mini-nutritional assessment short form. ESS, Exton Smith scale. FI, frailty index. GSS, geriatric status scale. MMSE, mini-mental state examination. BMI, body mass index. ISAR, identification of seniors at risk. FORCAST, frailty predicts death one year after elective cardiac surgery test. CFS, clinical frailty scale. NA, not applicable.

For peer review only

Table 2. Results of meta-analysis and GRADE assessment**

| Effects | | | | | GRADE assessment | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| # included study | Frailty measures* | # individuals | # events | Estimate (95% CI) | Study design | Risk of bias | Inconsistency | Indirectness | Imprecision | Other considerations | Certainty |
| Procedural death | | | | | | | | | | | |
| 6 | All | 9586 | 654 | 7.60% (4.41%-12.79%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 30-day mortality | | | | | | | | | | | |
| 13 | All | 23628 | 1236 | 7.32% (5.66%-9.42%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 8 | Multi | 1352 | 113 | 8.58% | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (7.18%-10.22%) | | | | | | | |
| 3 | Modified Fried | 407 | 31 | 7.86% (5.20%-11.70%) | Observational | Serious | Not serious | Strongly serious | Serious | None | Very low |
| Cardiovascular death at 30 days | | | | | | | | | | | |
| 2 | Single | 6453 | 259 | 3.37% (1.93%-5.81%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| 6-month mortality | | | | | | | | | | | |
| 2 | Multi | 187 | 30 | 16.12% (11.50%-22.13%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| 1-year mortality | | | | | | | | | | | |
| 10 | All | 15471 | 3151 | 23.98% (20.71%-27.58%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 6 | Multi | 845 | 191 | 22.75% (20.03%-25.71%) | Observational | Serious | Serious | Strongly serious | Serious | None | Very low |
| 2 | Fried and modified Fried | 223 | 60 | 26.91% (21.50%-33.11%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Survival | | | | | | | | | | | |
| 17 | All | 48258 | NA | 1-year survival: 75.6% (75.2%-76.0%) 2-year survival: 65.0% (63.3%-66.7%) 3-year survival: 48.7% (43.3%-54.7%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 4 | Fried and modified Fried | 484 | NA | 1-year survival: 73% (68.8%-77.5%) 2-year survival: 64.5% (56.4%-73.9%) 3-year survival: 58.9% | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (49%-70.9%) | | | | | | | |
| Procedural acute kidney injury | | | | | | | | | | | |
| 4 | Single | 6548 | 458 | 11.34% (6.43%-19.22%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural cardiac tamponade | | | | | | | | | | | |
| 3 | Single | 553 | 17 | 3.19% (1.99%-5.07%) | Observational | Not serious | Serious | Strongly serious | Not serious | None | Very low |
| Convert to open heart surgery | | | | | | | | | | | |
| 2 | All | 4259 | 300 | 2.29% (0.49%-9.91%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural life-threatening bleeding | | | | | | | | | | | |
| 5 | All | 653 | 63 | 9.75% (7.69%-12.29%) | Observational | Not serious | Serious | Strongly serious | Not serious | None | Very low |
| Procedural major bleeding | | | | | | | | | | | |
| 5 | Single | 830 | 104 | 8.53% (3.53%-19.19%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural minor bleeding | | | | | | | | | | | |
| 4 | Single | 774 | 147 | 18.34% (10.66%-29.73%) | Observational | Not serious | Strongly serious | Strongly serious | Serious | None | Very low |
| Procedural major vascular complications | | | | | | | | | | | |
| 3 | Single | 647 | 63 | 10.49% (4.76%-21.54%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 30-day major vascular complications | | | | | | | | | | | |
| 2 | All | 189 | 7 | 2.97% (0.34%-21.67%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Procedural minor vascular complications | | | | | | | | | | | |
| 2 | Single | 591 | 43 | 7.37% (3.24%-15.93%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural major access-site complications | | | | | | | | | | | |
| 3 | Single | 148 | 15 | 9.44% (4.04%-20.51%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Procedural permanent pacemaker | | | | | | | | | | | |
| 7 | All | 3660 | 365 | 8.12% (5.79%-11.26%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Readmission within 30 days | | | | | | | | | | | |
| 3 | Multi | 248 | 27 | 10.37% (3.75%-25.59%) | Observational | Serious | Strongly serious | Strongly serious | Strongly serious | None | Very low |
| Procedural stroke | | | | | | | | | | | |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 8 | All | 1756 | 39 | 2.94% (1.76%-4.88%) | Observational | Strongly serious | Serious | Strongly serious | Not serious | None | Very low |
| Stroke within 30 days | | | | | | | | | | | |
| 2 | Single | 6185 | 132 | 2.14% (1.81%-2.53%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Transfusion | | | | | | | | | | | |
| 3 | All | 458 | 191 | 41.01% (34.02%-48.39%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| 2-valve implantation | | | | | | | | | | | |
| 2 | Single | 409 | 10 | 2.46% (1.33%-4.51%) | Observational | Not serious | Serious | Strongly serious | Not serious | None | Very low |
| Length of hospitalization | | | | | | | | | | | |
| 6 | All | 308 | NA | 8.25 (6.62-10.27) | Observational | Strongly serious | Strongly serious | Strongly serious | Strongly serious | None | Very low |

**Meta-analyses conducted using random effects model

*Frailty measures are categorized as single, multi-measures, administrative data based, Fried, modified Fried and all.

Single indicates single measures

Multi indicates multi-measures

Fried indicates the Fried phenotype

Modified Fried indicates the modified Fried phenotype

Fried and modified Fried includes the Fried phenotype and modified Fried phenotype

All includes all single- and multi-measures

CI indicates confidence interval.

GRADE indicates Grading of Recommendations Assessment, Development and Evaluation

For peer review only

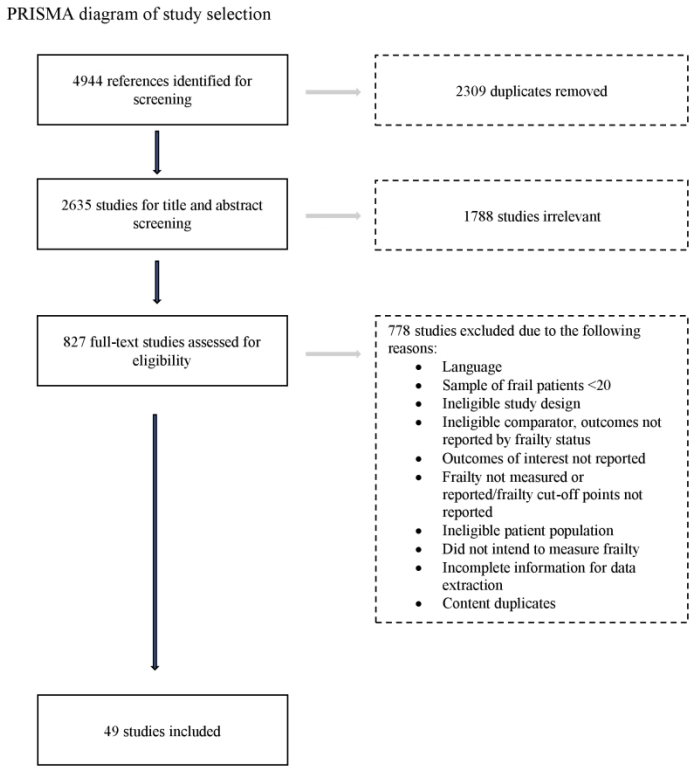



Figure 1. PRISMA flow diagram of included studies
PRISMA, reporting in systematic reviews and meta-analyses. N, number of studies.

215x279mm (300 x 300 DPI)

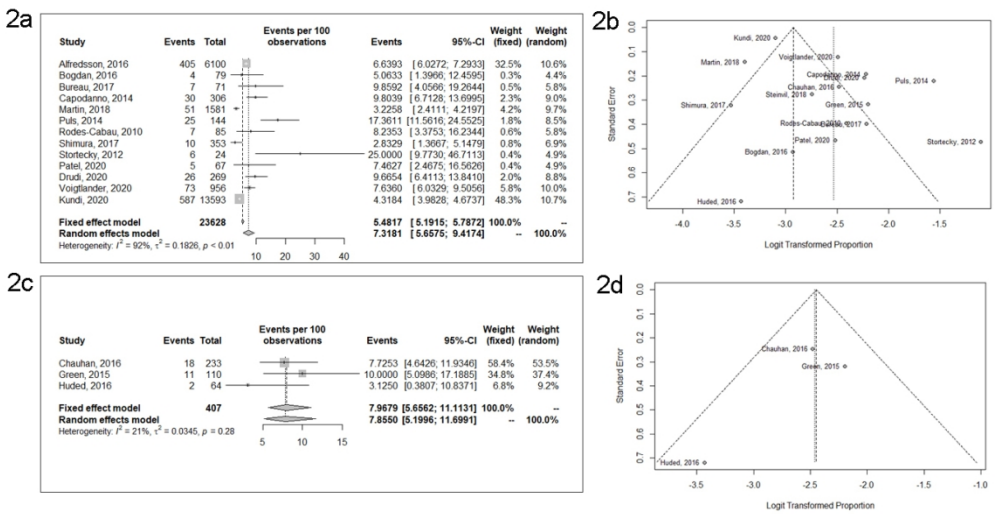



Figure 2a. Meta-analysis of 30-day mortality in frail patients after TAVI.
Frailty was measured using sing and multi-dimensional measures.
The squares indicate the 30-day mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 30-day mortality

Figure 2b. Funnel plots, using data from all studies that reported 30-day mortality
y-axis is the standard error of the 30-day mortality
x-axis is the logit of 30-day mortality

Figure 2c. Meta-analysis of 30-day mortality in frail patients after TAVI.
Frailty was measured using modified Fried frailty phenotype.
The squares indicate the 30-day mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 30-day mortality

Figure 2d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.
y-axis is the standard error of the 30-day mortality
x-axis is the logit of 30-day mortality

480x250mm (72 x 72 DPI)

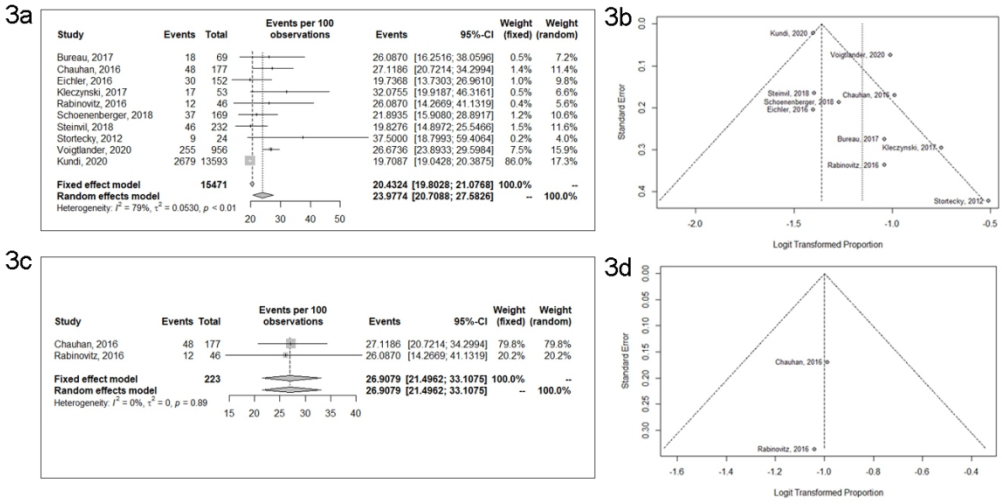



Figure 3a. Meta-analysis of 1-year mortality in frail patients after TAVI.
Frailty was measured using single and multi-dimensional measures.
The squares indicate the 1-year mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 1-year mortality

Figure 3b. Funnel plots, using data from all studies that reported 30-day mortality
y-axis is the standard error of the 1-year mortality
x-axis is the logit of 1-year mortality

Figure 3c. Meta-analysis of 1-year mortality in frail patients after TAVI.
Frailty was measured using the Fried frailty phenotype.
The squares indicate the 1-year mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 1-year mortality

Figure 3d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.
y-axis is the standard error of the 1-year mortality
x-axis is the logit of 1-year mortality

469x250mm (72 x 72 DPI)

## Supplementary Appendix A Literature Search

### Search: TAVI in Frail Patients with Aortic Stenosis

Database: Embase Classic+Embase 1947 to September 23, 2020

| # | Searches |
|---|---|
| 1 | frailty/ |
| 2 | frail elderly/ |
| 3 | geriatric assessment/ |
| 4 | frail*.mp. |
| 5 | fragile.mp. |
| 6 | or/1-5 |
| 7 | geriatric patient/ |
| 8 | very elderly/ |
| 9 | aged/ |
| 10 | aged hospital patient/ |
| 11 | geriatrician/ |
| 12 | *geriatrics/ |
| 13 | (aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*).mp. |
| 14 | or/7-13 |
| 15 | exhaustion/ |
| 16 | limited mobility/ |
| 17 | "timed up and go test"/ |
| 18 | exp walk test/ |
| 19 | walking speed/ |
| 20 | gait/ |
| 21 | physical activity/ |
| 22 | Performance Oriented Mobility Assessment/ |
| 23 | grip strength/ |
| 24 | hand strength/ |
| 25 | muscle strength/ |
| 26 | daily life activity/ |

For peer review only

| | |
|---|---|
| 27 | exp "activity of daily living assessment"/ |
| 28 | exp ADL disability/ |
| 29 | exp disability/ |
| 30 | exp functional status assessment/ |
| 31 | exp neuropsychological test/ |
| 32 | mental function assessment/ |
| 33 | cognition assessment/ |
| 34 | exp cognition/ |
| 35 | exp cognitive defect/ |
| 36 | memory assessment/ |
| 37 | nutritional assessment/ |
| 38 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) adj7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)).mp. |
| 39 | (chair adj2 (rise or stand)).mp. |
| 40 | ((grip* or grasp* or hand* or musc*) adj2 strength).mp. |
| 41 | (tug adj3 test).mp. |
| 42 | (timed* adj6 (up* or go)).mp. |
| 43 | timedupandgo.mp. |
| 44 | weight loss.mp. |
| 45 | (activit* adj3 (living or life)).mp. |
| 46 | (adl or iadl or badl).mp. |
| 47 | or/15-46 |
| 48 | 14 and 47 |
| 49 | 6 or 48 |
| 50 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) adj3 aortic valv*).mp. |
| 51 | transcatheter aortic valve implantation/ |

For peer review only

| | |
|---|---|
| 52 | or/50-51 |
| 53 | 49 and 52 |
| 54 | limit 53 to yr="2006 -Current" |

For peer review only

Database: Ovid MEDLINE(R) ALL 1946 to September 23, 2020

| # | Searches |
|---|---|
| 1 | Frailty/ |
| 2 | Frail Elderly/ |
| 3 | Geriatric Assessment/ |
| 4 | frail*.mp. |
| 5 | fragile.mp. |
| 6 | or/1-5 |
| 7 | "Aged, 80 and over"/ |
| 8 | Aged/ |
| 9 | Geriatricians/ |
| 10 | *Geriatrics/ |
| 11 | (aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*).mp. |
| 12 | or/7-11 |
| 13 | Physical Exertion/ |
| 14 | Physical Endurance/ |
| 15 | Walk Test/ |
| 16 | Exercise Test/ |
| 17 | Exercise Tolerance/ |
| 18 | Walking Speed/ |
| 19 | Gait/ |
| 20 | Exercise/ |
| 21 | mobility limitation/ |
| 22 | Hand Strength/ |
| 23 | Muscle Strength/ |
| 24 | exp "Activities of Daily Living"/ |
| 25 | "Quality of Life"/ |
| 26 | exp Neuropsychological Tests/ |
| 27 | exp Cognition/ |
| 28 | Cognitive Dysfunction/ |

For peer review only

| | |
|---|---|
| 29 | Cognition Disorders/ |
| 30 | Nutritional Assessment/ |
| 31 | Nutritional Status/ |
| 32 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) adj7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)).mp. |
| 33 | (chair adj2 (rise or stand)).mp. |
| 34 | ((grip* or grasp* or hand* or musc*) adj2 strength).mp. |
| 35 | (tug adj3 test).mp. |
| 36 | (timed* adj6 (up* or go)).mp. |
| 37 | timedupandgo.mp. |
| 38 | weight loss.mp. |
| 39 | (activit* adj3 (living or life)).mp. |
| 40 | (adl or iadl or badl).mp. |
| 41 | or/13-40 |
| 42 | 12 and 41 |
| 43 | 6 or 42 |
| 44 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) adj3 aortic valv*).mp. |
| 45 | Transcatheter Aortic Valve Replacement/ |
| 46 | or/44-45 |
| 47 | 43 and 46 |
| 48 | limit 47 to yr="2006 -Current" |

For peer review only

Database: Cochrane Central Register of Controlled Trials

| # | Searches |
|---|---|
| 1 | MeSH descriptor: [Frailty] this term only |
| 2 | MeSH descriptor: [Frail Elderly] this term only |
| 3 | MeSH descriptor: [Geriatric Assessment] this term only |
| 4 | frail*:ti,ab,kw (Word variations have been searched) |
| 5 | fragile:ti,ab,kw (Word variations have been searched) |
| 6 | {OR #1-#5} |
| 7 | MeSH descriptor: [Aged, 80 and over] this term only |
| 8 | MeSH descriptor: [Aged] this term only |
| 9 | MeSH descriptor: [Geriatrics] this term only |
| 10 | aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*:ti,ab,kw (Word variations have been searched) |
| 11 | {or #7-#10} |
| 12 | MeSH descriptor: [Mobility Limitation] this term only |
| 13 | MeSH descriptor: [Physical Exertion] this term only |
| 14 | MeSH descriptor: [Physical Endurance] this term only |
| 15 | MeSH descriptor: [Walk Test] this term only |
| 16 | MeSH descriptor: [Exercise Test] this term only |
| 17 | MeSH descriptor: [Exercise Tolerance] this term only |
| 18 | MeSH descriptor: [Walking Speed] this term only |
| 19 | MeSH descriptor: [Gait] this term only |
| 20 | MeSH descriptor: [Exercise] this term only |
| 21 | MeSH descriptor: [Hand Strength] this term only |
| 22 | MeSH descriptor: [Muscle Strength] this term only |
| 23 | MeSH descriptor: [Activities of Daily Living] explode all trees |
| 24 | MeSH descriptor: [Quality of Life] this term only |
| 25 | MeSH descriptor: [Neuropsychological Tests] explode all trees |
| 26 | MeSH descriptor: [Cognition] explode all trees |
| 27 | MeSH descriptor: [Cognition Disorders] this term only |

For Peer review only

| | |
|---|---|
| 28 | MeSH descriptor: [Cognitive Dysfunction] this term only |
| 29 | MeSH descriptor: [Nutrition Assessment] this term only |
| 30 | MeSH descriptor: [Nutritional Status] this term only |
| 31 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) next/7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)):ti,ab,kw (Word variations have been searched) |
| 32 | (chair next/2 (rise or stand)):ti,ab,kw (Word variations have been searched) |
| 33 | ((grip* or grasp* or hand* or musc*) next/2 strength):ti,ab,kw (Word variations have been searched) |
| 34 | (tug next/3 test):ti,ab,kw (Word variations have been searched) |
| 35 | (timed* next/6 (up* or go)):ti,ab,kw (Word variations have been searched) |
| 36 | "weight loss":ti,ab,kw (Word variations have been searched) |
| 37 | (activit* next/3 (living or life)):ti,ab,kw (Word variations have been searched) |
| 38 | adl or iadl or badl:ti,ab,kw (Word variations have been searched) |
| 39 | {or #12-#38} |
| 40 | #11 and #39 |
| 41 | #6 or #40 |
| 42 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) next/3 aortic valv*):ti,ab,kw (Word variations have been searched) |
| 43 | MeSH descriptor: [Transcatheter Aortic Valve Replacement] this term only |
| 44 | #42 or #43 |
| 45 | #41 and #44 |

For peer review only

Database: PsycINFO (ProQuest) - updated weekly

| # | Searches |
|---|---|
| 1 | MAINSUBJECT.EXACT("Health Impairments" OR "Geriatric Assessment") |
| 2 | AB,TI,SU(frail* OR fragile) |
| 3 | 1 OR 2 |
| 4 | MAINSUBJECT.EXACT("Geriatric Patients" OR "Geriatrics" OR "Gerontology") |
| 5 | AB,TI,SU(aged OR aging OR older OR elderly OR senior* OR geriatric* OR centenarian* OR nonagenarian* OR octogenarian* OR septuagenarian* OR sexagenarian*) |
| 6 | 4 OR 5 |
| 7 | MAINSUBJECT.EXACT("Fatigue" OR "Physical Mobility" OR "Locomotion" OR "Physical Agility" OR "Physical Activity" OR "Physical Endurance" OR "Physical Fitness" OR "Physical Strength" OR "Exercise" OR "Ability Level" OR "Activity Level" OR "Walking" OR "Gait" OR "Activities of Daily Living" OR "Daily Activities" OR "Quality of Life" OR "Neuropsychological Assessment" OR "Cognition" OR "Cognitive Impairment" OR "Memory Disorders" OR "Nutritional Deficiencies") |
| 8 | AB,TI,SU(exhaustion OR fatigue* OR tired* OR mobility OR gait OR walk* OR stand OR balance OR bath* OR dress* OR toilet* OR continence OR feeding OR cognition OR memory OR mental OR disability OR NYHA OR Karnofsky OR CSHA OR functional* OR Katz OR Fried OR Rockwood OR frailty OR nutrition*) AND AB,TI,SU(assess* OR phenotype* OR eval* OR test* OR exam* OR instrument* OR index OR indices OR scale* OR score* OR tool* OR declin* OR dependenc* OR impair*) |
| 9 | chair NEAR/2 (rise OR stand) |
| 10 | strength NEAR/2 (grip* OR grasp* OR hand* OR musc*) |
| 11 | tug NEAR/3 test |
| 12 | timed* NEAR/6 (up* OR go) |
| 13 | activit* NEAR/3 (living OR life) |
| 14 | AB,TI,SU(timedupandgo OR weight loss OR adl OR iadl OR badl) |
| 15 | 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 |
| 16 | 6 AND 15 |
| 17 | 3 OR 16 |
| 18 | (transfemoral* OR trans-femoral* OR transapical* OR trans-apical* OR transaxillary OR trans-axillary OR transarterial* OR trans-arterial* OR subclavian* OR sub-clavian* OR transcatheter* OR trans-catheter* OR transcutaneous* OR trans-cutaneous* OR per-cutaneous* OR percutaneous* OR transcaval* OR trans-caval* OR "direct aortic" OR tavi OR tavr OR pavi OR pavr OR sapien OR cribier OR revalv* OR lotus OR "direct flow" OR jenavalve OR portico OR engager OR evolut) AND (aortic valv*) |
| 19 | TAVI |

For peer review only

| | |
|---|---|
| 20 | 18 OR 19 |
| 21 | 17 AND 20 |

For peer review only

Database: Web of Science Core Collection (Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI Timespan=All years) updated daily

| # | Searches |
|---|---|
| 1 | TS=(frail*) |
| 2 | TS=(fragile) |
| 3 | #1 OR #2 |
| 4 | TS=(aged OR aging OR older OR elderly OR senior* OR geriatric* OR centenarian* OR nonagenarian* OR octogenarian* OR septuagenarian* OR sexagenarian*) |
| 5 | TS=((exhaustion OR fatigue* OR tired* OR mobility OR gait OR walk* OR stand OR balance OR bath* OR dress* OR toilet* OR continence OR feeding OR cognition OR memory OR mental OR disability OR NYHA OR Karnofsky OR CSHA OR functional* OR Katz OR Fried OR Rockwood OR frailty OR nutrition*) NEAR/7 (assess* OR phenotype* OR eval* OR test* OR exam* OR instrument* OR index OR indices OR scale* OR score* OR tool* OR declin* OR dependenc* OR impair*)) |
| 6 | TS=(chair NEAR/2 (rise OR stand)) |
| 7 | TS=((grip* OR grasp* OR hand* OR musc*) NEAR/2 strength) |
| 8 | TS=(tug NEAR/3 test) |
| 9 | TS=(timed* NEAR/6 (up OR go)) |
| 10 | TS=(weight loss) |
| 11 | TS=(activit* NEAR/3 (living OR life)) |
| 12 | TS=(ADL OR IADL OR BADL) |
| 13 | #12 OR #11 OR #10 OR #9 OR #8 OR #7 OR #6 OR #5 |
| 14 | #13 AND #4 |
| 15 | #14 OR #3 |
| 16 | TS=((transfemoral* OR trans-femoral* OR transapical* OR trans-apical* OR transaxillary OR trans-axillary OR transarterial* OR trans-arterial* OR subclavian* OR sub-clavian* OR transcatheter* OR trans-catheter* OR transcutaneous* OR trans-cutaneous* OR per-cutaneous* OR percutaneous* OR transcaval* OR trans-caval* OR "direct aortic" OR tavi OR tavr OR pavi OR pavr OR sapien OR cribier OR revalv* OR lotus OR "direct flow" OR jenavalve OR portico OR engager OR evolut) NEAR/3 aortic valv*) |
| 17 | #16 AND #15 |

For peer review only

## Supplementary appendix B. Baseline demographics and population characteristics

| Study | Recruitment period | N (frail) | Age (mean±SD) | Female (n,%) | NYHA III/IV (n,%) | STS (mean±SD) | Logistic Euro score (mean±SD) | TF-approach (n, %) | TA-approach (n,%) | AVA, $cm^2$ (mean±SD) | Creatinine (mg/dL) (mean±SD) | Haemoglobin, g/dL (mean±SD) | LVEF (mean±SD) | Aortic gradient, mmHg (mean±SD) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | 2011-2014 | 6100 | NR | 3421 (56.08%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bureau (2017) | 2013-2015 | 71 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | 127 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bogdan (2016) | 2009-2012 | 79 | 82±5 | 43 (55%) | 46 (58.2%) | 5.8±3.12 | 19.44±14.98 | NR | NR | 0.66±0.14 | 1.35±0.93 | 11.27±1.58 | 53.22±12.64 | 43.54±15.19 |
| Capodanno (2014) | 2010-2012 | 306 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | 2012-2015 | 233 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2016) | 2008-2014 | 65 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Eichler (2017) | 2013-2015 | 152 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Ghatak (2012) | NR | 22 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Green (2015) | PARTNER[1] | 110 | 87.1** (82.7-90.3) | 58 (53%) | NR | 11.3** (9.6-13.8) | NR | 57 (52%) | NR | 0.62** (0.51-0.72) | NR | NR | 55** (35-60) | 45.2** (34.9-59.7) |
| Green (2012) | PARTNER[1] | 76 | 87.1±6.6 | 40 (53%) | NR | 11.9±4 | NR | NR | 30 (39%) | 0.6±0.2 | NR | 11.1±1.6 | 47±15 | 45±15 |
| Grossman (2017) | 2008-2014 | 192 | 84.3±7.5 | 109 (56.8) | 122 (66%) | 6.2±4.5 | 20.8±16 | 132 (73%) | 42 (23%) | NR | NR | NR | NR | NR |
| Huded (2016) | 2012-2015 | 64 | 83.1±7.5 | 42 (66%) | 53 (90%) | 6.8±2.8 | NR | 57 (89%) | NR | NR | 1.19±0.45 | 11.8±1.8 | NR | NR |
| Koifman (2015) | 2007-2014 | 238 | 84±8 | 81 (21%) | 210 (94%) | 10±4.7 | NR | 181 (76%) | NR | 0.64±0.13 | 1.3±0.9 | 11±1.4 | NR | NR |
| Kleczynski (2017) | NR | 53 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kobe (2016) | 2011-2014 | 71 | NR | 37 (52.11%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | 2013-2014 | 1466 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | NR | 293 | 81.3±7.5 | 105 (36%) | 240 (82%) | 7.1±4 | NR | 174 (59.4%) | 119 (40.6%) | 0.68±0.21 | NR | NR | 54±14 | NR |
| Maniar (2016) | PARTNER II[2] | 73 | 84±7 | 49 (67.6%) | NR | 12.3±5.7 | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | NR | 30 | 92±2 | 24 (80%) | 24 (80%) | 8.4±5.4 | NR | 20 (67%) | NR | NR | NR | NR | NR | NR |
| Patel (2016) | 2012-2015 | 31 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | 2008-2012 | 144 | 83.1±5.2 | 106 (74%) | 136 (94.4%) | 8.8±6.7 | 28.8±16.3 | NR | NR | NR | NR | NR | NR | NR |
| Rabinovitz (2016) | 2009-2013 | 46 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | 2005-2009 | 85 | 83±7 | 58 (68.2%) | NR | 11.6±8.3 | NR | 42 (49.4%) | 43 (50.6%) | NR | 112±73 (umol/L) | NR | NR | NR |

For peer review only

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Rodriguez-Pascual (2016) | 2010-2015 | 68 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | Since 2007 | 242 | 83±7.6 | 140 (58%) | NR | 8.8±6.1 | NR | NR | NR | NR | NR | NR | NR | NR |
| Schoenenberger (2018) | 2009-2013 | 169 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2017) | 2013-2016 | 353 | NR | 276 (78.2%) | 208 (58.92%) | NR | NR | 285 (80.74%) | 58 (16.43%) | NR | NR | NR | NR | NR |
| Steinvil (2018) | 2011-2016 | 232 | 83±7 | 132 (57%) | 193 (83%) | 8.8±6.1 | NR | NR | NR | 0.7 ±0.2 | NR | NR | NR | 43±13 |
| Shi (2018) | 2014-2016 | 116 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | 2011-2015 | 34 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | 2007-2010 | 24 | 84±4 | NR | NR | NR | 24.8±16 | NR | NR | 0.5±0.2 | NR | NR | 52±19 | NR |
| Traynor (2017) | 2008-2015 | 60 | 85.4±7.4 | 38 (63.3%) | 49 (81.7%) | 12.7±11.2 | NR | 34 (56.67%) | NR | NR | 1.4±1.1 | NR | 56±14 | NR |
| Yamamoto (2015) | 2010-2011 | 56 | 85±6.7 | 38 (67.90%) | 47 (83.9%) | 10.9±7.3 | 23.6±11.2 | 41 (73.2%) | NR | 0.58±0.19 | NR | NR | NR | 53.3±21.5 |
| Zajarias (2016) | PARTNER II[2] | 265 | 85±6 | 157 (59.7%) | NR | 11.8±5.8 | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | 2014-2018 | 97 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | 2013-2018 | 70 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Welle (2020) | 2012-2019 | 483 | 82±8.2 | 222 (46.0%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Patel (2020) | 2012-2018 | 74 | 81±9.1 | 37 (50%) | 65 (88%) | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mach (2020) | 2017-2018 | 39 | NR | 17 (43.59%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Drudi (2018) | 2012-2017 | 254 | NR | NR | NR | NR | NR | 193 (75.98%) | 61 (24.02%) | NR | NR | NR | NR | NR |
| Kiani (2020) | 2011-2016 | 11316 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Malik (2020) | 2016 | 8419 | NR | 3884 (46.13%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Morris (2020) | 2012-2019 | 234 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Sami (2020) | 2012-2015 | 2865 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Gharibeh (2019) | 2012-2016 | 186 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Voigtlander (2020) | 2011-2014 | 956 | 82.5±6.8 | 715 (74.8%) | NR | 8.7±6.3 | NR | 681 (71.2%) | 247 (25.8%) | NR | NR | NR | NR | NR |
| Kundi (2020) | 2016 | 13593 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2020) | 2013-2017 | 284 | NR | 201 (70.77%) | 186 (65.49%) | NR | NR | 234 (81.25%) | NR | NR | NR | NR | NR | NR |

Baseline demographics and population characteristics (continued)

For peer review only

| Study | AF (n, %) | Active cancer (n, %) | BMI (mean±SD) | COPD (n,%) | Cerebrovascular Disease (n,%) | CAD (n, %) | Chronic Lung disease (n,%) | Chronic Dialysis (n,%) | Chronic Kidney disease (n, %) | dyslipidemia (n,%) | Diabetes Mellitus (n,%) | Hyperlipidemia (n,%) | Hypertension (n,%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson, (2016) | NR | NR | NR | NR | NR | NR | 1611 (26.41%) | 186 (3.05%) | NR | NR | 2137 (35.03%) | NR | 5510 (90.33%) |
| Bureau (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bogdan (2016) | 14 (17%) | NR | NR | 23 (29%) | NR | NR | NR | NR | NR | 39 (49%) | 27 (34%) | NR | 56 (70%) |
| Capodanno (2014) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2015) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Eichler (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Ghatak (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Green (2015) | NR | NR | NR | 46 (42%) | 26 (26%) | 91 (83%) | NR | NR | NR | NR | 28 (26%) | NR | 95 (86%) |
| Green (2012) | NR | NR | 25±6.4 | NR | NR | NR | NR | NR | NR | NR | 19 (25%) | 38 (50%) | 58 (76%) |
| Grossman (2017) | 57 (32%) | NR | NR | NR | NR | 100 (56%) | NR | NR | NR | NR | 71 (39%) | 136 (74%) | 160 (86%) |
| Huded (2016) | 26 (41%) | NR | 29.1±6.9 | 5 (8%) | NR | 54 (84%) | NR | NR | 14 (22%) | NR | 24 (38%) | 37 (58%) | 50 (78%) |
| Koifman (2015) | 101 (44%) | NR | NR | 77 (34%) | 32 (16%) | NR | NR | NR | NR | NR | 79 (35%) | NR | 211 (92%) |
| Kleczynski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kobe (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | 107 (37%) | NR | 25.4±4.9 | NR | NR | 206 (71%) | 94 (32%) | NR | NR | 208 (71%) | 85 (29%) | NR | 246 (84%) |
| Maniar (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | 11 (37%) | NR | 25.4±3.7 | NR | NR | 22 (73%) | 7 (23%) | 1 (3%) | NR | 17 (57%) | NR | NR | 28 (93%) |
| Patel (2016) | 19 (61.3%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | NR | NR | 26.9±5.1 | NR | NR | 101 (70%) | NR | NR | NR | NR | 50 (34%) | NR | NR |
| Rabinovitz (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | NR | NR | 25±5 | 23 (27.1%) | 15 (17.7%) | 57 (67.1%) | NR | 3 (3.5%) | NR | 55 (65.5%) | 21 (26.6%) | NR | 58 (68.2%) |
| Rodriguez-Pascual (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers | 109 | NR | 29±11 | 116 | NR | NR | NR | 15 | 90 | NR | 82 | 198 | 223 |

For peer review only

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (2018) | (45%) | | | (48%) | | | | (6%) | (37%) | | (34%) | (82%) | (92%) |
| Schoenenberger (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2017) | NR | 8 (2.3%) | NR | NR | NR | NR | NR | NR | 203 (57.5%) | NR | 94 (26.6%) | NR | 258 (73.1%) |
| Steinvil (2018) | 102 (44%) | NR | 29±11 | 114 (49%) | NR | NR | NR | 14 (6%) | 86 (37%) | NR | 79 (34%) | 190 (82%) | 216 (93%) |
| Shi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Traynor (2017) | NR | NR | NR | NR | 18 (30%) | NR | 32 (55.2%) | 2 (3.3%) | NR | NR | 28 (46.7%) | NR | 57 (95%) |
| Yamamoto (2015) | NR | NR | NR | 17 (30.40%) | NR | NR | NR | 1 (1.80%) | 42 (75%) | 19 (33.90%) | 5 (8.90%) | NR | 37 (66.10%) |
| Zajarias (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Welle (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Patel (2020) | 41 (55%) | NR | 29±6 | NR | NR | 29 (39%) | 26 (35%) | NR | NR | NR | 31 (42%) | NR | 67 (91%) |
| Mach (2020) | 18 (46.15%) | NR | NR | 13 (33.33%) | NR | NR | NR | NR | NR | NR | 17 (43.59%) | NR | 36 (92.31%) |
| Drudi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kiani (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Malik (2020) | 4055 (48.17%) | NR | NR | NR | NR | 520 (6.17%) | NR | NR | 4381 (52.04%) | NR | 3187 (37.86%) | NR | 7363 (87.46%) |
| Morris (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Sami (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Gharibeh (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Voigtlander (2020) | 262 (27.4%) | NR | 18.6±1.2 | NR | NR | 450 (47.1%) | NR | NR | NR | NR | NR | NR | 205 (21.9%) |
| Kundi (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2020) | 69 (24.30%) | 36 (12.68%) | NR | NR | NR | 114 (40.14%) | NR | NR | 215 (75.70%) | NR | 53 (18.66%) | NR | NR |

Baseline demographics and population characteristics (continued)

For peer review only

| Study | Liver disease (n,%) | Previous stroke (n, %) | Previous MI (n, %) | Previous BVA (n, %) | Previous PCI (n, %) | Previous CABG (n,%) | Pulmonary disease (n,%) | Peripheral arterial disease (n,%) | Peripheral Vascular disease (n, %) | Permanent pacemaker (n,%) | Valve type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | NR | 753 (12.34%) | 1422 (23.31%) | 774 (12.69%) | 2103 (34.48%) | 1800 (29.51%) | NR | 1930 (31.64%) | NR | NR | NR |
| Bureau (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve<br>Edwards Sapien<br>Jena<br>Lotus<br>NVT |
| Bogdan (2016) | 1 (1%) | NR | NR | NR | NR | NR | NR | NR | 9 (11%) | NR | CoreValve: 60 (75.94%)<br>Edwards Sapien: 19 (24.06%) |
| Capodanno (2014) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2015) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve: 277 (89%)<br>Edwards Sapien: 24 (7.7%)<br>Lotus: 10 (3.2%) |
| Eichler (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve (classic, Evolut R)<br>Edwards Sapien<br>Edwards Sapien XT |
| Ghatak (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien |
| Green (2015) | 9 (8%) | NR | NR | 35 (32%) | 45 (41%) | 47 (43%) | NR | NR | 46 (42%) | 27 (25%) | Edwards Sapien |
| Green (2012) | NR | 9 (12%) | NR | NR | NR | 27 (36%) | 22 (29%) | NR | 16 (21%) | 24 (32%) | Edwards Sapien |
| Grossman (2017) | NR | 6 (3%) | NR | NR | NR | NR | NR | NR | 19 (11%) | 13 (7%) | CoreValve<br>Edwards Sapien XT |
| Huded (2016) | NR | 13 (20%) | NR | NR | NR | 17 (27%) | NR | 6 (9%) | NR | NR | Edwards Sapien<br>Edwards Sapien XT<br>Edwards Sapien 3<br>CoreValve<br>Direct Flow Medical |
| Koifman (2015) | NR | NR | NR | NR | 60 (26%) | 67 (29%) | NR | NR | 76 (34%) | 37 (21%) | |
| Kleczynski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien<br>Edwards Sapien XT<br>Edwards Sapien 3<br>CoreValve<br>JenaValve |
| Kobe (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | NR | 56 (19%) | 84 (29%) | NR | NR | 57 (20%) | NR | NR | 87 (30%) | NR | NR |
| Maniar (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Okoh (2017) | NR | NR | NR | 1 (3%) | 4 (13%) | 4 (13%) | NR | NR | 7 (23%) | 7 (23%) | |
| Patel (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | NR | NR | NR | NR | 40 (28%) | 16 (11%) | 43 (30%) | NR | 42 (29%) | NR | CoreValve<br>Edwards Sapien |
| Rabinovitz (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | NR | NR | 39 (45.9%) | NR | 26 (30.6%) | 20 (23.5%) | NR | NR | 29 (34.1%) | NR | Cribier-Edwards<br>Edwards Sapien<br>Edwards Sapien XT |
| Rodriguez-Pascual (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | NR | 17 (7%) | 41 (17%) | 53 (22%) | 80 (33%) | 58 (24%) | NR | NR | 58 (24%) | 65 (27%) | Balloon expandable (67%)<br>Self-expanding (33%) |
| Schoenenberger (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve<br>Edwards Sapien XT |
| Shimura (2017) | 17 (4.8%) | 65 (18.4%) | 28 (7.9%) | NR | 94 (26.6%) | 21 (5.9%) | 108 (30.6%) | 67 (19%) | NR | NR | NR |
| Steinvil (2018) | NR | 16 (7%) | 42 (18%) | 49 (21%) | 79 (34%) | 58 (25%) | NR | NR | 53 (23%) | 67 (29%) | NR |
| Shi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | NR | NR | NR | NR | NR | 1 (4.2%) | NR | NR | NR | NR | NR |
| Traynor (2017) | NR | NR | 18 (30%) | NR | NR | 15 (25%) | NR | NR | 38 (63.3%) | 12 (20%) | NR |
| Yamamoto (2015) | NR | 6 (10.70%) | 5 (8.90%) | NR | NR | NR | NR | 8 (14.30%) | NR | 10 (17.90%) | CoreValve<br>Edwards Sapien |
| Zajarias (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien XT |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Welle (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Patel (2020) | NR | NR | NR | NR | NR | NR | NR | 17 (23%) | NR | NR | NR |
| Mach (2020) | NR | NR | 4 (10.26%) | NR | 9 (23.08%) | 3 (7.69%) | NR | NR | 18 (46.15%) | NR | Spaien XT<br>Spaien 3<br>Symetis accurate<br>Core Valve Evolut |
| Drudi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kiani (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Malik (2020) | 340 (4.04%) | NR | 1107 (13.15%) | NR | NR | NR | 3010 (35.75%) | NR | 2369 (28.14%) | NR | NR |
| Morris (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Sami (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Gharibeh (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Voigtlander (2020) | NR | 97 (10.1%) | 128 (13.4%) | NR | NR | NR | NR | NR | NR | NR | NR |
| Kundi (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2020) | 14 (4.93%) | 32 (11.27%) | NR | NR | NR | 19 (6.69%) | 62 (21.83%) | 37 (13.03%) | NR | NR | Balloon expandable valve |

N, sample size of frail TAVI recipients. SD, standard deviation. NYHA, New York Heart Association. STS, Society of Thoracic Surgeons. TF, transfemoral approach. TA, transapical approach. AVA, aortic valve area. LVEF, left ventricular ejection fraction. AF, atrial fibrillation. BMI, body mass index. COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. MI, myocardial infarction. BVA, balloon aortic valvuloplasty. PCI, percutaneous coronary intervention. CABG, coronary artery bypass grafting.

NR, not reported

§ = Euro score II

**= median and IQR

1, recruitment period, 2007-2010

2, recruitment period, 2011-2016

For peer review only

| Study | Study Participation | Study Attrition | Prognostic Factor Measurement | Outcome Measurement | Statistical Analysis Reporting | Overall Risk of Bias | Study Design | Abstract |
|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | high | moderate | low | low | low | moderate | prospective | no |
| Bureau (2017) | high | moderate | low | low | low | moderate | prospective | no |
| Bagienski (2017) | moderate | moderate | low | low | low | moderate | prospective | no |
| Bogdan (2016) | moderate | high | low | low | moderate | moderate | retrospective | no |
| Capodanno (2014) | moderate | moderate | low | low | low | moderate | prospective | no |
| Chauhan (2016) | moderate | high | low | low | low | moderate | retrospective | no |
| Cockburn (2015) | moderate | low | moderate | low | moderate | moderate | prospective | no |
| Eichler (2017) | high | low | low | low | low | low | prospective | no |
| Ghatak (2012) | high | high | moderate | low | moderate | high | retrospective | yes |
| Grossman (2017) | moderate | high | moderate | low | low | moderate | retrospective | no |
| Green (2012) | moderate | low | low | low | low | low | prospective | no |
| Green (2015) | moderate | low | low | low | low | low | prospective | no |
| Huded (2016) | high | high | low | low | low | high | retrospective | no |
| Kleczynski (2017) | low | low | low | low | low | low | prospective | no |
| Kobe (2016) | moderate | moderate | low | low | low | moderate | prospective | no |
| Koifman (2015) | moderate | high | low | low | low | moderate | retrospective | no |
| Martin (2018) | high | high | low | low | low | high | prospective | no |
| Maniar (2016) | high | moderate | moderate | low | low | moderate | retrospective | yes |
| Mok (2016) | high | low | low | low | low | low | retrospective | no |
| Okoh (2017) | Low | high | moderate | low | low | moderate | prospective | no |
| Patel (2016) | moderate | high | low | low | low | moderate | retrospective | no |
| Puls (2014) | moderate | low | low | low | low | low | prospective | no |
| Rabinovitz (2016) | high | high | moderate | low | low | high | retrospective | no |
| Rodes-Cabau (2010) | low | low | high | low | low | low | prospective | no |
| Rodriguez-Pascual (2016) | low | low | low | low | low | low | prospective | no |
| Rogers (2018) | moderate | moderate | low | low | low | moderate | retrospective | no |
| Schoenenberger (2018) | high | low | low | low | low | low | prospective | no |
| Shimura (2017) | low | moderate | low | low | low | low | prospective | no |
| Steinvil (2018) | moderate | moderate | low | low | low | moderate | prospective | no |
| Shi (2018) | moderate | high | low | low | low | moderate | prospective | no |
| Skaar (2018) | low | low | low | low | low | low | prospective | no |
| Stortecky (2012) | high | moderate | moderate | moderate | high | high | prospective | yes |
| Traynor (2017) | low | low | moderate | moderate | low | moderate | retrospective | no |
| Yamamoto (2015) | moderate | low | low | low | low | low | prospective | no |
| Zajarias (2016) | high | moderate | low | low | low | moderate | prospective | yes |
| Goudzwaard (2020) | low | moderate | low | low | low | low | prospective | no |

For peer review only

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Goudzwaard (2020) | low | moderate | low | low | low | low | prospective | no |
| Welle (2020) | moderate | high | low | moderate | low | moderate | prospective | yes |
| Patel (2020) | low | low | low | low | low | low | retrospective | no |
| Mach (2020) | low | high | low | low | low | low | Prospective | no |
| Drudi (2018) | low | low | low | low | low | low | prospective | no |
| Kiani (2020) | low | high | moderate | low | low | moderate | prospective | no |
| Malik (2020) | low | high | low | low | low | low | retrospective | no |
| Morris (2020) | high | high | low | low | moderate | high | prospective | yes |
| Sami (2020) | moderate | high | low | low | low | moderate | retrospective | yes |
| Gharibeh (2020) | high | high | moderate | low | moderate | high | retrospective | yes |
| Voigtlander (2020) | low | high | low | low | low | low | prospective | no |
| Kundi (2020) | low | high | low | low | low | low | retrospective | no |
| Shimura (2020) | low | high | low | low | low | low | prospective | no |

## Supplementary appendix C. Assessment of quality in individual studies

*Risk of bias was assessed using the Quality in Prognostic Studies tool (QUIPS)
Low risk= studies with four or five low risk domains
High risk=studies with two or more high risk domains
Moderate risk=remaining studies

Reference. Hayden JA, van der Windt DA, Cartwright JL, Côté P, Bombardier C. Assessing bias in studies of prognostic factors. Annals of internal medicine. 2013 Feb 19;158(4):280-6.

For peer review only

# Supplementary appendix D. Prognosis of frail TAVI recipients

| Study | Procedural/in-hospital | 30 days | 6 months | 8 months | 1 year | 2 year |
|---|---|---|---|---|---|---|
| ***Primary outcomes*** | | | | | | |
| ***Death*** | | | | | | |
| Alfredsson, 2016 | 300/6100 (4.9%) | 405/6100 (6.6%) | | | | |
| Bureau, 2017 | | 7/71 (9.9%) | 13/71 (18.3%) | | 18/69 (26.1%) | |
| Bogdan, 2016 | | 4/79 (5.1%) | | | | 35.4% (2.1 years) |
| Capodanno, 2014 | | 30/306 (9.8%) | | | | |
| Chauhan, 2016 | | 18/233 (7.7%) | | | 48/177 (27.1%) | |
| Eichler, 2017 | | | | | 30/152 (19.7%) | |
| Green, 2015 | | 11/110 (10%) | | | 32.7% | |
| Huded, 2016 | | 2/64 (3.13%) | | | | |
| Kleczynski, 2017 | | | | | 17/53 (32.1%) | |
| Martin, 2018 | | 51/1581(3.23%) | | | | |
| Maniar, 2016 | | 12.3% | | | 36.5% | |
| Okoh, 2017 | | | | | | 30% |
| Puls, 2014 | 33/144 (23%) | 25/144 (17%) | | | | |
| Rabinovitz, 2016 | | | | | 12/46 (26.1%) | |
| Rodriguez-Pascual, 2016 | | | | | | 39.7%*(98weeks) |
| Rodes-Cabau, 2010 | 2/85 (2.4%) | 7/85 (8.2%) | | 19/85 (22.4%) | | |
| Schoenenberger, 2018 | | | | | 37/169(21.9%) | |
| Shimura, 2017 | 15/353 (4.25%) | 10/353 (2.83%) | | | | |
| Steinvil, 2018 | | 14/232 (6.03%) | | | 46/232 (19.8%) | |
| Shi, 2018 | | | 17/116 (15%) | | | |
| Skaar, 2018 | | | | | | 9/34 (26.47%) |
| Stortecky, 2012 | | 6/24 (25%) | | | 9/24 (37.5%) | |
| Traynor, 2017 | 2/60 (3.3%) | | | | 14.8% | |
| Zajarias, 2016 | | 17/265 (6.4%) | | | 88/265 (33.2%) | |
| Patel, 2020 | | 5/67 (6.7%) | | | | |
| Mach, 2020 | 3/39 (7.69%) | | | | | |
| Drudi, 2020 | | 26/269 (9.67%) | | | | |
| Kiani, 2020 | | 4.55% | | | 18.07% | |
| Sami, 2020 | 301/2865 (10.5%) | | | | | |
| Gharibeh, 2020 | | | | | | 25.3% |
| Voigtlander, 2020 | | 73/956 (7.64%) | | | 255/956 (26.67%) | |
| Kundi, 2020 | | 587/13593 (4.32%) | | | 2679/13593 (19.71%) | |
| ***Death (cardiovascular cause)*** | | | | | | |
| Alfredsson, 2016 | | 251/6100 (4.1%) | | | | |
| Green, 2015 | | 8/110 (7%) | | | | |
| Shimura, 2017 | 4/353(1.13%) | 8/353 (2.27%) | | | | |
| ***Secondary outcomes*** | | | | | | |
| ***Acute kidney injury*** | | | | | | |
| Alfredsson, 2016 | 397/6100 (6.5%) | | | | | |
| Bogdan, 2016 | | 25/79 (31.6%) | | | | |
| Green, 2012 | 3/76 (3.95%) | | | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Okoh, 2017 | 2/30 (6.67%) | | | | | |
| Shimura, 2017 | 47/353 (13.31%) | | | | | |
| Yamamoto, 2015 | 6/56 (10.7%) | | | | | |
| Mach, 2020 | 8/39 (20.51%) | | | | | |
| Shimura, 2020 | 43/288 (14.93%) | | | | | |
| ***Arrhythmia*** | | | | | | |
| Rodes-Cabau, 2010 | 9/85 (10.6%) | | | | | |
| ***Annulus rupture*** | | | | | | |
| Yamamoto, 2015 | 1/56 (1.8%) | | | | | |
| ***Bleeding*** | | | | | | |
| Zajarias, 2016 | 15/265 (43.4%) | | | | | |
| Kiani, 2020 | | 11.80% | | | 22.20% | |
| Shimura, 2020 | 71/288 (24.65%) | | | | | |
| ***Cardiac tamponade*** | | | | | | |
| Puls, 2014 | 3/144 (2%) | | | | | |
| Shimura, 2017 | 11/353 (3.11%) | | | | | |
| Yamamoto, 2015 | 3/56 (5.4%) | | | | | |
| Shimura, 2020 | 6/288 (2.08%) | | | | | |
| ***Conversion to open surgery*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| Shimura, 2017 | 8/353 (2.27%) | | | | | |
| Sami, 2020 | 284/2865 (9.9%) | | | | | |
| Voigtlander, 2020 | 7/956 (0.9%) | | | | | |
| Shimura, 2020 | 0/288 (0%) | | | | | |
| ***Coronary obstruction*** | | | | | | |
| Shimura, 2017 | 3/353 (0.85%) | | | | | |
| ***Delirium*** | | | | | | |
| Bagienski, 2017 | 105/127 (82.68%) | | | | | |
| Goudzwaard, 2020 | 24/97 (24.7%) | | | | | |
| ***Life-threatening bleeding*** | | | | | | |
| Alfredsson, 2016 | | 391/6100 (6.4%) | | | | |
| Green, 2012 | 7/76 (9.21%) | | | | | |
| Puls, 2014 | 15/144 (10%) | | | | | |
| Stortecky, 2012 | 4/24 (16.7%) | | | | | |
| Shimura, 2017 | 33/353 (9.35%) | | | | | |
| Yamamoto, 2015 | 4/56 (7.1%) | | | | | |
| Shimura, 2020 | 14/288 (4.86%) | | | | | |
| ***Major bleeding*** | | | | | | |
| Green, 2015 | | 10/110 (9%) | | | | |
| Koifman, 2015 | 6/238 (3%) | | | | | |
| Puls, 2014 | 12/144 (8%) | | | | | |
| Shimura, 2017 | 77/353 (21.81%) | | | | | |
| Yamamoto, 2015 | 9/56 (16.1%) | | | | | |
| Mach, 2020 | 1/39 (2.56%) | | | | | |
| Shimura, 2020 | 33/288 (11.46%) | | | | | |
| ***Minor bleeding*** | | | | | | |
| Koifman, 2015 | 31/238 (13%) | | | | | |
| Puls, 2014 | 50/144 (35%) | | | | | |
| Shimura, 2017 | 62/353 (17.56%) | | | | | |
| Mach, 2020 | 4/39 (10.26%) | | | | | |
| ***Major vascular complications*** | | | | | | |
| Bogdan, 2016 | | 0/79 (0%) | | | | |
| Green, 2015 | | 7/110 (6%) | | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Koifman, 2015 | 37/238 (16%) | | | | | |
| Shimura, 2017 | 18/353 (5.1%) | | | | | |
| Yamamoto, 2015 | 8/56 (14.3%) | | | | | |
| Shimura,2020 | 9/288 (3.13%) | | | | | |
| ***Minor vascular complications*** | | | | | | |
| Bogdan, 2016 | | 2/79 (2.5%) | | | | |
| Koifman, 2015 | 26/238 (11%) | | | | | |
| Shimura, 2017 | 17/353 (4.82%) | | | | | |
| ***Major access site complications*** | | | | | | |
| Rodes-Cabau, 2010 | 12/85 (14.1%) | | | | | |
| Stortecky, 2012 | 2/24 (8.3%) | | | | | |
| Mach, 2020 | 1/39 (2.56%) | | | | | |
| ***Major vascular injury*** | | | | | | |
| Huded, 2016 | | 4/64 (6%) | | | | |
| ***Myocardial infarction*** | | | | | | |
| Rodes-Cabau, 2010 | | 2/85 (2.4%) | | | | |
| Yamamoto, 2015 | 0/56 (0%) | | | | | |
| Mach, 2020 | 2/39 (5.13%) | | | | | |
| Kiani, 2020 | | 0.63% | | | 1.98% | |
| Voigtlander, 2020 | 3/956 (0.3%) | | | | | |
| ***Need for second valve*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| ***Permanent pacemaker*** | | | | | | |
| Green, 2015 | | 10/110 (9%) | | | | |
| Koifman, 2015 | 12/238 (5%) | | | | | |
| Okoh, 2017 | 2/30 (6.67%) | | | | | |
| Puls, 2014 | 3/144 (2%) | | | | | |
| Rodes-Cabau, 2010 | | 7/85 (8.2%) | | | | |
| Yamamoto, 2015 | 5/56 (8.9%) | | | | | |
| Mach, 2020 | 5/39 (12.82%) | | | | | |
| Kiani, 2020 | | 11.75% | | | | |
| Sami, 2020 | 309/2865 (10.8%) | | | | | |
| Shimura, 2020 | 29/288 (10.07%) | | | | | |
| ***Percutaneous closure device failure*** | | | | | | |
| Bogdan, 2016 | | 10/79 (12.7%) | | | | |
| ***Renal failure*** | | | | | | |
| Green, 2015 | | 9/110 (8%) | | | | |
| ***Respiratory failure*** | | | | | | |
| Huded, 2016 | | 1/64 (2%) | | | | |
| ***Readmission*** | | | | | | |
| Green, 2015 | | 4/110(4%) | | | | |
| Huded, 2016 | | 7/64 (11%) | | | | |
| Patel, 2020 | | 16/74 (21.6%) | | | | |
| Kiani, 2020 | | 3.69% | | | | |
| Kundi, 2020 | | | | | 6492/13593 (47.76%) | |
| Shimura, 2020 | | | | | 64/288 (22.22%) | |
| ***Stroke*** | | | | | | |
| Alfredsson, 2016 | | 129/6100 (2.1%) | | | | |
| Green, 2015 | | 1/110 (1%) | | | | |
| Huded, 2016 | | 4/64 (6%) | | | | |
| Koifman, 2015 | 11/238 (5%) | | | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Rodes-Cabau, 2010 | | 3/85 (3.5%) | | | | |
| Okoh, 2017 | 0/30 (0%) | | | | | |
| Shimura, 2017 | 6/353 (1.7%) | | | | | |
| Stortecky, 2012 | 2/24 (8.3%) | | | | | |
| Traynor, 2017 | 2/60 (3.3%) | | | | | |
| Yamamoto, 2015 | 3/56 (5.4%) | | | | | |
| Mach, 2020 | 1/39 (2.56%) | | | | | |
| Kiani, 2020 | | 2.56% | | | 5.70% | |
| Voigtlander, 2020 | 14/956 (1.5%) | | | | | |
| Shimura, 2020 | 2/288 (0.07%) | | | | | |
| ***Sepsis*** | | | | | | |
| Rodes-Cabau, 2010 | | 4/85 (4.7%) | | | | |
| Voigtlander, 2020 | 22/956 (2.3%) | | | | | |
| ***Transfusion*** | | | | | | |
| Green, 2012 | 24/76 (32.9%) | | | | | |
| Koifman, 2015 | 100/238 (43%) | | | | | |
| Puls, 2014 | 67/144 (47%) | | | | | |
| ***Valve embolization*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| ***Vascular complications*** | | | | | | |
| Bogdan, 2016 | | 12/79 (15.18%) | | | | |
| Voigtlander, 2020 | 43/956 (4.5%) | | | | | |
| Shimura, 2020 | 22/288 (7.64%) | | | | | |
| ***2-valve implantation*** | | | | | | |
| Shimura, 2017 | 9/353 (2.55%) | | | | | |
| Yamamoto, 2015 | 1/56 (1.8%) | | | | | |
| ***Length of hospital stay for frail TAVI recipients, days*** | | | | | | |
| Ghatak, 2012 | 12.1±8.4 | | | | | |
| Green, 2012 | 9±6 | | | | | |
| Huded, 2016 | 6±3.5 | | | | | |
| Okoh, 2017 | 7±8 | | | | | |
| Patel, 2016 | 9** | | | | | |
| Traynor, 2017 | 6.8±5.2 | | | | | |
| Yamamoto, 2015 | 10.3±7.5 | | | | | |
| Kiani, 2020 | 5 (3-8) | | | | | |
| ***Intensive care unit (ICU) stay, days*** | | | | | | |
| Patel, 2016 | 2.7** | | | | | |
| Yamamoto, 2015 | 4.9±4 | | | | | |
| Voigtlander, 2020 | 3.58±5.0 | | | | | |

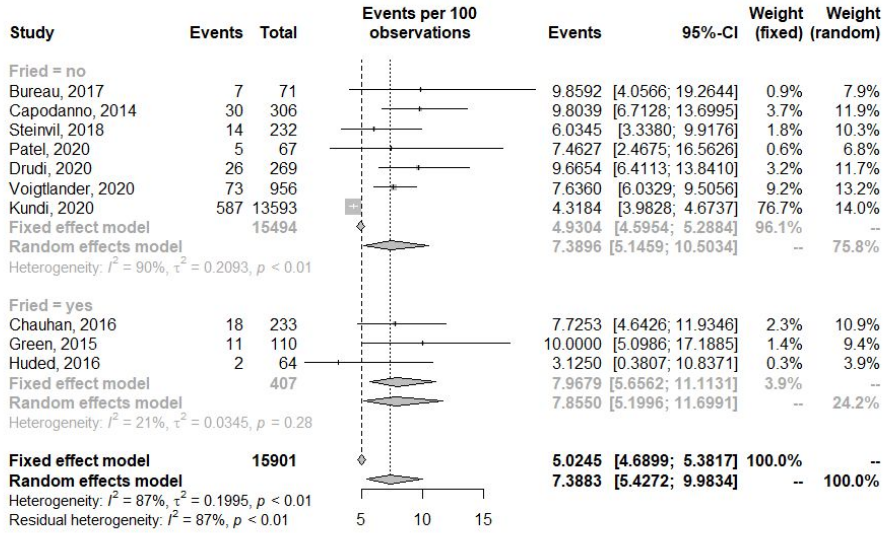
*= median + IQR

**=mean without standard deviation reported

For peer review only

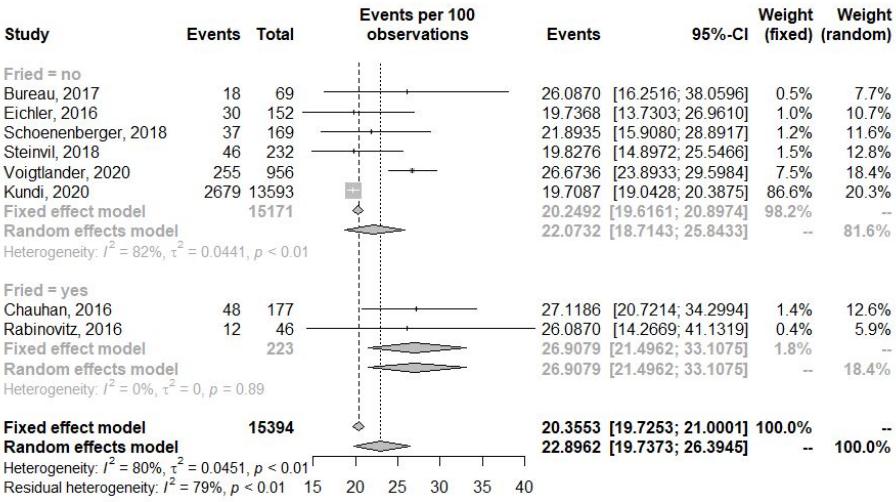
**Supplementary appendix G. Subgroup analyses of studies reporting frailty measurement using the Fried phenotype compared to non-Fried phenotype. The squares indicate the mortality reported by each study. The diamonds indicate the pooled estimates for mortality.**

## Subgroup analysis of 30-day mortality




Test for subgroup differences: Q=0.05 (p=0.8250)

## Subgroup analysis of 1-year mortality




Test for subgroup differences: Q=2.02 (p=0.1551)

## Supplementary appendix G. Quality of life among frail TAVI recipients

| Study | QoL measures | Baseline (SD) | Post-TAVI (SD) | Change in QoL post-TAVI | 30-day (SD) | Change in QoL at 30 days | 12-months (SD) |
|---|---|---|---|---|---|---|---|
| Okoh, 2017 | KCCQ overall score | | | | | -5.86 (p=.62) | |
| | KCCQ physical limitation | | | | | -8.34 (p=0.46) | |
| | KCCQ symptoms | | | | | -5.68 (p=0.65) | |
| | KCCQ quality of life | | | | | 6.36 (p=0.63) | |
| | KCCQ social limitation | | | | | -7.57 (p=0.58) | |
| Kobe, 2016 Subgroup of frail patients with a frailty score 4-6 | SF-36 physical functioning | 55.8 (15.3) | | | 49.6 (22.2) | | |
| | SF-36 role physical | 37.9 (32.8) | | | 33.8 (43.1) | | |
| | SF-36 bodily pain | 88.1 (23.2) | | | 74.4 (23.7) | | |
| | SF-36 role emotional | 88.2 (23.6) | | | 38.3 (42.3) | | |
| | SF-36 general health | 73.4 (18.7) | | | 62 (15.5) | | |
| | SF-36 vitality | 45.5 (24.6) | | | 43.4 (15.5) | | |
| | SF-36 social functioning | 91.5 (17.8) | | | 68.8 (26.4) | | |
| | SF-36 mental health | 80.4 (16.8) | | | 67.2 (23.2) | | |
| | SF-36 physical component summary | 42.9 (6.54) | | | 42.3 (8.02) | | |
| | SF-36 mental component summary | 55 (8.79) | | | 42.8 (9.86) | | |
| Kobe, 2016 Subgroup of frail patients with a frailty score 6-14 | SF-36 physical functioning | 40.9 (20.5) | | | 46.5 (30.2) | | |
| | SF-36 role physical | 25.6 (33.2) | | | 27.5 (39.7) | | |
| | SF-36 bodily pain | 69.1 (35.6) | | | 60.4 (34.8) | | |
| | SF-36 role emotional | 82.5 (32.9) | | | 45 (46.2) | | |
| | SF-36 general health | 63.1 (22.7) | | | 50.6 (20.9) | | |
| | SF-36 vitality | 35.4 (25.8) | | | 44.1 (24.9) | | |
| | SF-36 social functioning | 79.7 (28.7) | | | 67.9 (23.9) | | |
| | SF-36 mental health | 75 (16.3) | | | 66 (23.3) | | |
| | SF-36 physical component | 36.1 (9.57) | | | 38.6 (8.68) | | |

For peer review only

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | summary | | | | | | |
| | SF-36 mental component summary | 53 (9.23) | | | 45.2 (11.6) | | |
| Goudzwaard, 2020 | EQ-5D index | 0.55 (0.26) | | | | | 0.44 (0.33) |
| | EQ-VAS | 58.4 (16.8) | | | | | 63.4 (14.6) |
| Patel, 2020 | KCCQ 12 score | 36 (25) | 75 (70-80) | 37 (31-43) | | | |
| Morris, 2020 | KCCQ 12 score | 37 | 66 | | | | |

KCCQ, Kansas City Cardiomyopathy Questionnaire. SF-36, Short Form-36 questionnaire. EQ-5D, Euro-Qol-5-dimensions. VAS, visual analogue scale.

For peer review only

# MOOSE Checklist for Meta-analyses of Observational Studies

| Item No | Recommendation | Reported on Page No |
|---|---|---|
| Reporting of background should include | | |
| 1 | Problem definition | 4 |
| 2 | Hypothesis statement | - |
| 3 | Description of study outcome(s) | 5 |
| 4 | Type of exposure or intervention used | 5 |
| 5 | Type of study designs used | 5 |
| 6 | Study population | 5 |
| Reporting of search strategy should include | | |
| 7 | Qualifications of searchers (eg, librarians and investigators) | 1, 5 |
| 8 | Search strategy, including time period included in the synthesis and key words | 5, Supplementary appendix A |
| 9 | Effort to include all available studies, including contact with authors | 5 |
| 10 | Databases and registries searched | 5 |
| 11 | Search software used, name and version, including special features used (eg, explosion) | Protocol |
| 12 | Use of hand searching (eg, reference lists of obtained articles) | Protocol |
| 13 | List of citations located and those excluded, including justification | Fig 1 |
| 14 | Method of addressing articles published in languages other than English | - |
| 15 | Method of handling abstracts and unpublished studies | Protocol |
| 16 | Description of any contact with authors | - |
| Reporting of methods should include | | |
| 17 | Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested | 5-6, protocol |
| 18 | Rationale for the selection and coding of data (eg, sound clinical principles or convenience) | 5-6, protocol |
| 19 | Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability) | 5-6, protocol |
| 20 | Assessment of confounding (eg, comparability of cases and controls in studies where appropriate) | - |
| 21 | Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results | 6, protocol |
| 22 | Assessment of heterogeneity | 7 |
| 23 | Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated | 6-7, protocol |
| 24 | Provision of appropriate tables and graphics | Tables 1-2, Figs 1 Supplementary appendix A-G |
| Reporting of results should include | | |
| 25 | Graphic summarizing individual study estimates and overall estimate | Table 1-2, Supplementary appendix D-G |
| 26 | Table giving descriptive information for each study included | Table 1, Supplementary |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | |
|---|---|---|
| | | appendix B |
| 27 | Results of sensitivity testing (eg, subgroup analysis) | 9-10 Table 2, Supplementary appendix E, F |
| 28 | Indication of statistical uncertainty of findings | 9-12 |

| Item No | Recommendation | Reported on Page No |
|---|---|---|
| Reporting of discussion should include | | |
| 29 | Quantitative assessment of bias (eg, publication bias) | 8, Supplementary appendix C |
| 30 | Justification for exclusion (eg, exclusion of non-English language citations) | 5, protocol |
| 31 | Assessment of quality of included studies | 6 |
| Reporting of conclusions should include | | |
| 32 | Consideration of alternative explanations for observed results | 12-16 |
| 33 | Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review) | 17 |
| 34 | Guidelines for future research | 16 |
| 35 | Disclosure of funding source | 17 |

*From*: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

For peer review only

BMJ Open

# Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis



| | |
|---|---|
| Journal: | *BMJ Open* |
| Manuscript ID | bmjopen-2020-040459.R2 |
| Article Type: | Original research |
| Date Submitted by the Author: | 09-Dec-2020 |
| Complete List of Authors: | Li, Zhe; University of Western Ontario, Epidemiology and Biostatistics<br>Dawson , Emily ; Western University, Centre for Medical Evidence, Decision Integrity & Clinical Impact (MEDICI)<br>Moodie, Jessica; University of Western Ontario, Centre for Medical Evidence, Decision Integrity and Clinical Impact (MEDICI)<br>Martin, Janet; University of Western Ontario, High Impact Technology Evaluation Centre<br>Bagur, Rodrigo; Western University, Department of Epidemiology and Bio statistics<br>Cheng, Davy; London Health Sciences Centre University Hospital, Anesthesia & Perioperative Medicine<br>Kiaii, Bob; Western University, Department of Cardiology, Division of Cardiac Surgery<br>Hashi, Adam; Faculty of Sciences<br>Bi, Ran; Schulich Interfaculty Program in Public Health<br>Yeschin, Michelle ; Department of Epidemiology and Biostatistics, Western University<br>John-Baptiste, Ava; University of Western Ontario, Anesthesia & Perioperative Medicine |
| Primary Subject Heading: | Cardiovascular medicine |
| Secondary Subject Heading: | Cardiovascular medicine, Epidemiology |
| Keywords: | Valvular heart disease < CARDIOLOGY, EPIDEMIOLOGY, Cardiac surgery < SURGERY, VASCULAR SURGERY |
| | |



SCHOLARONE™
Manuscripts



*I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.*

*The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.*

*Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.*

For peer review only

# Measurement and prognosis of frail patients undergoing transcatheter aortic valve implantation: a systematic review and meta-analysis

Zhe Li [a, b, c], Emily Dawson [a, b, c], Jessica Moodie [b, c], Janet Martin [a, b, c], Rodrigo Bagur [a], Davy Cheng [b, c, d], Bob Kiaii [e], Adam Hashi [f], Ran Bi [g], Michelle Yeschin [a], Ava John-Baptiste [a, b, c, g]

**Affiliations**:

a. Department of Epidemiology and Biostatistics, Schulich School of Medicine and Dentistry,

Western University, London, ON, Canada.

b. Centre for Medical Evidence, Decision Integrity & Clinical Impact (MEDICI), Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

c. Department of Anesthesia & Perioperative Medicine, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

d. Department of Medicine, Division of Critical Care Medicine, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

e. Department of Surgery, Division of Cardiac Surgery, Schulich School of Medicine and Dentistry, Western University, London, ON, Canada.

f. Faculty of Science, Western University, London, ON, Canada.

g. Schulich Interfaculty Program in Public Health, Western University, London, ON, Canada.

**Corresponding Author:** Dr. Ava John-Baptiste

The Western Centre for Public Health and Family Medicine

Rm 4111, 1465 Richmond St.

London, ON, Canada, N6G 2M1

e. ajohnbap@uwo.ca

t. 519.661.2111 (86265)

**Running head**: Prognosis of frail patients after TAVI

## Abstract



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Objectives**: Our objectives were to review the literature to identify frailty instruments in use for TAVI recipients and synthesize prognostic data from these studies, in order to inform clinical management of frail patients undergoing TAVI. **Methods**: We systematically reviewed the literature published in 2006 or later. We included studies of patients with aortic stenosis, diagnosed as frail, who underwent a TAVI procedure that reported mortality or clinical outcomes. We categorized the frailty instruments and reported on the prevalence of frailty in each study. We summarized the frequency of clinical outcomes and pooled outcomes from multiple studies. We explored heterogeneity and performed sub-group analysis, where possible. We also used GRADE to assess the overall certainty of the estimates. **Results**: Of 49 included studies, 21 used single-dimension measures to assess frailty, 3 used administrative data-based measures, and 25 used multi-dimensional measures. Prevalence of frailty ranged from 5.67% to 90.07%. Albumin was the most commonly used single-dimension frailty measure and the Fried or modified Fried phenotype were the most commonly used multi-dimensional measures. Meta-analyses of studies that used either the Fried or modified Fried phenotype showed a 30-day mortality of 7.86% (95% confidence interval,



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

5.20% to 11.70%) and a 1-year mortality of 26.91% (95% confidence interval, 21.50% to 33.11%). The GRADE system suggests very low certainty of the respective estimates.

**Conclusions**: Frailty instruments varied across studies, leading to a wide range of frailty prevalence estimates for TAVI recipients and substantial heterogeneity. The results provide clinicians, patients, and health care administrators, with potentially useful information on the prognosis of frail patients undergoing TAVI. This review highlights the need for standardization of frailty measurement to promote consistency.

## Strengths and limitations

- This study examines the heterogeneity across different frailty assessment tool and determines the frequency of adverse outcomes and pool the prognosis after transcatheter aortic valve implantation in frail patients.
- This study uses a comprehensive literature search strategy and includes frail patients from randomized controlled trials and observational studies.
- This study excluded studies in which dimensions of frailty were assessed without reference to the goal of frailty assessment.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Keywords**: frailty; transcatheter aortic valve implantation; systematic review; meta-analysis

Transcatheter aortic valve implantation (TAVI) has become an alternative, less invasive treatment option for patients with severe symptomatic aortic stenosis.[1] The evidence continues to accumulate and synthesis of the evidence to better understand the prognosis of frail patients who undergo TAVI may be helpful.[2]

Frailty is a biological syndrome characterized by an increased vulnerability to stressors.[3] When exposed to stressors, such as chronic illness and surgery, frail patients are susceptible to adverse events, procedural complications, prolonged recovery, functional decline, and reduced survival.[4] Clinical research has identified frailty as an important risk factor for mortality and morbidity following TAVI.[5] Health economics research has shown that compared to non-frail patients, frail older adults undergoing cardiac surgery incurred substantially higher hospitalization costs.[6] Given the clinical and economic implications of TAVI, searching for and synthesizing outcomes of frail patients undergoing TAVI may provide information that can help optimize the selection of TAVI candidates and ultimately improve decision-making related to treatment of aortic stenosis.[2]

When considering valve procedures, clinical practice guidelines recommend assessing frailty as one component of risk.[7] We performed a systematic review of the literature to identify studies reporting the prognosis of frail patients undergoing TAVI. With no single standard method of measuring frailty and a diversity of frailty measurements, the optimal approach to assessing frailty in patients undergoing TAVI is unclear.[2,5] We catalogued frailty measures used in identified studies, to perform sub-group analyses for studies using the most common measures.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

## Methods

This systematic review and meta-analysis is reported according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses (PRISMA) guidelines[8] and follows the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) guidelines.[9] The protocol was registered in PROSPERO (CRD42018090597) and published in *BJM Open*.[10]

### Literature search and eligibility criteria

We searched PubMed, EMBASE, PsycINFO, Cochrane Library, Web of Science and ClinicalTrials.gov for articles published between January 2006 and September 23, 2020 (Supplementary appendix A). Conference abstracts from relevant conferences held in the last 3 years were also searched. The detailed inclusion and exclusion criteria were described in detail in the protocol.[10] We included patients with aortic stenosis, diagnosed as frail, who underwent a TAVI procedure. We only included studies that intended to measure frailty with a defined method of frailty assessment. Studies were excluded if baseline frailty status was measured after the TAVI procedure. We included all forms of TAVI, regardless of procedural approach and types of valves. Outcome measures included mortality, clinical outcomes or quality of life. We included studies describing non-comparative cohorts of patients undergoing TAVI who have been diagnosed with frailty and studies describing comparative cohorts of frail and non-frail patients undergoing TAVI in which outcomes were reported separately for frail patients. Authors (ZL, ED, AH, BR and MY) independently assessed study eligibility. Disagreements were resolved by consulting a third reviewer.

### Risk of bias assessment

The risk of bias in individual studies was appraised independently by two authors (ZL and ED) using the Quality in Prognosis Studies (QUIPS) tool.[11] We classified studies with four or five low risk domains as having a low risk of bias overall, studies with two or more high risk domains as high risk of bias overall, and the remaining studies as moderate risk of bias overall.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Data synthesis and meta-analysis**

Pre-specified statistical details were described in the protocol.[10] We summarized the method of measuring frailty used in each study including the frailty tool used, dimensions of frailty measured, the cut-off for frail status, and the prevalence of frailty in the study population as measured by the frailty tool. We only extracted data from the most commonly used frailty instruments if multiple frailty instruments were applied in the same patient group. We categorized clinical outcomes and reported the frequency at each time point. Heterogeneity across studies was assessed using the Cochrane Q-statistic ($I^2$).[12]

For adverse clinical outcomes, we pooled proportions using the inverse-variance weighted DerSimonian and Laird model and incorporated the Freeman-Tukey double arcsine transformation.[13,14] A funnel plot was used to plot the effect estimates from individual studies against the standard error of each study. In the absence of bias and heterogeneity, the funnel plot will be symmetrical.[15] For the length of hospitalization, we pooled the values, estimating the mean and standard deviation using the random effects model for continuous variables.[16] For studies presenting Kaplan-Meier curves with time to death, we collected the information on numbers at risk and total number of events, and then created a single pooled Kaplan-Meier curve. We pooled time to death data from individual studies to obtain an overall estimate of survival, based on an algorithm developed by Guyot et al.[17] All analyses were conducted using R software (version 3.5.0). A two-sided p-value of 0.05 or less was considered statistically significant.

**Subgroup analysis**

We conducted a sub-group analysis to see if the estimates of mortality rates differed for studies that used the Fried phenotype, the most common multi-dimensional measure, compared to studies that did not use the Fried phenotype.

**Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) assessment**

We used the GRADE system to conduct an evaluation of the overall estimates based on considerations of risk of bias, consistency, precision, directness, and publication bias.[18] Given



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

that cohort studies of prognosis exclude randomized controlled trial study designs, we did not downgrade the certainty of evidence due to observational study design.

**Patient and Public Involvement**

No patient involved.

**Results**

**Characteristics of included studies**

Our search identified 4,944 records with 2,635 articles remaining after removing duplicates. After screening, 49 studies[19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53,54,55,56,57,58,59,60,61,62,63,64,65,66,67] were identified as eligible for inclusion in the review (figure 1).

The characteristics of the included studies are summarized in supplementary appendix B. Three studies[40,43,53] enrolled patients from the Placement of Aortic Transcatheter Valves (PARTNER) trial reporting separately on outcomes of frail patients; the remaining studies reported on patients from a single cohort or registry. Most studies collected patient data prospectively; seventeen studies[21,23,24,26,32,35,38,41,43,45,46,48,57,60,61,62,65] were conducted retrospectively.

Supplementary appendix C summarizes the risk of bias assessment of individual studies. Of the 49 studies, 21 [25,26,28,29,31,33,37,39,40,47,49,52,55,58,59,60,62,63,64,65,66] were rated at overall low risk of bias, 21 [19-24, 32,35,36, 42-45, 48, 50,51, 53,54,56,61,] at moderate risk and 7 [27, 38, 41, 46, 50, 57,67] at high risk of bias.

**Measurement of frailty in patients undergoing TAVI**

Table 1 summarizes frailty assessment in patients undergoing TAVI. Twenty one studies [19-33, 54-59] used single-dimension measures, 3 studies [60-62] used administrative data-based measures, and 25 studies [34-53, 63-67] used multi-dimensional measures. The prevalence of frailty varied widely among studies that assessed frailty with single dimension measures, ranging from 5.67% to 90.07%. Albumin, body mass index, and Katz Activity of Daily Living were the three most commonly used single-dimension measures when assessing frailty in TAVI patients. However, even with the same measure, different cut-points or definitions of frailty were used.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

For example, four studies [21,23,24,59] used albumin to assess frailty; two [21,23] defined frailty as albumin level below 4g/dL, and two [24, 59] as albumin level below 3.5g/dL.

Among studies that used frailty indices based on administrative data, the prevalence of frailty ranged from 5.54% to 47.64%. Two studies [60, 62] used the Hospital Frailty Risk Score, a frailty algorithm calculated based on a list of predefined International Statistical Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10) diagnostic codes. Frailty prevalence reported among the two studies was 41.06% and 47.64%, respectively. One study [61] used Johns Hopkins Adjusted Clinical Groups frailty-defining diagnosis indicator that was based on 10 clusters of frailty-defining diagnoses.

The prevalence of frailty reported by studies that assessed frailty using multi-dimensional measures ranged from 15.23% to 84.67%. Most of these studies assessed frailty based on the Fried frailty phenotype; one study[34] assessed frailty based on the accumulated deficits frailty index. Of the 25 studies reporting multi-dimensional measures, four [46-48, 51] used the original Fried frailty phenotype and eight [35,38-41,43,44,53] modified the Fried frailty phenotype by examining fewer dimensions, altering cut-off values or measuring the same domains with different criteria. Among the eight studies [35,38-41,43,44,53] reporting the modified Fried frailty phenotype, measures used to assess mobility and disability were identical. Measures used to assess nutrition were different; seven studies [35,38-40,43,44,53] measured serum albumin, and 1 study [41] measured weight loss.

**Prognosis of frail TAVI recipients**

Supplementary appendix D summarized prognosis of frail TAVI recipients reported for each study. Twenty studies [19,21,27-31,34-36,40,41,43,50,53,56,58,62,65,66] reported 30-day mortality, which ranged from 2.83% to 25%; the combined 30-day mortality estimate was 7.32% (95% confidence interval 5.66% to 9.42%, table 2, figure 2). Combining three studies [35,40,41] that measured frailty using the modified Fried frailty phenotype, we estimated a 30-day mortality of 7.86% (5.20% to 11.70%, table 2, figure 2).

Fifteen studies[25,30,31,34,35,37,40,43,46,49,50,53,56,58,62] reported 1-year mortality, ranging from 14.8% to 37.5%. The combined 1-year mortality estimate was 23.98% (20.71% to 27.58%, table 2, figure 3).When pooling two studies[35,46] that used the Fried or modified Fried frailty phenotype to assess frailty, the estimated 1-year mortality was 26.91% (21.50% to 33.11%, table 2, figure



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

3). Subgroup analyses of studies reporting frailty measurement using the Fried phenotype compared to non-Fried phenotype did not find statistical differences in effect estimates on 30-day and 1-year mortality (supplementary appendix E).

Seventeen studies [22,23,25-29,31,33,34,40,41,47,48,52,56,58,60,62]reported survival of frail patients after TAVI using a Kaplan-Meier curve. The combined survival estimates at 1, 2, and 3 years were 75.6% (75.2% to 76.0%, table 2), 65.0% (63.3% to 66.7%, table 2), and 48.7% (43.3% to 54.7%, table 2), respectively. Combining the studies that used the Fried or modified Fried phenotype, we found survival estimates at 1, 2, and 3 years were 73% (68.8% to 77.5%, table 2), 64.5% (56.4% to 73.9%, table 2), and 58.9% (49% to 70.9%, table 2), respectively. Details of survival are provided in supplementary appendix F.

Five studies [42,44,63,65,67] measured health-related quality of life (supplementary appendix G). Three studies [44,65,67] assessed quality of life preoperatively using the 12-item Kansas City Cardiomyopathy Questionnaire (KCCQ). Two studies [65,67] assessed quality of life post-TAVI; both studies found improved quality of life overall. Okoh et al [44] assessed quality of life at 30 days following TAVI, and found that at 30 days, frail patients reported worsening in two domains, KCCQ-symptoms and KCCQ physical limitation, but quality of life improved slightly overall. Kobe et al [42] assessed quality of life before and 30 days after TAVI using the Short Form-36 questionnaire; they found that at 30-day follow-up, the mean scores of all but role physical and social functioning were significantly lower for frail patients. Goudzwaard et al [63] assessed quality of life using the Euro-QoL-5-dimension (EQ-5D) scale; they found that at 12 months follow-up, the mean EQ-5D decreased while the mean EQ-visual analogue scale increased.

Other commonly reported outcomes measuring the prognosis of frail TAVI recipients include procedural acute kidney injury (ranging from 3.95% to 20.51%), conversion to open heart surgery (ranging from 0% to 9.9%), life-threatening bleeding (ranging from 4.86% to 16.7%), major bleeding (ranging from 2.56% to 21.81%), permanent pacemaker implantation (ranging from 2% to 12.82%), and stroke (ranging from 0% to 8.3%). Eight studies[32,33,38,39,41,44,45,56] reported the mean length of hospitalization, ranging from 5 days to 12.1 days.

**GRADE assessment**



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The GRADE certainty assessment per outcome, together with the pooled effects, is provided in table 2. Due to inconsistency as influenced by heterogeneity of estimates and indirectness of frailty measures as influenced by lack of homogeneity across the TAVI populations, confidence in the overall estimates was very low.

## Discussion

### Principal findings

We found that multi-dimensional measures are more commonly used than single-dimension measures. Even with the same frailty measure, different definitions or cut-offs were used. The most frequently used frailty measure in TAVI the studies we identified was the modified Fried phenotype, in which disability, muscle strength, mobility, and nutrition were assessed. Approaches to modifying the Fried phenotype included measuring fewer domains, using different cut-offs, or using different tools to assess the same domain.

Greater heterogeneity of meta-analyses that included single measures suggests single measures did not measure the same frailty construct and did not reliably measure frailty. Single measures included a mix of biological variables (albumin and BMI) or single performance measures (gait speed or activities of daily living), which address only a single component of the frailty construct. Thus, our study suggests that frailty is a multi-dimensional phenomenon that cannot be captured by a single construct.

The variety of frailty definitions and the diversity of TAVI populations in the studies contribute to the wide range and substantial heterogeneity of patient outcomes after TAVI. Using GRADE to assess confidence in prognosis estimates from the meta-analyses, we found very low confidence in the overall estimates, mainly due to inconsistency as influenced by heterogeneity of estimates and indirectness of frailty measures as influenced by lack of homogeneity across the TAVI populations identified in the studies.

### Comparison with other studies

Previous studies demonstrated that the assessment of frailty significantly enhances prediction of mortality after TAVI when combined with the European system for cardiac operative risk evaluation (EuroSCORE) or the Society of Thoracic Surgeons (STS) score.[49] There have been several studies reviewing frailty in cardiac surgical populations. Kim et al[5]



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

conducted a systematic review of frailty instruments in older adults undergoing cardiac surgical procedures. Kim et al[5] found high quality evidence that used mobility assessment as a single frailty measure and found mobility to be the most frequently assessed domain. Sepehri et al[68] performed a systematic review to demonstrate the association of frailty with negative postoperative outcomes in patients undergoing cardiac surgery. Our study adds to the existing literature as we investigate the frequency of adverse outcomes and pool estimates of survival after TAVI in frail patients from multiple studies.

The FRAILTY-AVR study[69] examined the validity of frailty measures in predicting mortality amongst TAVI recipients. The study added value to the literature by selecting frailty elements with the greatest predictive value, finding that the Essential Frailty Toolset (EFT) consisting of chair rise, cognition measured by the Mini-Mental State Examination, hemoglobin and serum albumin, performed best for predicting one-year mortality.[69] Due to the focus on predictive validity, the FRAILTY-AVR study[69] did not report outcomes separately for frail patients. As a result, the study[69] did not meet the inclusion criteria for our systematic review, which was focused on prognostic information amongst frail patients only. The FRAILTY-AVR study[69] makes important efforts to define a standard frailty assessment tool. Although the Fried and modified Fried were the most commonly used instruments amongst studies included in our meta-analysis, the FRAILTY-AVR showed the Fried did not perform as well as the EFT in predicting mortality amongst TAVI patients.[69] We suggest use of a standard measure, such as the EFT, can enhance the quality of frailty research in the TAVI patient population. We also recognize that use of a standard frailty measure is unlikely as researchers and clinicians may value use of diverse measures which reflect different aspects of frailty. If the EFT emerges as a standard, it may be used by clinicians to exclude frail patients from treatment, due to concerns about increased mortality. This would limit the opportunity to better understand the prognosis of frail patients undergoing TAVI, which was the primary goal of our study.

**Strengths and limitations**

This review has several unique strengths. We performed a comprehensive literature search to identify both published and unpublished studies, in addition to searching citations from previous reviews. We included prognostic data from randomized controlled trials and observational studies. Using the QUIPS tool, two reviewers independently assessed the risk of



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

bias, and the use of the GRADE system to assess the certainty of evidence offers a structured and transparent evaluation of our findings. We systematically reviewed the operationalization of frailty assessment in TAVI patients, and pooled clinical outcomes of frail TAVI recipients. We tested for heterogeneity and attempted to address heterogeneity by performing sensitivity analysis and sub-group analysis.

This review has some important limitations. Given the limited data reported by the included studies, we were unable to perform meta-regression to further investigate the potential sources of heterogeneity and to determine the influence of mean age on outcomes. We therefore explored the causes and types of heterogeneity relying on the investigation of the $I^2$ statistic, which may be imprecise when the number of studies is small.[70] When extracting data, we encountered several studies that applied multiple frailty instruments in the same patient group, and in this situation, we only extracted data from the most commonly used frailty instrument, and this may introduce selection bias. Some studies defined an intermediate 'pre-frail' group, but we did not find sufficient data to synthesize outcomes for this important sub-group. Though less vulnerable than the frail group, pre-frail patients may be at higher risk than robust patients for experiencing adverse outcomes.[71,72] Individual-patient level data were not available, precluding adjustment for any study level differences in clinical or procedural variables that may have influenced prognosis across the cohorts. Therefore, clinical heterogeneity could not be ruled-out and along with high levels of heterogeneity, resulted in lower GRADE evaluations. The aim of this study was to characterize prognosis for frail patients undergoing TAVI, therefore, we did not directly compare prognosis to other groups of patients or to frail patients undergoing different therapies, nor were we able to determine which frailty measures perform best as prognostic tools for TAVI recipients.

**Implications**

When selecting candidates to undergo TAVI, several multivariate risk scores have been widely used to estimate operative mortality based on patient characteristics. The STS score and the EuroSCORE are the most commonly used scoring systems.[73,74] However, a disadvantage of both scores is that the main variables for scoring perioperative risk are medical diagnoses and comorbidities, which may not reflect the true 'biological status' of the patient.[73,74] When considering valve procedures for patients, clinical practice guidelines recommend assessing



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

frailty as one component of risk.[5,7] Although a large number of frailty measures exist, there is currently little consensus on the optimal approach to assessing frailty in patients undergoing TAVI.[2] Frailty has consistently been shown to significantly predict mortality[68] and post-operative delirium,[75] even after controlling for other risk factors, suggesting that use of any frailty assessment is better than none when selecting patients for TAVI. Systematically reviewing the operationalization of frailty assessment in TAVI patients and pooling clinical outcomes of frail TAVI recipients will help better understand how frailty is assessed among TAVI patients, provide information on the prognosis of frail patients after TAVI, and can ultimately improve decisions related to treatment of AS.

To help achieve consensus on frailty measures to be applied in TAVI recipients, future studies should evaluate the prognostic value of frailty measures in TAVI recipients and determine the additional prognostic value of frailty measurement in addition to these established risk scores. Future studies should also compare prognosis of frail patients undergoing TAVI to frail patients undergoing surgical intervention or medical therapy. Few studies reported quality of life measures. In order to address the gaps in the literature future studies should measure quality of life before and after TAVI with use of standardized quality of life measurement tools such as the Short-Form 36.

**Conclusion**

In conclusion, frailty instruments for TAVI recipients varied across studies, leading to a range of frailty prevalence estimates and substantial heterogeneity. The results of this systematic review provide clinicians, patients, and health care administrators, with potentially useful evidence on the prognosis of frail patients.

**Acknowledgement:** We gratefully thank Dr. Farid Foroutan for his assistance with meta-analysis of survival data.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Ethics committee approval:** Due to the nature of the study, there are no ethical concerns and informed consent was not required.

**Contributorship statement:** Zhe Li, Emily Dawson, Jessica Moodie, Janet Martin, Rodrigo Bagur, Davy Cheng, Bob Kiaii, and Ava John-Baptiste proposed and designed the purpose, review questions, and methods of this study. Searching strategy was developed by Jessica Moodie, and revised by Zhe Li, Emily Dawson, and Ava John-Baptiste. Zhe Li, Emily Dawson, Ava John-Baptiste, Adam Hashi, Ran Bi and Michelle Yeschin screened articles and verified data abstraction. Zhe Li and Emily Dawson abstracted data from articles. Zhe Li, Emily Dawson and Ava John-Baptiste critically appraised the articles. Data analysis was primarily conducted by Zhe Li. Ava John-Baptiste provided primary academic supervision. Zhe Li drafted the manuscript. All co-authors have contributed to the revision of the manuscript.

**Competing interests:** There is no competing interest to declare.

**Funding:** This work was supported by the Academic Medical Organization of Southwestern Ontario (grant number: INN 17-001).



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Data sharing statement**: There is no additional unpublished data from the study. All data are provided in full in the main manuscript document.

## References


1. Siontis GCM, Overtchouk P, Cahill TJ, et al. Transcatheter aortic valve implantation vs. surgical aortic valve replacement for treatment of symptomatic severe aortic stenosis: An updated meta-analysis. *Eur Heart J*. 2019;40(38):3143-3153. doi:10.1093/eurheartj/ehz275
2. Puri R, Iung B, Cohen DJ, Rodés-Cabau J. TAVI or No TAVI: Identifying patients unlikely to benefit from transcatheter aortic valve implantation. *Eur Heart J*. 2016;37(28):2217-2225. doi:10.1093/eurheartj/ehv756
3. Fried LP, Tangen CM, Walston J, et al. Frailty in older adults: Evidence for a phenotype. *Journals Gerontol - Ser A Biol Sci Med Sci*. 2001;56(3):146-157. doi:10.1093/gerona/56.3.m146
4. Shamliyan T, Talley KMC, Ramakrishnan R, Kane RL. Association of frailty with survival: A systematic literature review. *Ageing Res Rev*. 2013;12(2):719-736. doi:10.1016/j.arr.2012.03.001
5. Kim DH, Kim CA, Placide S, Lipsitz LA, Marcantonio E. Preoperative frailty assessment and outcomes at 6 months or later in older adults undergoing cardiac surgical procedures: A systematic review. *Ann Intern Med*. 2016;165(9):650-660. doi:10.7326/M16-0652.Preoperative
6. Goldfarb M, Bendayan M, Rudski LG, et al. Cost of Cardiac Surgery in Frail Compared With Nonfrail Older Adults. *Can J Cardiol*. 2017;33:1020-1026. doi:10.1016/j.cjca.2017.03.019
7. Nishimura RA, Otto CM, Bonow RO, et al. 2017 AHA/ACC focused update of the 2014 AHA/ACC guideline for the management of patients with valvular heart disease: a report of the American College of Cardiology/American Heart




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Association Task Force on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2017;70(2):252-289. doi:10.1016/j.jacc.2017.03.011

8. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M SP, LA. S. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev*. 2015;4(1):1-19.
9. Stroup DF, Berlin JA, Morton SC, Olkin I, Williamson GD, Rennie D, Moher D, Becker BJ, Sipe TA TS. Meta-analysis of observational studies in Epidemiology. *Mod Methods Epidemiol*. Published online 2012:173-189. doi:10.1007/978-94-007-3024-3_10
10. Li Z, Dawson E, Moodie J, et al. Frailty in patients undergoing transcatheter aortic valve implantation : a protocol for a systematic review. 2019;(1):1-7. doi:10.1136/bmjopen-2018-024163
11. Hayden J a, Windt D a Van Der, Cartwright JL, Co P. Assessing Bias in Studies of Prognostic Factors. *Ann Intern Med*. 2013;144:427-437. doi:10.7326/0003-4819-158-4-201302190-00009
12. Higgins JPT, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *Br Med J*. 2003;327(7414):557-560. doi:10.1136/bmj.327.7414.557
13. Jackson D, White IR, Thompson SG. Extending DerSimonian and Laird's methodology to perform multivariate random effects meta-analyses. *Stat Med*. 2010;29(12):1282-1297. doi:10.1002/sim.3602
14. Nyaga VN, Arbyn M, Aerts M. Metaprop: A Stata command to perform meta-analysis of binomial data. *Arch Public Heal*. 2014;72(1):1-10. doi:10.1186/2049-3258-72-39
15. Sterne JAC, Sutton AJ, Ioannidis JPA, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ*. 2011;343(7818):1-8. doi:10.1136/bmj.d4002
16. Marín-Martínez F, Sánchez-Meca J. Weighting by Inverse Variance or by Sample Size in Random-Effects Meta-Analysis. *Educ Psychol Meas*. 2010;70(1):56-73. doi:10.1177/0013164409344534
17. Guyot P, Ades AE, Ouwens MJNM, Welton NJ. Enhanced secondary analysis of



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

survival data: Reconstructing the data from published Kaplan-Meier survival curves. *BMC Med Res Methodol*. 2012;12. doi:10.1186/1471-2288-12-9

18. Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P SH. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *Rev Colomb Obstet Ginecol*. 2016;67(1):26-35. doi:10.18597/rcog.371
19. Alfredsson J, Stebbins A, Brennan JM, et al. Gait Speed Predicts 30-Day Mortality After Transcatheter Aortic Valve Replacement Cardiology Transcatheter Valve Therapy Registry. *Ciculation*. 2016;133:1351-1359. doi:10.1161/CIRCULATIONAHA.115.020279
20. Bagienski M, Kleczynski P, Dziewierz A, et al. Incidence of Postoperative Delirium and Its Impact on Outcomes After Transcatheter Aortic Valve Implantation. *Am J Cardiol*. 2017;120(7):1187-1192. doi:10.1016/j.amjcard.2017.06.068
21. Bogdan A, Barbash IM, Segev A, Fefer P, Bogdan SN, Asher E, Fink N, Hamdan A, Spiegelstein D, Raanani E G V. Albumin correlates with all-cause mortality in elderly patients undergoing transcatheter aortic valve implantation. *EuroIntervention*. 2016;12(e1057-64).
22. Cockburn J, Singh MS, Rafi NHM, et al. Poor mobility predicts adverse outcome better than other frailty indices in patients undergoing transcatheter aortic valve implantation. *Catheter Cardiovasc Interv*. 2015;86(7):1271-1277. doi:10.1002/ccd.25991
23. Grossman Y, Barbash IM, Fefer P, et al. Addition of albumin to Traditional Risk Score Improved Prediction of Mortality in Individuals Undergoing Transcatheter Aortic Valve Replacement. *J Am Geriatr Soc*. 2017;65(11):2413-2417. doi:10.1111/jgs.15070
24. Koifman E, Magalhaes MA, Ben-Dor I, et al. Impact of pre-procedural serum albumin levels on outcome of patients undergoing transcatheter aortic valve replacement. *Am J Cardiol*. 2015;115(9):1260-1264. doi:10.1016/j.amjcard.2015.02.009
25. Kleczynski P, Dziewierz A, Bagienski M, et al. Impact of frailty on mortality after transcatheter aortic valve implantation. *Am Heart J*. 2017;185:52-58.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

doi:10.1016/j.ahj.2016.12.005

26. Mok M, Allende R, Leipsic J, et al. Prognostic value of fat mass and skeletal muscle mass determined by computed tomography in patients who underwent transcatheter aortic valve implantation. *Am J Cardiol*. 2016;117(5):828-833. doi:10.1016/j.amjcard.2015.12.015
27. Martin GP, Sperrin M, Ludman PF, et al. Do frailty measures improve prediction of mortality and morbidity following transcatheter aortic valve implantation? An analysis of the UK TAVI registry. *BMJ Open*. 2018;8(6):1-9. doi:10.1136/bmjopen-2018-022543
28. Puls M, Sobisiak B, Bleckmann A, Jacobshagen C, Danner BC, Huenlich M, Beissbarth T, Schoendube F, Hasenfuss G, Seipelt R SW. Impact of frailty on short-and long-term morbidity and mortality after transcatheter aortic valve implantation: risk assessment by Katz Index of activities of daily living. *EuroIntervention J Eur Collab with Work Gr Interv Cardiol Eur Soc Cardiol*. 2014;10(609-19).
29. Rodés-cabau J, Webb JG, Cheung A, et al. Transcatheter Aortic Valve Implantation for the Treatment of Severe Symptomatic Aortic Stenosis in Patients at Very High or Prohibitive Surgical Risk Acute and Late Outcomes of the Multicenter Canadian Experience. *JAC*. 2010;55(11):1080-1090. doi:10.1016/j.jacc.2009.12.014
30. Stortecky S, Windecker S, Pilgrim T et al. The association of body mass index with clinical outcome in patients with severe aortic stenosis undergoing transcatheter aortic valve implantation. *EuroIntervention*. 2012;8.
31. Shimura T, Yamamoto M, Kano S, et al. Impact of the clinical frailty scale on outcomes after transcatheter aortic valve replacement. *Circulation*. 2017;135(21):2013-2024. doi:10.1161/CIRCULATIONAHA.116.025630
32. Traynor MM, Greason KL, Nkomo VT, et al. Assisted care as a baseline patient risk characteristic affecting the outcome of transcatheter aortic valve insertion. *Catheter Cardiovasc Interv*. 2017;89(7):1268-1272. doi:10.1002/ccd.26580
33. Yamamoto M, Hayashida K, Watanabe Y, et al. Effect of body mass index <20 kg/m2 on events in patients who underwent transcatheter aortic valve



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

replacement. *Am J Cardiol*. 2015;115(2):227-233. doi:10.1016/j.amjcard.2014.10.026

34. Bureau ML, Liuu E, Christiaens L, et al. Using a multidimensional prognostic index (MPI) based on comprehensive geriatric assessment (CGA) to predict mortality in elderly undergoing transcatheter aortic valve implantation. *Int J Cardiol*. 2017;236:381-386. doi:10.1016/j.ijcard.2017.02.048
35. Chauhan D, Haik N, Merlo A, Haik BJ, Chen C. Quantitative increase in frailty is associated with diminished survival after transcatheter aortic valve replacement. Published online 2015. doi:10.1016/j.ahj.2016.06.028
36. Capodanno D, Barbanti M, Tamburino C, et al. A simple risk tool (the OBSERVANT score) for prediction of 30-day mortality after transcatheter aortic valve replacement. *Am J Cardiol*. 2014;113(11):1851-1858. doi:10.1016/j.amjcard.2014.03.014
37. Eichler S, Salzwedel A, Harnath A, et al. Nutrition and mobility predict all-cause mortality in patients 12 months after transcatheter aortic valve implantation. *Clin Res Cardiol*. 2018;107(4):304-311. doi:10.1007/s00392-017-1183-1
38. Abhijit Ghatak, Brian O'Neill, Ruchi Patel, Sachil Shah, Claudia Martinez, Mauro Moscucci, Pedro Clark, Mauricio Cohen, Alan Heldman WO and CA. Frailty is a Major Determinant of Length of Stay After Transcatheter Aortic Valve Replacement. *J Am Coll Cardiol*. 2012;60(17). doi:10.1016/j.jacc.2012.08.921
39. Green P, Woglom AE, Genereux P, et al. The impact of frailty status on survival after transcatheter aortic valve replacement in older adults with severe aortic stenosis: A single-center experience. *JACC Cardiovasc Interv*. 2012;5(9):974-981. doi:10.1016/j.jcin.2012.06.011
40. Green P, Arnold S V., Cohen DJ, et al. Relation of Frailty to Outcomes After Transcatheter Aortic Valve Replacement (from the PARTNER Trial). *Am J Cardiol*. 2015;116(2):264-269. doi:10.1016/j.amjcard.2015.03.061
41. Huded CP, Huded JM, Friedman JL, et al. Frailty Status and Outcomes After Transcatheter Aortic Valve Implantation. Published online 2016. doi:10.1016/j.amjcard.2016.03.044
42. Kobe AR, Meyer A, Elmubarak H, et al. Frailty assessed by the FORECAST is a



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

valid tool to predict short-term outcome after transcatheter aortic valve replacement. *Innov Technol Tech Cardiothorac Vasc Surg*. 2016;11(6):407-413. doi:10.1097/IMI.0000000000000321

43. Maniar H, Zajarias A, Quader N, Arnold S, Cohen D, Mack M, Melby S, Alu M, Ayele G, Lindman B LJ. THE ASSOCIATION OF FRAILTY WITH MORTALITY IN HIGH RISK PATIENTS UNDERGOING TRANSCATHETER AORTIC VALVE REPLACEMENT: AN ANALYSIS OF THE PARTNER II ALTERNATIVE ACCESS COHORTS. *J Am Coll Cardiol*. 2016;67 (13 Sup:241.
44. Okoh AK, Chauhan D, Kang N, et al. The impact of frailty status on clinical and functional outcomes after transcatheter aortic valve replacement in nonagenarians with severe aortic stenosis. *Catheter Cardiovasc Interv*. 2017;90(6):1000-1006. doi:10.1002/ccd.27083
45. Patel J, Banga S, Kim MC, Wattanakit K, Barzallo MA MS. TCT-684 Frailty Profile is Independently Associated with Higher Cost for Patients undergoing Transcatheter Aortic Valve Replacement for Symptomatic Severe Aortic Stenosis: a Single Center Experience. *J Am Coll Cardiol*. 2016;68(18 Supp:B277.
46. Rabinovitz E, Finkelstein A, Ben E, et al. Norton scale for predicting prognosis in elderly patients undergoing trans-catheter aortic valve implantation : A historical prospective study. *J Cardiol*. 2016;67(6):519-525. doi:10.1016/j.jjcc.2016.01.017
47. Rodríguez-Pascual C, Paredes-Galán E, Ferrero-Martínez AI, et al. The frailty syndrome and mortality among very old patients with symptomatic severe aortic stenosis under different treatments. *Int J Cardiol*. 2016;224:125-131. doi:10.1016/j.ijcard.2016.09.020
48. Rogers T, Alraies MC, Moussa Pacha H, et al. Clinical Frailty as an Outcome Predictor After Transcatheter Aortic Valve Implantation. *Am J Cardiol*. 2018;121(7):850-855. doi:10.1016/j.amjcard.2017.12.035
49. Schoenenberger AW, Bertschi D, Wenaweser P, et al. Improvement of Risk Prediction After Transcatheter Aortic Valve Replacement by Combining Frailty With Conventional Risk Scores. 2018;11(4). doi:10.1016/j.jcin.2017.11.012
50. Steinvil A, Buchanan KD, Kiramijyan S, et al. Utility of an additive frailty tests index score for mortality risk assessment following transcatheter aortic valve



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

replacement. *Am Heart J*. 2018;200:11-16. doi:10.1016/j.ahj.2018.01.007

51. Shi S, Afilalo J, Lipsitz LA, et al. Frailty phenotype and deficit accumulation frailty index in predicting recovery after transcatheter and surgical aortic valve replacement. *Journals Gerontol - Ser A Biol Sci Med Sci*. 2019;74(8):1249-1256. doi:10.1093/gerona/gly196
52. Skaar E, Eide LSP, Norekvål TM, et al. A novel geriatric assessment frailty score predicts 2-year mortality after transcatheter aortic valve implantation. *Eur Hear J - Qual Care Clin Outcomes*. 2019;5(2):153-160. doi:10.1093/ehjqcco/qcy044
53. Zajarias A, Maniar H, Arnold S, Quader N, Cohen D, Melby S, Alu M, Ayele G, Lindman B, Lasala J MR. The association of frailty with mortality in high risk patients undergoing transcatheter aortic valve replacement: an analysis of the partner ii b transfemoral cohort. *J Am Coll Cardiol*. 2016;67(13 Supp:249.
54. Welle GA, Mahowald M, Alqahtani F, Fatema K, Alkhouli M. Five Meter Walk Test Is an Independent Predictor of Mortality in Transcutaneous Aortic Valve Replacement. *J Am Coll Cardiol*. 2020;75(11):1439. doi:10.1016/s0735-1097(20)32066-0
55. Mach M, Watzal V, Hasan W, et al. Fitness-Tracker Assisted Frailty-Assessment Before Transcatheter Aortic Valve Implantation: Proof-of-Concept Study. *JMIR mHealth uHealth*. 2020;8(10):e19227. doi:10.2196/19227
56. Kiani S, Stebbins A, Thourani VH, et al. The Effect and Relationship of Frailty Indices on Survival After Transcatheter Aortic Valve Replacement. *JACC Cardiovasc Interv*. 2020;13(2):219-231. doi:10.1016/j.jcin.2019.08.015
57. Gharibeh S, Zweiker D, Binder J, et al. A routine nursing questionnaire adds predictive value to conventional risk scores for TAVI ( Transcatheter Aortic Valve Implantation ) outcome. *https://academic.oup.com/eurheartj/article/40/Supplement_1/ehz7450596/5594816*. 2019;(September):3743.
58. Voigtländer L, Twerenbold R, Schäfer U, et al. Prognostic Impact of Underweight (Body Mass Index <20 kg/m2) in Patients With Severe Aortic Valve Stenosis Undergoing Transcatheter Aortic Valve Implantation or Surgical Aortic Valve Replacement (from the German Aortic Valve Registry [GARY]). *Am J Cardiol*.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2020;129(1):79-86. doi:10.1016/j.amjcard.2020.05.002

59. Shimura T, Yamamoto M, Kano S, et al. Predictors and prognostic impact of nutritional changes after transcatheter aortic valve replacement. *Cardiovasc Revascularization Med*. 2020;(xxxx). doi:10.1016/j.carrev.2020.08.031
60. Malik AH, Yandrapalli S, Zaid S, et al. Impact of Frailty on Mortality, Readmissions, and Resource Utilization After TAVI. *Am J Cardiol*. 2020;127:120-127. doi:10.1016/j.amjcard.2020.03.047
61. Sami F, Ranka S, Shah A, Torres C, Villablanca P. Impact of Frailty on Outcomes in Patients Undergoing Transcatheter Aortic Valve Replacement: a Report From National Inpatient Sample. *J Am Coll Cardiol*. 2020;75(11):1487. doi:10.1016/s0735-1097(20)32114-8
62. Kundi H, Popma JJ, Reynolds MR, et al. Frailty and related outcomes in patients undergoing transcatheter valve therapies in a nationwide cohort. *Eur Heart J*. 2019;40(27):2231-2239. doi:10.1093/eurheartj/ehz187
63. Goudzwaard JA, de Ronde-Tillmans MJAG, van Hoorn FED, et al. Impact of frailty on health-related quality of life 1 year after transcatheter aortic valve implantation. *Age Ageing*. Published online 2020:1-6. doi:10.1093/ageing/afaa071
64. Goudzwaard JA, de Ronde-Tillmans MJAG, de Jager TAJ, et al. Incidence, determinants and consequences of delirium in older patients after transcatheter aortic valve implantation. *Age Ageing*. 2020;49(3):389-394. doi:10.1093/ageing/afaa001
65. Patel JN, Ahmad M, Kim M, et al. Relation of Frailty to Cost for Patients Undergoing Transcatheter Aortic Valve Implantation. *Am J Cardiol*. 2020;125(3):469-474. doi:10.1016/j.amjcard.2019.10.021
66. Drudi LM, Ades M, Asgar A, et al. Interaction Between Frailty and Access Site in Older Adults Undergoing Transcatheter Aortic Valve Replacement. *JACC Cardiovasc Interv*. 2018;11(21):2185-2192. doi:10.1016/j.jcin.2018.06.037
67. Morris J, Bruckel J, Welch T, et al. Subjective Versus Objective Frailty Assessment As a Predictor of Outcomes With Transcatheter Aortic Valve Replacement. *J Am Coll Cardiol*. 2020;75(11):1176. doi:10.1016/s0735-1097(20)31803-9



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

68. Sepehri A, Beggs T, Hassan A. The impact of frailty on outcomes after cardiac surgery : A systematic review. *J Thorac Cardiovasc Surg*. 2014;148(6):3110-3117. doi:10.1016/j.jtcvs.2014.07.087
69. Afilalo J, Lauck S, Kim DH, et al. Frailty in older adults undergoing aortic valve replacement: the FRAILTY-AVR study. *J Am Coll Cardiol*. 2017;70(6):689-700. doi:10.1016/j.jacc.2017.06.024
70. Hippel PT Von. The heterogeneity statistic I 2 can be biased in small meta-analyses. Published online 2015:1-8. doi:10.1186/s12874-015-0024-z
71. Bouillon K, Kivimaki M, Hamer M, et al. Measures of frailty in population-based studies: An overview. *BMC Geriatr*. 2013;13(1). doi:10.1186/1471-2318-13-64
72. Frost R, Belk C, Jovicic A, et al. Health promotion interventions for community-dwelling older people with mild or pre-frailty : a systematic review and meta-analysis. *BMC Geriatr*. 2017;1:157. doi:10.1186/s12877-017-0547-8
73. Sündermann S, Dademasch A, Praetorius J, Kempfert J, Dewey T, Falk V, Mohr FW WT. Comprehensive assessment of frailty for elderly high-risk patients undergoing cardiac surgery. *Eur J Cardio-thoracic Surg*. 2011;39:33-37. doi:10.1016/j.ejcts.2010.04.013
74. Vahanian A, Otto CM. Risk stratification of patients with aortic stenosis. *Eur Heart J*. 2010;31(4):416-423. doi:10.1093/eurheartj/ehp575
75. Jung P, Pereira MA, Hiebert B, et al. The impact of frailty on postoperative delirium in cardiac surgery patients. *J Thorac Cardiovasc Surg*. 2015;149(3):869-875.e2. doi:10.1016/j.jtcvs.2014.10.118

## Figure legends

**Figure 1. PRISMA flow diagram of included studies**

PRISMA, reporting in systematic reviews and meta-analyses. N, number of studies.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Figure 2a. Meta-analysis of 30-day mortality in frail patients after TAVI.**

Frailty was measured using sing and multi-dimensional measures.

The squares indicate the 30-day mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 30-day mortality

**Figure 2b. Funnel plots, using data from all studies that reported 30-day mortality**

y-axis is the standard error of the 30-day mortality

x-axis is the logit of 30-day mortality

**Figure 2c. Meta-analysis of 30-day mortality in frail patients after TAVI.**

Frailty was measured using modified Fried frailty phenotype.

The squares indicate the 30-day mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 30-day mortality



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

**Figure 2d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.**

y-axis is the standard error of the 30-day mortality

x-axis is the logit of 30-day mortality

**Figure 3a. Meta-analysis of 1-year mortality in frail patients after TAVI.**

Frailty was measured using single and multi-dimensional measures.

The squares indicate the 1-year mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 1-year mortality

**Figure 3b. Funnel plots, using data from all studies that reported 30-day mortality**

y-axis is the standard error of the 1-year mortality

x-axis is the logit of 1-year mortality

**Figure 3c. Meta-analysis of 1-year mortality in frail patients after TAVI.**



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Frailty was measured using the Fried frailty phenotype.

The squares indicate the 1-year mortality reported by each study

The horizontal lines indicate the magnitude of the confidence interval

The diamond indicates the pooled estimate for 1-year mortality

**Figure 3d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.**

y-axis is the standard error of the 1-year mortality

x-axis is the logit of 1-year mortality

Table 1. Frailty assessment in patients undergoing TAVI

| Studies that used a single dimension to assess frailty | | | | | |
|---|---|---|---|---|---|
| **Study, year** | **Measure** | **Dimensions** | **Definition** | **Total N** | **Frail n (%)** |
| Alfredsson, 2016 [19 a] | Gait speed (5 meters) | mobility | <0.83m/s or >6 s | 8,039 | 6,100 (75.88%) |
| Bagienski, 2017 [20 b] | Katz ADL | disability | <6 points | 141 | 127 (90.07%) |
| Bogdan, 2016 [21] | Albumin | nutrition | ≤4g/dl | 150 | 79 (52.67%) |
| Cockburn, 2015 [22] | Brighton mobility index | mobility | Poor mobility | 312 | 65 (20.83%) |
| Grossman, 2017 [23] | Albumin | nutrition | <4 g/dl | 426 | 192 (45.07%) |
| Koifman, 2015 [24 c] | Albumin | nutrition | <3.5g/dl | 476 | 238 (50%) |
| Kleczynski, 2017 [25] | ISAR | unclear | ≥2 points | 101 | 53 (52.48%) |
| Mok, 2016 [26] | Sarcopenia | nutrition | skeletal muscle mass index 2 SDs less than the mean SMM of young, healthy gender-specific reference ranges | 460 | 293 (63.70%) |
| Martin, 2018 [27] | CSHA score (1-7) | physical function | Scores 5-7 | 2624 | 1043 (39.75%) |



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | |
|---|---|---|---|---|---|
| Puls, 2014 [28] | Katz ADL | disability | <6 points | 300 | 144 (48%) |
| Rodes-Cabau, 2010 [29] | Clinical judgment | subjective | Unclear | 339 | 85 (25.07%) |
| Stortecky, 2012 [30] | BMI | nutrition | <20kg/m$^2$ | 256 | 24 (9.38%) |
| Shimura, 2017 [31 d] | CFS | subjective | ≥5 points (score ranges 0-9) | 1215 | 353 (29.05%) |
| Traynor, 2017 [32] | Assisted care | unclear | Need assisted care | 597 | 60 (10.05%) |
| Yamamoto, 2015 [33] | BMI | nutrition | <20kg/m$^2$ | 777 | 56 (7.21%) |
| Welle, 2020 [54] | Gait speed (5 meters) | mobility | ≥6 s | 723 | 483 (66.8%) |
| Mach, 2020 [55] | Fitness-tracker assisted frailty score | unclear | ≥1 point | 50 | 39 (78%) |
| Kiani, 2020 [56 a] | Gait speed (5 meters) | mobility | <0.83m/s or >6 s (including unable to perform the test) | 56,500 | 11,316 (20.03%) |
| Gharibeh, 2019 [57] | Clinical judgment | subjective | Indicators for limited self-dependence | 461 | 186 (40.35%) |
| Voigtlander, 2020 [58] | BMI | nutrition | <20kg/m$^2$ | 16,865 | 956 (5.67%) |
| Shimura, 2020 [59 d] | Albumin | nutrition | <3.5 g/dl | 1,524 | 284 (18.64%) |

**Studies that used administrative database algorithms to assess frailty**

| Study, year | Measure | Definition/ cut-off points | Total N | Frail n (%) |
|---|---|---|---|---|
| Malik, 2020 [60] | Hospital Frailty Risk Score | Hospital Frailty Risk Score ≥ 5 points | 20,504 | 8,419 (41.06%) |
| Sami, 2020 [61] | Johns-Hopkins ACG frailty indicator | A dichotomous indicator defined based on 10 clusters of frailty-defining diagnoses | 51,685 | 2,865 (5.54%) |
| Kundi, 2019 [62] | Hospital Frailty Risk Score | Hospital Frailty Risk Score ≥ 5 points | 28,531 | 13,593 (47.64%) |

**Studies that used multiple dimensions to assess frailty**

| Study, year | Name | Measures | Dimensions | Definition | Total N | Frail n (%) |
|---|---|---|---|---|---|---|
| Bureau, 2017 [34] | MPI | ADL | disability | MPI≥0.34 (the sum of all domain values is divided by 8 to obtain the MPI score between 0 and 1) | 116 | 71 (61.21%) |
| | | IADL | disability | | | |
| | | SPMSQ | cognition | | | |
| | | CIRS-CI | medical | | | |
| | | MNA-SF | nutrition | | | |
| | | ESS | medical | | | |
| | | Number of medications | medical | | | |
| | | Social support network | living status | | | |
| Chauhan, 2016 [35 e] | Modified Fried phenotype | ADL | disability | Presence of 2 or more criteria | 343 | 233 (67.93%) |
| | | Hand strength | muscle strength | | | |
| | | Gait speed | mobility | | | |
| | | Albumin | nutrition | | | |
| Capodanno, 2014 [36] | GSS | Not reported | not reported | Value of 2 or 3 | 1256 | 306 (24.36%) |
| Eichler, 2017 [37] | FI | MMSE | cognition | ≥3 points (score ranges 0-7) | 333 | 152 (45.65%) |
| | | MNA | nutrition | | | |
| | | ADL | disability | | | |
| | | IADL | disability | | | |
| | | Time up and go test | mobility | | | |
| | | Subjective mobility disability | mobility | | | |
| Ghatak, 2012 [38] | Modified Fried phenotype | Albumin | nutrition | Presence of 3 or more criteria | 45 | 22 (48.89%) |
| | | Katz ADL | disability | | | |
| | | 5MWT | mobility | | | |
| | | Grip strength | muscle strength | | | |
| Green, 2015 [39] | Modified Fried phenotype | Gait speed | mobility | Frailty score ≥6 | 244 | 110 (45.08%) |
| | | Grip strength | muscle strength | | | |
| | | Albumin | nutrition | | | |
| | | ADL | disability | | | |
| Green, 2012 [40] | Modified Fried phenotype | Gait speed | mobility | Frailty score ≥5 points | 159 | 76 (47.80%) |
| | | Grip strength | muscle strength | | | |
| | | Albumin | nutrition | | | |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | | |
|---|---|---|---|---|---|---|
| | | ADL | disability | | | |
| Huded, 2016 [41] | Modified Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 191 | 64 (33.51%) |
| | | Grip strength | muscle strength | | | |
| | | 5MWT | mobility | | | |
| | | Katz ADL | disability | | | |
| Kobe, 2016 [42] | FORCAST | Chair rise | muscle strength | ≥ 4 points (score ranges 0-12) | 130 | 71 (54.62%) |
| | | Weakness | muscle strength | | | |
| | | Stair | mobility | | | |
| | | CFS | subjective | | | |
| | | Creatinine level | medical | | | |
| Maniar, 2016 [43] | Modified Fried phenotype | Serum albumin | nutrition | ≥6 points (score ranges 0-12) | 219 | 73 (33.3%) |
| | | Gait speed | mobility | | | |
| | | Grip strength | muscle strength | | | |
| | | Katz ADL | disability | | | |
| Okoh, 2017 [44 e] | Modified Fried phenotype | Hand grip strength | muscle strength | FI ≥3/4 | 75 | 30 (40%) |
| | | Gait speed | mobility | | | |
| | | Serum albumin | nutrition | | | |
| | | ADL | disability | | | |
| Patel, 2016 [45] | NA | Gait speed | mobility | Gait speed≥6s or/and albumin<3.5g/dl | 117 | 31 (26.50%) |
| | | Albumin | nutrition | | | |
| Rabinovitz, 2016 [46] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 302 | 46 (15.23%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | physical activity | | | |
| Rodriguez-Pascual, 2016 [47] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 109 | 68 (62.39%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | physical activity | | | |
| Rogers, 2018 [48] | Fried phenotype | Unintentional weight loss | nutrition | Presence of 3 or more criteria | 544 | 242 (44.49%) |
| | | Exhaustion | exhaustion | | | |
| | | Weakness | muscle strength | | | |
| | | Walk speed | mobility | | | |
| | | Low physical activity | disability | | | |
| Schoenenberger, 2018 [49] | NA | MMSE | cognition | ≥3 points (score ranges 0-7) | 330 | 169 (51.21%) |
| | | Time up and go | mobility | | | |
| | | MNA | nutrition | | | |
| | | Basic ADL | disability | | | |
| | | Incremental ADL | disability | | | |
| Steinvil, 2018 [50,] | NA | BMI | nutrition | Presence of 3 or more criteria | 498 | 232 (46.59%) |
| | | Albumin | nutrition | | | |
| | | Katz ADL | disability | | | |
| | | Grip strength | muscle strength | | | |
| | | Walk test | mobility | | | |
| Shi, 2018 [51] | Fried phenotype | Weight loss | nutrition | Presence of 3 or more criteria | 137 | 116 (84.67%) |
| | | Exhaustion | Exhaustion | | | |
| | | Minnesota leisure time activity | Physical activity | | | |
| | | 5m walk test | mobility | | | |
| | | Grip strength | muscle strength | | | |
| Skaar, 2018 [52] | Geriatric assessment tool (0-9) | MMSE | cognition | Scores≥4 | 142 | 34 (23.94%) |
| | | Nottingham extended ADL | disability | | | |



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | | |
|---|---|---|---|---|---|---|
| | | BMI<20.5 | nutrition | | | |
| | | Low energy | exhaustion | | | |
| | | Weight loss | nutrition | | | |
| | | Chair stand | muscle strength | | | |
| | | Charlson comorbidity index | comorbidity | | | |
| | | Hospital anxiety and depression scale | psychological | | | |
| Zajarias, 2016 [53] | Modified Fried phenotype | Albumin | nutrition | ≥6 points (score ranges 0-12) | 553 | 265 (47.92%) |
| | | Gait speed | mobility | | | |
| | | Grip strength | muscle strength | | | |
| | | Katz ADL | disability | | | |
| Goudzwaard, 2020 [63] | Erasmus Frailty Score | MMSE | cognition | Presence of 3 or more criteria | 330 | 97 (29.50%) |
| | | Hand grip test | Muscle strength | | | |
| | | Malnutrition universal screening tool | Nutrition | | | |
| | | Katz ADL | Inactivity in basic activities of daily living | | | |
| | | Lawton and Brody index | Inactivity in instrumental activities of daily living | | | |
| Goudzwaard, 2020 [64] | Erasmus Frailty Score | MMSE | cognition | Presence of 3 or more criteria | 239 | 70 (29.3%) |
| | | Hand grip test | Muscle strength | | | |
| | | Malnutrition universal screening tool | Nutrition | | | |
| | | Katz ADL | Inactivity in basic activities of daily living | | | |
| | | Lawton and Brody index | Inactivity in instrumental activities of daily living | | | |
| Patel, 2020 [65] | A composite of two frailty markers | Gait speed | mobility | Presence of both criteria | 407 | 74 (18.18%) |
| | | Serum albumin | nutrition | | | |
| Drudi, 2018 [66] | Essential frailty toolset | Muscle weakness | Muscle strength | ≥3 scores (out of 5) | 723 | 254 (35.13%) |
| | | Cognitive impairment | Cognition | | | |
| | | Anemia | Nutrition | | | |
| | | Hypoalbuminemia | Nutrition | | | |
| Morris, 2020 [67] | Essential frailty toolset | Muscle weakness | Muscle strength | ≥3 scores (out of 5) | 559 | 234 (41.86%) |
| | | Cognitive impairment | Cognition | | | |
| | | Anemia | Nutrition | | | |
| | | Hypoalbuminemia | Nutrition | | | |

a. Alfredsson (2016) and Kiani (2020) enrolled patient populations from the STS/ACC registry. Chauhan (2016), Green (2012), Green (2015), Huded (2016), Okoh (2017), Rogers (2018), Steinvil (2018), Traynor (2017), and Bagienski (2017) enrolled patients from the participating centres of STS/ACC registry.

b. Bagienski (2017) and Kleczynski (2017) enrolled patients from the same medical centre but used different frailty definitions.

c. Koifman (2015), Rogers (2018), and Steinvil (2018) enrolled patients from the same medical centre but used different frailty definitions.

d. Shimura (2020) and Shimura (2017) enrolled patients from the same registry but used different frailty definitions.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

e. Chauhan (2016) and Okoh (2017) enrolled patients from the same medical centre but used different frailty definitions.

N, the number of patients. SMM, skeletal muscle mass. 5MWT, 5-meter walk test. ADL, activities of daily living. MPI, multidimensional prognostic index. SPMSQ, short portable mental status questionnaire. CIRS-CI, cumulative illness rating scale comorbidity index. MNA-SF, mini-nutritional assessment short form. ESS, Exton Smith scale. FI, frailty index. GSS, geriatric status scale. MMSE, mini-mental state examination. BMI, body mass index. ISAR, identification of seniors at risk. FORCAST, frailty predicts death one year after elective cardiac surgery test. CFS, clinical frailty scale. NA, not applicable.

For peer review only

Table 2. Results of meta-analysis and GRADE assessment**



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| Effects | | | | | GRADE assessment | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| # included study | Frailty measures* | # individuals | # events | Estimate (95% CI) | Study design | Risk of bias | Inconsistency | Indirectness | Imprecision | Other considerations | Certainty |
| Procedural death | | | | | | | | | | | |
| 6 | All | 9586 | 654 | 7.60% (4.41%-12.79%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 30-day mortality | | | | | | | | | | | |
| 13 | All | 23628 | 1236 | 7.32% (5.66%-9.42%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 8 | Multi | 1352 | 113 | 8.58% (7.18%-10.22%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| 3 | Modified Fried | 407 | 31 | 7.86% (5.20%-11.70%) | Observational | Serious | Not serious | Strongly serious | Serious | None | Very low |
| Cardiovascular death at 30 days | | | | | | | | | | | |
| 2 | Single | 6453 | 259 | 3.37% (1.93%-5.81%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| 6-month mortality | | | | | | | | | | | |
| 2 | Multi | 187 | 30 | 16.12% (11.50%-22.13%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| 1-year mortality | | | | | | | | | | | |
| 10 | All | 15471 | 3151 | 23.98% (20.71%-27.58%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 6 | Multi | 845 | 191 | 22.75% (20.03%-25.71%) | Observational | Serious | Serious | Strongly serious | Serious | None | Very low |
| 2 | Fried and modified Fried | 223 | 60 | 26.91% (21.50%-33.11%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Survival | | | | | | | | | | | |
| 17 | All | 48258 | NA | 1-year survival: 75.6% (75.2%-76.0%) 2-year survival: 65.0% (63.3%-66.7%) 3-year survival: 48.7% (43.3%-54.7%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | Fried and modified Fried | 484 | NA | 1-year survival: 73% (68.8%-77.5%) 2-year survival: 64.5% (56.4%-73.9%) 3-year survival: 58.9% (49%-70.9%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Procedural acute kidney injury | | | | | | | | | | | |
| 4 | Single | 6548 | 458 | 11.34% (6.43%-19.22%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural cardiac tamponade | | | | | | | | | | | |
| 3 | Single | 553 | 17 | 3.19% (1.99%-5.07%) | Observational | Not serious | Serious | Strongly serious | Not serious | None | Very low |
| Convert to open heart surgery | | | | | | | | | | | |
| 2 | All | 4259 | 300 | 2.29% (0.49%-9.91%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural life-threatening bleeding | | | | | | | | | | | |
| 5 | All | 653 | 63 | 9.75% (7.69%-12.29%) | Observational | Not serious | Serious | Strongly serious | Not serious | None | Very low |
| Procedural major bleeding | | | | | | | | | | | |
| 5 | Single | 830 | 104 | 8.53% (3.53%-19.19%) | Observational | Not serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural minor bleeding | | | | | | | | | | | |
| 4 | Single | 774 | 147 | 18.34% (10.66%-29.73%) | Observational | Not serious | Strongly serious | Strongly serious | Serious | None | Very low |
| Procedural major vascular complications | | | | | | | | | | | |
| 3 | Single | 647 | 63 | 10.49% (4.76%-21.54%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| 30-day major vascular complications | | | | | | | | | | | |
| 2 | All | 189 | 7 | 2.97% (0.34%-21.67%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Procedural minor vascular complications | | | | | | | | | | | |
| 2 | Single | 591 | 43 | 7.37% (3.24%-15.93%) | Observational | Serious | Strongly serious | Strongly serious | Not serious | None | Very low |
| Procedural major access-site complications | | | | | | | | | | | |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 3 | Single | 148 | 15 | 9.44% (4.04%-20.51%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| Procedural permanent pacemaker | | | | | | | | | | | |
| 7 | All | 3660 | 365 | 8.12% (5.79%-11.26%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Readmission within 30 days | | | | | | | | | | | |
| 3 | Multi | 248 | 27 | 10.37% (3.75%-25.59%) | Observational | Serious | Strongly serious | Strongly serious | Strongly serious | None | Very low |
| Procedural stroke | | | | | | | | | | | |
| 8 | All | 1756 | 39 | 2.94% (1.76%-4.88%) | Observational | Strongly serious | Serious | Strongly serious | Not serious | None | Very low |
| Stroke within 30 days | | | | | | | | | | | |
| 2 | Single | 6185 | 132 | 2.14% (1.81%-2.53%) | Observational | Serious | Serious | Strongly serious | Not serious | None | Very low |
| Transfusion | | | | | | | | | | | |
| 3 | All | 458 | 191 | 41.01% (34.02%-48.39%) | Observational | Serious | Serious | Strongly serious | Strongly serious | None | Very low |
| 2-valve implantation | | | | | | | | | | | |
| 2 | Single | 409 | 10 | 2.46% (1.33%-4.51%) | Observational | Not serious | Serious | Strongly serious | Not serious | None | Very low |
| Length of hospitalization | | | | | | | | | | | |
| 6 | All | 308 | NA | 8.25 (6.62-10.27) | Observational | Strongly serious | Strongly serious | Strongly serious | Strongly serious | None | Very low |

**Meta-analyses conducted using random effects model

*Frailty measures are categorized as single, multi-measures, administrative data based, Fried, modified Fried and all.

Single indicates single measures

Multi indicates multi-measures

Fried indicates the Fried phenotype

Modified Fried indicates the modified Fried phenotype

Fried and modified Fried includes the Fried phenotype and modified Fried phenotype

All includes all single- and multi-measures

CI indicates confidence interval.

GRADE indicates Grading of Recommendations Assessment, Development and Evaluation

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

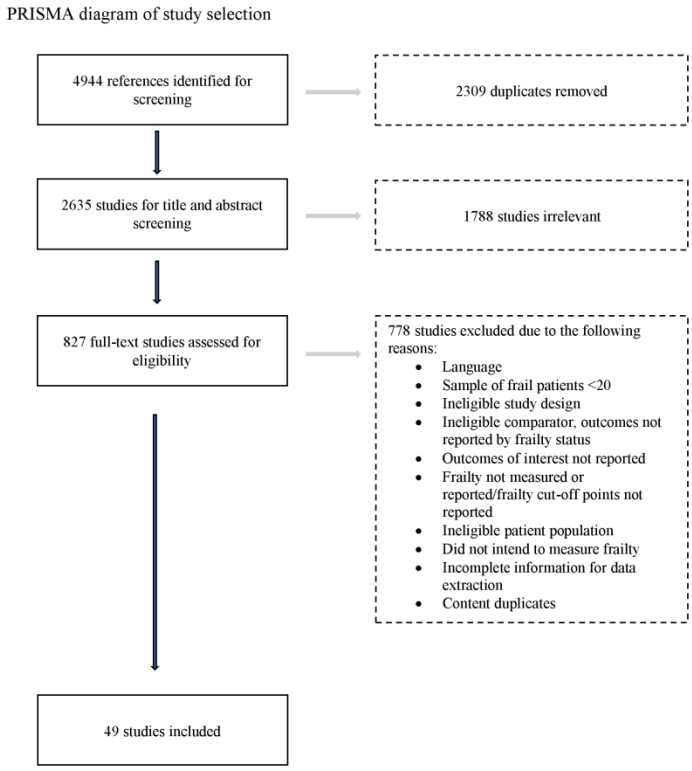



Figure 1. PRISMA flow diagram of included studies
PRISMA, reporting in systematic reviews and meta-analyses. N, number of studies.

215x279mm (300 x 300 DPI)

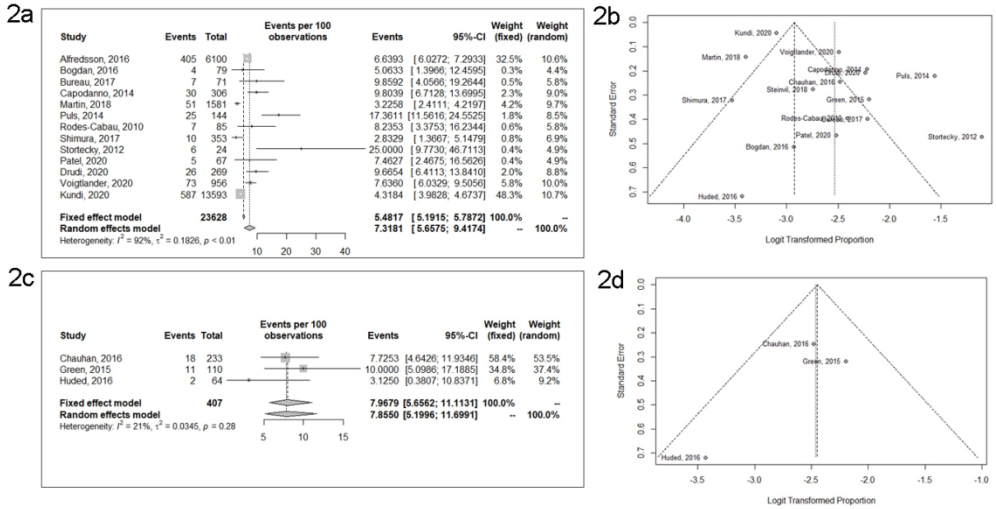



Figure 2a. Meta-analysis of 30-day mortality in frail patients after TAVI.
Frailty was measured using sing and multi-dimensional measures.
The squares indicate the 30-day mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 30-day mortality

Figure 2b. Funnel plots, using data from all studies that reported 30-day mortality
y-axis is the standard error of the 30-day mortality
x-axis is the logit of 30-day mortality

Figure 2c. Meta-analysis of 30-day mortality in frail patients after TAVI.
Frailty was measured using modified Fried frailty phenotype.
The squares indicate the 30-day mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 30-day mortality

Figure 2d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.
y-axis is the standard error of the 30-day mortality
x-axis is the logit of 30-day mortality

480x250mm (72 x 72 DPI)

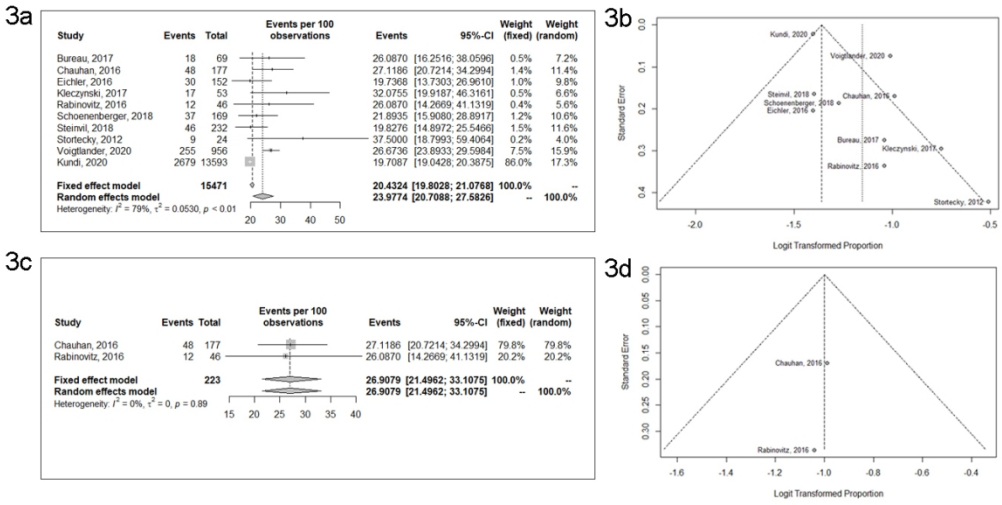



Figure 3a. Meta-analysis of 1-year mortality in frail patients after TAVI.
Frailty was measured using single and multi-dimensional measures.
The squares indicate the 1-year mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 1-year mortality

Figure 3b. Funnel plots, using data from all studies that reported 30-day mortality
y-axis is the standard error of the 1-year mortality
x-axis is the logit of 1-year mortality

Figure 3c. Meta-analysis of 1-year mortality in frail patients after TAVI.
Frailty was measured using the Fried frailty phenotype.
The squares indicate the 1-year mortality reported by each study
The horizontal lines indicate the magnitude of the confidence interval
The diamond indicates the pooled estimate for 1-year mortality

Figure 3d. Funnel plots, using data from studies that frailty was measured using modified Fried frailty phenotype.
y-axis is the standard error of the 1-year mortality
x-axis is the logit of 1-year mortality

469x250mm (72 x 72 DPI)

**Supplementary Appendix A Literature Search**

**Search: TAVI in Frail Patients with Aortic Stenosis**

Database: Embase Classic+Embase 1947 to September 23, 2020

| # | Searches |
|---|---|
| 1 | frailty/ |
| 2 | frail elderly/ |
| 3 | geriatric assessment/ |
| 4 | frail*.mp. |
| 5 | fragile.mp. |
| 6 | or/1-5 |
| 7 | geriatric patient/ |
| 8 | very elderly/ |
| 9 | aged/ |
| 10 | aged hospital patient/ |
| 11 | geriatrician/ |
| 12 | *geriatrics/ |
| 13 | (aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*).mp. |
| 14 | or/7-13 |
| 15 | exhaustion/ |
| 16 | limited mobility/ |
| 17 | "timed up and go test"/ |
| 18 | exp walk test/ |
| 19 | walking speed/ |
| 20 | gait/ |
| 21 | physical activity/ |
| 22 | Performance Oriented Mobility Assessment/ |
| 23 | grip strength/ |
| 24 | hand strength/ |
| 25 | muscle strength/ |
| 26 | daily life activity/ |

For peer review only

| | |
|---|---|
| 27 | exp "activity of daily living assessment"/ |
| 28 | exp ADL disability/ |
| 29 | exp disability/ |
| 30 | exp functional status assessment/ |
| 31 | exp neuropsychological test/ |
| 32 | mental function assessment/ |
| 33 | cognition assessment/ |
| 34 | exp cognition/ |
| 35 | exp cognitive defect/ |
| 36 | memory assessment/ |
| 37 | nutritional assessment/ |
| 38 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) adj7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)).mp. |
| 39 | (chair adj2 (rise or stand)).mp. |
| 40 | ((grip* or grasp* or hand* or musc*) adj2 strength).mp. |
| 41 | (tug adj3 test).mp. |
| 42 | (timed* adj6 (up* or go)).mp. |
| 43 | timedupandgo.mp. |
| 44 | weight loss.mp. |
| 45 | (activit* adj3 (living or life)).mp. |
| 46 | (adl or iadl or badl).mp. |
| 47 | or/15-46 |
| 48 | 14 and 47 |
| 49 | 6 or 48 |
| 50 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) adj3 aortic valv*).mp. |
| 51 | transcatheter aortic valve implantation/ |

For peer review only

| | |
|---|---|
| 52 | or/50-51 |
| 53 | 49 and 52 |
| 54 | limit 53 to yr="2006 -Current" |

For peer review only

Database: Ovid MEDLINE(R) ALL 1946 to September 23, 2020

| # | Searches |
|---|---|
| 1 | Frailty/ |
| 2 | Frail Elderly/ |
| 3 | Geriatric Assessment/ |
| 4 | frail*.mp. |
| 5 | fragile.mp. |
| 6 | or/1-5 |
| 7 | "Aged, 80 and over"/ |
| 8 | Aged/ |
| 9 | Geriatricians/ |
| 10 | *Geriatrics/ |
| 11 | (aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*).mp. |
| 12 | or/7-11 |
| 13 | Physical Exertion/ |
| 14 | Physical Endurance/ |
| 15 | Walk Test/ |
| 16 | Exercise Test/ |
| 17 | Exercise Tolerance/ |
| 18 | Walking Speed/ |
| 19 | Gait/ |
| 20 | Exercise/ |
| 21 | mobility limitation/ |
| 22 | Hand Strength/ |
| 23 | Muscle Strength/ |
| 24 | exp "Activities of Daily Living"/ |
| 25 | "Quality of Life"/ |
| 26 | exp Neuropsychological Tests/ |
| 27 | exp Cognition/ |
| 28 | Cognitive Dysfunction/ |

For peer review only

| | |
|---|---|
| 29 | Cognition Disorders/ |
| 30 | Nutritional Assessment/ |
| 31 | Nutritional Status/ |
| 32 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) adj7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)).mp. |
| 33 | (chair adj2 (rise or stand)).mp. |
| 34 | ((grip* or grasp* or hand* or musc*) adj2 strength).mp. |
| 35 | (tug adj3 test).mp. |
| 36 | (timed* adj6 (up* or go)).mp. |
| 37 | timedupandgo.mp. |
| 38 | weight loss.mp. |
| 39 | (activit* adj3 (living or life)).mp. |
| 40 | (adl or iadl or badl).mp. |
| 41 | or/13-40 |
| 42 | 12 and 41 |
| 43 | 6 or 42 |
| 44 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) adj3 aortic valv*).mp. |
| 45 | Transcatheter Aortic Valve Replacement/ |
| 46 | or/44-45 |
| 47 | 43 and 46 |
| 48 | limit 47 to yr="2006 -Current" |

For peer review only

Database: Cochrane Central Register of Controlled Trials

| # | Searches |
|---|---|
| 1 | MeSH descriptor: [Frailty] this term only |
| 2 | MeSH descriptor: [Frail Elderly] this term only |
| 3 | MeSH descriptor: [Geriatric Assessment] this term only |
| 4 | frail*:ti,ab,kw (Word variations have been searched) |
| 5 | fragile:ti,ab,kw (Word variations have been searched) |
| 6 | {OR #1-#5} |
| 7 | MeSH descriptor: [Aged, 80 and over] this term only |
| 8 | MeSH descriptor: [Aged] this term only |
| 9 | MeSH descriptor: [Geriatrics] this term only |
| 10 | aged or aging or older or elderly or senior* or geriatric* or centenarian* or nonagenarian* or octogenarian* or septuagenarian* or sexagenarian*:ti,ab,kw (Word variations have been searched) |
| 11 | {or #7-#10} |
| 12 | MeSH descriptor: [Mobility Limitation] this term only |
| 13 | MeSH descriptor: [Physical Exertion] this term only |
| 14 | MeSH descriptor: [Physical Endurance] this term only |
| 15 | MeSH descriptor: [Walk Test] this term only |
| 16 | MeSH descriptor: [Exercise Test] this term only |
| 17 | MeSH descriptor: [Exercise Tolerance] this term only |
| 18 | MeSH descriptor: [Walking Speed] this term only |
| 19 | MeSH descriptor: [Gait] this term only |
| 20 | MeSH descriptor: [Exercise] this term only |
| 21 | MeSH descriptor: [Hand Strength] this term only |
| 22 | MeSH descriptor: [Muscle Strength] this term only |
| 23 | MeSH descriptor: [Activities of Daily Living] explode all trees |
| 24 | MeSH descriptor: [Quality of Life] this term only |
| 25 | MeSH descriptor: [Neuropsychological Tests] explode all trees |
| 26 | MeSH descriptor: [Cognition] explode all trees |
| 27 | MeSH descriptor: [Cognition Disorders] this term only |

For peer review only

| | |
|---|---|
| 28 | MeSH descriptor: [Cognitive Dysfunction] this term only |
| 29 | MeSH descriptor: [Nutrition Assessment] this term only |
| 30 | MeSH descriptor: [Nutritional Status] this term only |
| 31 | ((exhaustion or fatigue* or tired* or mobility or gait or walk* or stand or balance or bath* or dress* or toilet* or continence or feeding or cognition or memory or mental or disability or NYHA or Karnofsky or CSHA or functional* or Katz or Fried or Rockwood or frailty or nutrition*) next/7 (assess* or phenotype* or eval* or test* or exam* or instrument* or index or indices or scale* or score* or tool* or declin* or dependenc* or impair*)):ti,ab,kw (Word variations have been searched) |
| 32 | (chair next/2 (rise or stand)):ti,ab,kw (Word variations have been searched) |
| 33 | ((grip* or grasp* or hand* or musc*) next/2 strength):ti,ab,kw (Word variations have been searched) |
| 34 | (tug next/3 test):ti,ab,kw (Word variations have been searched) |
| 35 | (timed* next/6 (up* or go)):ti,ab,kw (Word variations have been searched) |
| 36 | "weight loss":ti,ab,kw (Word variations have been searched) |
| 37 | (activit* next/3 (living or life)):ti,ab,kw (Word variations have been searched) |
| 38 | adl or iadl or badl:ti,ab,kw (Word variations have been searched) |
| 39 | {or #12-#38} |
| 40 | #11 and #39 |
| 41 | #6 or #40 |
| 42 | ((transfemoral* or trans-femoral* or transapical* or trans-apical* or transaxillary or trans-axillary or transarterial* or trans-arterial* or subclavian* or sub-clavian* or transcatheter* or trans-catheter* or transcutaneous* or trans-cutaneous* or per-cutaneous* or percutaneous* or transcaval* or trans-caval* or "direct aortic" or tavi or tavr or pavi or pavr or sapien or cribier or revalv* or lotus or "direct flow" or jenavalve or portico or engager or evolut) next/3 aortic valv*):ti,ab,kw (Word variations have been searched) |
| 43 | MeSH descriptor: [Transcatheter Aortic Valve Replacement] this term only |
| 44 | #42 or #43 |
| 45 | #41 and #44 |

For peer review only

Database: PsycINFO (ProQuest) - updated weekly

| # | Searches |
|---|---|
| 1 | MAINSUBJECT.EXACT("Health Impairments" OR "Geriatric Assessment") |
| 2 | AB,TI,SU(frail* OR fragile) |
| 3 | 1 OR 2 |
| 4 | MAINSUBJECT.EXACT("Geriatric Patients" OR "Geriatrics" OR "Gerontology") |
| 5 | AB,TI,SU(aged OR aging OR older OR elderly OR senior* OR geriatric* OR centenarian* OR nonagenarian* OR octogenarian* OR septuagenarian* OR sexagenarian*) |
| 6 | 4 OR 5 |
| 7 | MAINSUBJECT.EXACT("Fatigue" OR "Physical Mobility" OR "Locomotion" OR "Physical Agility" OR "Physical Activity" OR "Physical Endurance" OR "Physical Fitness" OR "Physical Strength" OR "Exercise" OR "Ability Level" OR "Activity Level" OR "Walking" OR "Gait" OR "Activities of Daily Living" OR "Daily Activities" OR "Quality of Life" OR "Neuropsychological Assessment" OR "Cognition" OR "Cognitive Impairment" OR "Memory Disorders" OR "Nutritional Deficiencies") |
| 8 | AB,TI,SU(exhaustion OR fatigue* OR tired* OR mobility OR gait OR walk* OR stand OR balance OR bath* OR dress* OR toilet* OR continence OR feeding OR cognition OR memory OR mental OR disability OR NYHA OR Karnofsky OR CSHA OR functional* OR Katz OR Fried OR Rockwood OR frailty OR nutrition*) AND AB,TI,SU(assess* OR phenotype* OR eval* OR test* OR exam* OR instrument* OR index OR indices OR scale* OR score* OR tool* OR declin* OR dependenc* OR impair*) |
| 9 | chair NEAR/2 (rise OR stand) |
| 10 | strength NEAR/2 (grip* OR grasp* OR hand* OR musc*) |
| 11 | tug NEAR/3 test |
| 12 | timed* NEAR/6 (up* OR go) |
| 13 | activit* NEAR/3 (living OR life) |
| 14 | AB,TI,SU(timedupandgo OR weight loss OR adl OR iadl OR badl) |
| 15 | 7 OR 8 OR 9 OR 10 OR 11 OR 12 OR 13 OR 14 |
| 16 | 6 AND 15 |
| 17 | 3 OR 16 |
| 18 | (transfemoral* OR trans-femoral* OR transapical* OR trans-apical* OR transaxillary OR trans-axillary OR transarterial* OR trans-arterial* OR subclavian* OR sub-clavian* OR transcatheter* OR trans-catheter* OR transcutaneous* OR trans-cutaneous* OR per-cutaneous* OR percutaneous* OR transcaval* OR trans-caval* OR "direct aortic" OR tavi OR tavr OR pavi OR pavr OR sapien OR cribier OR revalv* OR lotus OR "direct flow" OR jenavalve OR portico OR engager OR evolut) AND (aortic valv*) |
| 19 | TAVI |

For peer review only

| | |
|---|---|
| 20 | 18 OR 19 |
| 21 | 17 AND 20 |

For peer review only

Database: Web of Science Core Collection (Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, ESCI Timespan=All years) updated daily

| # | Searches |
|---|---|
| 1 | TS=(frail*) |
| 2 | TS=(fragile) |
| 3 | #1 OR #2 |
| 4 | TS=(aged OR aging OR older OR elderly OR senior* OR geriatric* OR centenarian* OR nonagenarian* OR octogenarian* OR septuagenarian* OR sexagenarian*) |
| 5 | TS=((exhaustion OR fatigue* OR tired* OR mobility OR gait OR walk* OR stand OR balance OR bath* OR dress* OR toilet* OR continence OR feeding OR cognition OR memory OR mental OR disability OR NYHA OR Karnofsky OR CSHA OR functional* OR Katz OR Fried OR Rockwood OR frailty OR nutrition*) NEAR/7 (assess* OR phenotype* OR eval* OR test* OR exam* OR instrument* OR index OR indices OR scale* OR score* OR tool* OR declin* OR dependenc* OR impair*)) |
| 6 | TS=(chair NEAR/2 (rise OR stand)) |
| 7 | TS=((grip* OR grasp* OR hand* OR musc*) NEAR/2 strength) |
| 8 | TS=(tug NEAR/3 test) |
| 9 | TS=(timed* NEAR/6 (up OR go)) |
| 10 | TS=(weight loss) |
| 11 | TS=(activit* NEAR/3 (living OR life)) |
| 12 | TS=(ADL OR IADL OR BADL) |
| 13 | #12 OR #11 OR #10 OR #9 OR #8 OR #7 OR #6 OR #5 |
| 14 | #13 AND #4 |
| 15 | #14 OR #3 |
| 16 | TS=((transfemoral* OR trans-femoral* OR transapical* OR trans-apical* OR transaxillary OR trans-axillary OR transarterial* OR trans-arterial* OR subclavian* OR sub-clavian* OR transcatheter* OR trans-catheter* OR transcutaneous* OR trans-cutaneous* OR per-cutaneous* OR percutaneous* OR transcaval* OR trans-caval* OR "direct aortic" OR tavi OR tavr OR pavi OR pavr OR sapien OR cribier OR revalv* OR lotus OR "direct flow" OR jenavalve OR portico OR engager OR evolut) NEAR/3 aortic valv*) |
| 17 | #16 AND #15 |

For peer review only

Supplementary appendix B. Baseline demographics and population characteristics

| Study | Recruitment period | N (frail) | Age (mean±SD) | Female (n,%) | NYHA III/IV (n,%) | STS (mean±SD) | Logistic Euro score (mean±SD) | TF-approach (n, %) | TA-approach (n,%) | AVA, $cm^2$ (mean±SD) | Creatinine (mg/dL) (mean±SD) | Haemoglobin, g/dL (mean±SD) | LVEF (mean±SD) | Aortic gradient, mmHg (mean±SD) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | 2011-2014 | 6100 | NR | 3421 (56.08%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bureau (2017) | 2013-2015 | 71 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | 127 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bogdan (2016) | 2009-2012 | 79 | 82±5 | 43 (55%) | 46 (58.2%) | 5.8±3.12 | 19.44±14.98 | NR | NR | 0.66±0.14 | 1.35±0.93 | 11.27±1.58 | 53.22±12.64 | 43.54±15.19 |
| Capodanno (2014) | 2010-2012 | 306 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | 2012-2015 | 233 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2016) | 2008-2014 | 65 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Eichler (2017) | 2013-2015 | 152 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Ghatak (2012) | NR | 22 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Green (2015) | PARTNER[1] | 110 | 87.1** (82.7-90.3) | 58 (53%) | NR | 11.3** (9.6-13.8) | NR | 57 (52%) | NR | 0.62** (0.51-0.72) | NR | NR | 55** (35-60) | 45.2** (34.9-59.7) |
| Green (2012) | PARTNER[1] | 76 | 87.1±6.6 | 40 (53%) | NR | 11.9±4 | NR | NR | 30 (39%) | 0.6±0.2 | NR | 11.1±1.6 | 47±15 | 45±15 |
| Grossman (2017) | 2008-2014 | 192 | 84.3±7.5 | 109 (56.8) | 122 (66%) | 6.2±4.5 | 20.8±16 | 132 (73%) | 42 (23%) | NR | NR | NR | NR | NR |
| Huded (2016) | 2012-2015 | 64 | 83.1±7.5 | 42 (66%) | 53 (90%) | 6.8±2.8 | NR | 57 (89%) | NR | NR | 1.19±0.45 | 11.8±1.8 | NR | NR |
| Koifman (2015) | 2007-2014 | 238 | 84±8 | 81 (21%) | 210 (94%) | 10±4.7 | NR | 181 (76%) | NR | 0.64±0.13 | 1.3±0.9 | 11±1.4 | NR | NR |
| Kleczynski (2017) | NR | 53 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kobe (2016) | 2011-2014 | 71 | NR | 37 (52.11%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | 2013-2014 | 1466 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | NR | 293 | 81.3±7.5 | 105 (36%) | 240 (82%) | 7.1±4 | NR | 174 (59.4%) | 119 (40.6%) | 0.68±0.21 | NR | NR | 54±14 | NR |
| Maniar (2016) | PARTNER II[2] | 73 | 84±7 | 49 (67.6%) | NR | 12.3±5.7 | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | NR | 30 | 92±2 | 24 (80%) | 24 (80%) | 8.4±5.4 | NR | 20 (67%) | NR | NR | NR | NR | NR | NR |
| Patel (2016) | 2012-2015 | 31 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | 2008-2012 | 144 | 83.1±5.2 | 106 (74%) | 136 (94.4%) | 8.8±6.7 | 28.8±16.3 | NR | NR | NR | NR | NR | NR | NR |
| Rabinovitz (2016) | 2009-2013 | 46 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | 2005-2009 | 85 | 83±7 | 58 (68.2%) | NR | 11.6±8.3 | NR | 42 (49.4%) | 43 (50.6%) | NR | 112±73 (umol/L) | NR | NR | NR |
| Rodriguez-Pascual | 2010-2015 | 68 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |

For peer review only

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (2016) | | | | | | | | | | | | | | |
| Rogers (2018) | Since 2007 | 242 | 83±7.6 | 140 (58%) | NR | 8.8±6.1 | NR | NR | NR | NR | NR | NR | NR | NR |
| Schoenenberger (2018) | 2009-2013 | 169 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2017) | 2013-2016 | 353 | NR | 276 (78.2%) | 208 (58.92%) | NR | NR | 285 (80.74%) | 58 (16.43%) | NR | NR | NR | NR | NR |
| Steinvil (2018) | 2011-2016 | 232 | 83±7 | 132 (57%) | 193 (83%) | 8.8±6.1 | NR | NR | NR | 0.7 ±0.2 | NR | NR | NR | 43±13 |
| Shi (2018) | 2014-2016 | 116 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | 2011-2015 | 34 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | 2007-2010 | 24 | 84±4 | NR | NR | NR | 24.8±16 | NR | NR | 0.5±0.2 | NR | NR | 52±19 | NR |
| Traynor (2017) | 2008-2015 | 60 | 85.4±7.4 | 38 (63.3%) | 49 (81.7%) | 12.7±11.2 | NR | 34 (56.67%) | NR | NR | 1.4±1.1 | NR | 56±14 | NR |
| Yamamoto (2015) | 2010-2011 | 56 | 85±6.7 | 38 (67.90%) | 47 (83.9%) | 10.9±7.3 | 23.6±11.2 | 41 (73.2%) | NR | 0.58±0.19 | NR | NR | NR | 53.3±21.5 |
| Zajarias (2016) | PARTNER II[2] | 265 | 85±6 | 157 (59.7%) | NR | 11.8±5.8 | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | 2014-2018 | 97 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | 2013-2018 | 70 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Welle (2020) | 2012-2019 | 483 | 82±8.2 | 222 (46.0%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Patel (2020) | 2012-2018 | 74 | 81±9.1 | 37 (50%) | 65 (88%) | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mach (2020) | 2017-2018 | 39 | NR | 17 (43.59%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Drudi (2018) | 2012-2017 | 254 | NR | NR | NR | NR | NR | 193 (75.98%) | 61 (24.02%) | NR | NR | NR | NR | NR |
| Kiani (2020) | 2011-2016 | 11316 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Malik (2020) | 2016 | 8419 | NR | 3884 (46.13%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Morris (2020) | 2012-2019 | 234 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Sami (2020) | 2012-2015 | 2865 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Gharibeh (2019) | 2012-2016 | 186 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Voigtlander (2020) | 2011-2014 | 956 | 82.5±6.8 | 715 (74.8%) | NR | 8.7±6.3 | NR | 681 (71.2%) | 247 (25.8%) | NR | NR | NR | NR | NR |
| Kundi (2020) | 2016 | 13593 | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2020) | 2013-2017 | 284 | NR | 201 (70.77%) | 186 (65.49%) | NR | NR | 234 (81.25%) | NR | NR | NR | NR | NR | NR |

Baseline demographics and population characteristics (continued)

| Study | AF (n, %) | Active cancer | BMI (mean±SD) | COPD (n,%) | Cerebrovascular Disease | CAD (n, %) | Chronic Lung | Chronic Dialysis | Chronic Kidney | dyslipidemia (n,%) | Diabetes Mellitus | Hyperlipidemia (n,%) | Hypertension (n,%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | (n, %) | | | (n,%) | | disease (n,%) | (n,%) | disease (n, %) | | (n,%) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson, (2016) | NR | NR | NR | NR | NR | NR | 1611 (26.41%) | 186 (3.05%) | NR | NR | 2137 (35.03%) | NR | 5510 (90.33%) |
| Bureau (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bogdan (2016) | 14 (17%) | NR | NR | 23 (29%) | NR | NR | NR | NR | NR | 39 (49%) | 27 (34%) | NR | 56 (70%) |
| Capodanno (2014) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2015) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Eichler (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Ghatak (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Green (2015) | NR | NR | NR | 46 (42%) | 26 (26%) | 91 (83%) | NR | NR | NR | NR | 28 (26%) | NR | 95 (86%) |
| Green (2012) | NR | NR | 25±6.4 | NR | NR | NR | NR | NR | NR | NR | 19 (25%) | 38 (50%) | 58 (76%) |
| Grossman (2017) | 57 (32%) | NR | NR | NR | NR | 100 (56%) | NR | NR | NR | NR | 71 (39%) | 136 (74%) | 160 (86%) |
| Huded (2016) | 26 (41%) | NR | 29.1±6.9 | 5 (8%) | NR | 54 (84%) | NR | NR | 14 (22%) | NR | 24 (38%) | 37 (58%) | 50 (78%) |
| Koifman (2015) | 101 (44%) | NR | NR | 77 (34%) | 32 (16%) | NR | NR | NR | NR | NR | 79 (35%) | NR | 211 (92%) |
| Kleczynski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kobe (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | 107 (37%) | NR | 25.4±4.9 | NR | NR | 206 (71%) | 94 (32%) | NR | NR | 208 (71%) | 85 (29%) | NR | 246 (84%) |
| Maniar (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | 11 (37%) | NR | 25.4±3.7 | NR | NR | 22 (73%) | 7 (23%) | 1 (3%) | NR | 17 (57%) | NR | NR | 28 (93%) |
| Patel (2016) | 19 (61.3%) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls (2014) | NR | NR | 26.9±5.1 | NR | NR | 101 (70%) | NR | NR | NR | NR | 50 (34%) | NR | NR |
| Rabinovitz (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | NR | NR | 25±5 | 23 (27.1%) | 15 (17.7%) | 57 (67.1%) | NR | 3 (3.5%) | NR | 55 (65.5%) | 21 (26.6%) | NR | 58 (68.2%) |
| Rodriguez-Pascual (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | 109 (45%) | NR | 29±11 | 116 (48%) | NR | NR | NR | 15 (6%) | 90 (37%) | NR | 82 (34%) | 198 (82%) | 223 (92%) |
| Schoenenberger | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |

For peer review only

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (2018) | | | | | | | | | | | | | |
| Shimura (2017) | NR | 8 (2.3%) | NR | NR | NR | NR | NR | NR | 203 (57.5%) | NR | 94 (26.6%) | NR | 258 (73.1%) |
| Steinvil (2018) | 102 (44%) | NR | 29±11 | 114 (49%) | NR | NR | NR | 14 (6%) | 86 (37%) | NR | 79 (34%) | 190 (82%) | 216 (93%) |
| Shi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Traynor (2017) | NR | NR | NR | NR | 18 (30%) | NR | 32 (55.2%) | 2 (3.3%) | NR | NR | 28 (46.7%) | NR | 57 (95%) |
| Yamamoto (2015) | NR | NR | NR | 17 (30.40%) | NR | NR | NR | 1 (1.80%) | 42 (75%) | 19 (33.90%) | 5 (8.90%) | NR | 37 (66.10%) |
| Zajarias (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Welle (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Patel (2020) | 41 (55%) | NR | 29±6 | NR | NR | 29 (39%) | 26 (35%) | NR | NR | NR | 31 (42%) | NR | 67 (91%) |
| Mach (2020) | 18 (46.15%) | NR | NR | 13 (33.33%) | NR | NR | NR | NR | NR | NR | 17 (43.59%) | NR | 36 (92.31%) |
| Drudi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kiani (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Malik (2020) | 4055 (48.17%) | NR | NR | NR | NR | 520 (6.17%) | NR | NR | 4381 (52.04%) | NR | 3187 (37.86%) | NR | 7363 (87.46%) |
| Morris (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Sami (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Gharibeh (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Voigtlander (2020) | 262 (27.4%) | NR | 18.6±1.2 | NR | NR | 450 (47.1%) | NR | NR | NR | NR | NR | NR | 205 (21.9%) |
| Kundi (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Shimura (2020) | 69 (24.30%) | 36 (12.68%) | NR | NR | NR | 114 (40.14%) | NR | NR | 215 (75.70%) | NR | 53 (18.66%) | NR | NR |

Baseline demographics and population characteristics (continued)

| Study | Liver disease (n,%) | Previous stroke (n, %) | Previous MI (n, %) | Previous BVA (n, %) | Previous PCI (n, %) | Previous CABG (n,%) | Pulmonary disease (n,%) | Peripheral arterial disease (n,%) | Peripheral Vascular disease (n, %) | Permanent pacemaker (n,%) | Valve type |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | NR | 753 (12.34%) | 1422 (23.31%) | 774 (12.69%) | 2103 (34.48%) | 1800 (29.51%) | NR | 1930 (31.64%) | NR | NR | NR |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Bureau (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Bagienski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve<br>Edwards Sapien<br>Jena<br>Lotus<br>NVT |
| Bogdan (2016) | 1 (1%) | NR | NR | NR | NR | NR | NR | NR | 9 (11%) | NR | CoreValve: 60 (75.94%)<br>Edwards Sapien: 19 (24.06%) |
| Capodanno (2014) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Chauhan (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Cockburn (2015) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve: 277 (89%)<br>Edwards Sapien: 24 (7.7%)<br>Lotus: 10 (3.2%) |
| Eichler (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve (classic, Evolut R)<br>Edwards Sapien<br>Edwards Sapien XT |
| Ghatak (2012) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien |
| Green (2015) | 9 (8%) | NR | NR | 35 (32%) | 45 (41%) | 47 (43%) | NR | NR | 46 (42%) | 27 (25%) | Edwards Sapien |
| Green (2012) | NR | 9 (12%) | NR | NR | NR | 27 (36%) | 22 (29%) | NR | 16 (21%) | 24 (32%) | Edwards Sapien |
| Grossman (2017) | NR | 6 (3%) | NR | NR | NR | NR | NR | NR | 19 (11%) | 13 (7%) | CoreValve<br>Edwards Sapien XT |
| Huded (2016) | NR | 13 (20%) | NR | NR | NR | 17 (27%) | NR | 6 (9%) | NR | NR | Edwards Sapien<br>Edwards Sapien XT<br>Edwards Sapien 3<br>CoreValve<br>Direct Flow Medical |
| Koifman (2015) | NR | NR | NR | NR | 60 (26%) | 67 (29%) | NR | NR | 76 (34%) | 37 (21%) | |
| Kleczynski (2017) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien<br>Edwards Sapien XT<br>Edwards Sapien 3<br>CoreValve<br>JenaValve |
| Kobe (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Martin (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Mok (2016) | NR | 56 (19%) | 84 (29%) | NR | NR | 57 (20%) | NR | NR | 87 (30%) | NR | NR |
| Maniar (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Okoh (2017) | NR | NR | NR | 1 (3%) | 4 (13%) | 4 (13%) | NR | NR | 7 (23%) | 7 (23%) | |
| Patel (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Puls | NR | NR | NR | NR | 40 | 16 | 43 | NR | 42 | NR | CoreValve |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (2014) | | | | | (28%) | (11%) | (30%) | | (29%) | | Edwards Sapien |
| Rabinovitz (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rodes-Cabau (2010) | NR | NR | 39 (45.9%) | NR | 26 (30.6%) | 20 (23.5%) | NR | NR | 29 (34.1%) | NR | Cribier-Edwards<br>Edwards Sapien<br>Edwards Sapien XT |
| Rodriguez-Pascual (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Rogers (2018) | NR | 17 (7%) | 41 (17%) | 53 (22%) | 80 (33%) | 58 (24%) | NR | NR | 58 (24%) | 65 (27%) | Balloon expandable (67%)<br>Self-expanding (33%) |
| Schoenenberger (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | CoreValve<br>Edwards Sapien XT |
| Shimura (2017) | 17 (4.8%) | 65 (18.4%) | 28 (7.9%) | NR | 94 (26.6%) | 21 (5.9%) | 108 (30.6%) | 67 (19%) | NR | NR | NR |
| Steinvil (2018) | NR | 16 (7%) | 42 (18%) | 49 (21%) | 79 (34%) | 58 (25%) | NR | NR | 53 (23%) | 67 (29%) | NR |
| Shi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Skaar (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Stortecky (2012) | NR | NR | NR | NR | NR | 1 (4.2%) | NR | NR | NR | NR | NR |
| Traynor (2017) | NR | NR | 18 (30%) | NR | NR | 15 (25%) | NR | NR | 38 (63.3%) | 12 (20%) | NR |
| Yamamoto (2015) | NR | 6 (10.70%) | 5 (8.90%) | NR | NR | NR | NR | 8 (14.30%) | NR | 10 (17.90%) | CoreValve<br>Edwards Sapien |
| Zajarias (2016) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | Edwards Sapien XT |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Goudzwaard (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Welle (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Patel (2020) | NR | NR | NR | NR | NR | NR | NR | 17 (23%) | NR | NR | NR |
| Mach (2020) | NR | NR | 4 (10.26%) | NR | 9 (23.08%) | 3 (7.69%) | NR | NR | 18 (46.15%) | NR | Spaien XT<br>Spaien 3<br>Symetis accurate<br>Core Valve Evolut |
| Drudi (2018) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Kiani (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Malik (2020) | 340 (4.04%) | NR | 1107 (13.15%) | NR | NR | NR | 3010 (35.75%) | NR | 2369 (28.14%) | NR | NR |
| Morris (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Sami (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Gharibeh (2020) | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |
| Voigtlander (2020) | NR | 97 (10.1%) | 128 (13.4%) | NR | NR | NR | NR | NR | NR | NR | NR |
| Kundi | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR | NR |

For peer review only

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| (2020) | | | | | | | | | | | |
| Shimura (2020) | 14 (4.93%) | 32 (11.27%) | NR | NR | NR | 19 (6.69%) | 62 (21.83%) | 37 (13.03%) | NR | NR | Balloon expandable valve |

N, sample size of frail TAVI recipients. SD, standard deviation. NYHA, New York Heart Association. STS, Society of Thoracic Surgeons. TF, transfemoral approach. TA, transapical approach. AVA, aortic valve area. LVEF, left ventricular ejection fraction. AF, atrial fibrillation. BMI, body mass index. COPD, chronic obstructive pulmonary disease. CAD, coronary artery disease. MI, myocardial infarction. BVA, balloon aortic valvuloplasty. PCI, percutaneous coronary intervention. CABG, coronary artery bypass grafting.
NR, not reported
§ = Euro score II
**= median and IQR
1, recruitment period, 2007-2010
2, recruitment period, 2011-2016

For peer review only

| Study | Study Participation | Study Attrition | Prognostic Factor Measurement | Outcome Measurement | Statistical Analysis Reporting | Overall Risk of Bias | Study Design | Abstract |
|---|---|---|---|---|---|---|---|---|
| Alfredsson (2016) | high | moderate | low | low | low | moderate | prospective | no |
| Bureau (2017) | high | moderate | low | low | low | moderate | prospective | no |
| Bagienski (2017) | moderate | moderate | low | low | low | moderate | prospective | no |
| Bogdan (2016) | moderate | high | low | low | moderate | moderate | retrospective | no |
| Capodanno (2014) | moderate | moderate | low | low | low | moderate | prospective | no |
| Chauhan (2016) | moderate | high | low | low | low | moderate | retrospective | no |
| Cockburn (2015) | moderate | low | moderate | low | moderate | moderate | prospective | no |
| Eichler (2017) | high | low | low | low | low | low | prospective | no |
| Ghatak (2012) | high | high | moderate | low | moderate | high | retrospective | yes |
| Grossman (2017) | moderate | high | moderate | low | low | moderate | retrospective | no |
| Green (2012) | moderate | low | low | low | low | low | prospective | no |
| Green (2015) | moderate | low | low | low | low | low | prospective | no |
| Huded (2016) | high | high | low | low | low | high | retrospective | no |
| Kleczynski (2017) | low | low | low | low | low | low | prospective | no |
| Kobe (2016) | moderate | moderate | low | low | low | moderate | prospective | no |
| Koifman (2015) | moderate | high | low | low | low | moderate | retrospective | no |
| Martin (2018) | high | high | low | low | low | high | prospective | no |
| Maniar (2016) | high | moderate | moderate | low | low | moderate | retrospective | yes |
| Mok (2016) | high | low | low | low | low | low | retrospective | no |
| Okoh (2017) | Low | high | moderate | low | low | moderate | prospective | no |
| Patel (2016) | moderate | high | low | low | low | moderate | retrospective | no |
| Puls (2014) | moderate | low | low | low | low | low | prospective | no |
| Rabinovitz (2016) | high | high | moderate | low | low | high | retrospective | no |
| Rodes-Cabau (2010) | low | low | high | low | low | low | prospective | no |
| Rodriguez-Pascual (2016) | low | low | low | low | low | low | prospective | no |
| Rogers (2018) | moderate | moderate | low | low | low | moderate | retrospective | no |
| Schoenenberger (2018) | high | low | low | low | low | low | prospective | no |
| Shimura (2017) | low | moderate | low | low | low | low | prospective | no |
| Steinvil (2018) | moderate | moderate | low | low | low | moderate | prospective | no |
| Shi (2018) | moderate | high | low | low | low | moderate | prospective | no |
| Skaar (2018) | low | low | low | low | low | low | prospective | no |
| Stortecky (2012) | high | moderate | moderate | moderate | high | high | prospective | yes |
| Traynor (2017) | low | low | moderate | moderate | low | moderate | retrospective | no |
| Yamamoto (2015) | moderate | low | low | low | low | low | prospective | no |
| Zajarias (2016) | high | moderate | low | low | low | moderate | prospective | yes |
| Goudzwaard (2020) | low | moderate | low | low | low | low | prospective | no |

For Peer review only

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Goudzwaard (2020) | low | moderate | low | low | low | low | prospective | no |
| Welle (2020) | moderate | high | low | moderate | low | moderate | prospective | yes |
| Patel (2020) | low | low | low | low | low | low | retrospective | no |
| Mach (2020) | low | high | low | low | low | low | Prospective | no |
| Drudi (2018) | low | low | low | low | low | low | prospective | no |
| Kiani (2020) | low | high | moderate | low | low | moderate | prospective | no |
| Malik (2020) | low | high | low | low | low | low | retrospective | no |
| Morris (2020) | high | high | low | low | moderate | high | prospective | yes |
| Sami (2020) | moderate | high | low | low | low | moderate | retrospective | yes |
| Gharibeh (2020) | high | high | moderate | low | moderate | high | retrospective | yes |
| Voigtlander (2020) | low | high | low | low | low | low | prospective | no |
| Kundi (2020) | low | high | low | low | low | low | retrospective | no |
| Shimura (2020) | low | high | low | low | low | low | prospective | no |

Supplementary appendix C. Assessment of quality in individual studies

*Risk of bias was assessed using the Quality in Prognostic Studies tool (QUIPS)
Low risk= studies with four or five low risk domains
High risk=studies with two or more high risk domains
Moderate risk=remaining studies

Reference. Hayden JA, van der Windt DA, Cartwright JL, Côté P, Bombardier C. Assessing bias in studies of prognostic factors. Annals of internal medicine. 2013 Feb 19;158(4):280-6.

For peer review only

# Supplementary appendix D. Prognosis of frail TAVI recipients

| Study | Procedural/in-hospital | 30 days | 6 months | 8 months | 1 year | 2 year |
|---|---|---|---|---|---|---|
| ***Primary outcomes*** | | | | | | |
| ***Death*** | | | | | | |
| Alfredsson, 2016 | 300/6100 (4.9%) | 405/6100 (6.6%) | | | | |
| Bureau, 2017 | | 7/71 (9.9%) | 13/71 (18.3%) | | 18/69 (26.1%) | |
| Bogdan, 2016 | | 4/79 (5.1%) | | | | 35.4% (2.1 years) |
| Capodanno, 2014 | | 30/306 (9.8%) | | | | |
| Chauhan, 2016 | | 18/233 (7.7%) | | | 48/177 (27.1%) | |
| Eichler, 2017 | | | | | 30/152 (19.7%) | |
| Green, 2015 | | 11/110 (10%) | | | 32.7% | |
| Huded, 2016 | | 2/64 (3.13%) | | | | |
| Kleczynski, 2017 | | | | | 17/53 (32.1%) | |
| Martin, 2018 | | 51/1581(3.23%) | | | | |
| Maniar, 2016 | | 12.3% | | | 36.5% | |
| Okoh, 2017 | | | | | | 30% |
| Puls, 2014 | 33/144 (23%) | 25/144 (17%) | | | | |
| Rabinovitz, 2016 | | | | | 12/46 (26.1%) | |
| Rodriguez-Pascual, 2016 | | | | | | 39.7%*(98weeks) |
| Rodes-Cabau, 2010 | 2/85 (2.4%) | 7/85 (8.2%) | | 19/85 (22.4%) | | |
| Schoenenberger, 2018 | | | | | 37/169(21.9%) | |
| Shimura, 2017 | 15/353 (4.25%) | 10/353 (2.83%) | | | | |
| Steinvil, 2018 | | 14/232 (6.03%) | | | 46/232 (19.8%) | |
| Shi, 2018 | | | 17/116 (15%) | | | |
| Skaar, 2018 | | | | | | 9/34 (26.47%) |
| Stortecky, 2012 | | 6/24 (25%) | | | 9/24 (37.5%) | |
| Traynor, 2017 | 2/60 (3.3%) | | | | 14.8% | |
| Zajarias, 2016 | | 17/265 (6.4%) | | | 88/265 (33.2%) | |
| Patel, 2020 | | 5/67 (6.7%) | | | | |
| Mach, 2020 | 3/39 (7.69%) | | | | | |
| Drudi, 2020 | | 26/269 (9.67%) | | | | |
| Kiani, 2020 | | 4.55% | | | 18.07% | |
| Sami, 2020 | 301/2865 (10.5%) | | | | | |
| Gharibeh, 2020 | | | | | | 25.3% |
| Voigtlander, 2020 | | 73/956 (7.64%) | | | 255/956 (26.67%) | |
| Kundi, 2020 | | 587/13593 (4.32%) | | | 2679/13593 (19.71%) | |
| ***Death (cardiovascular cause)*** | | | | | | |
| Alfredsson, 2016 | | 251/6100 (4.1%) | | | | |
| Green, 2015 | | 8/110 (7%) | | | | |
| Shimura, 2017 | 4/353(1.13%) | 8/353 (2.27%) | | | | |
| ***Secondary outcomes*** | | | | | | |
| ***Acute kidney injury*** | | | | | | |
| Alfredsson, 2016 | 397/6100 (6.5%) | | | | | |
| Bogdan, 2016 | | 25/79 (31.6%) | | | | |
| Green, 2012 | 3/76 (3.95%) | | | | | |
| Okoh, 2017 | 2/30 (6.67%) | | | | | |
| Shimura, 2017 | 47/353 (13.31%) | | | | | |
| Yamamoto, 2015 | 6/56 (10.7%) | | | | | |
| Mach, 2020 | 8/39 (20.51%) | | | | | |
| Shimura, 2020 | 43/288 (14.93%) | | | | | |
| ***Arrhythmia*** | | | | | | |
| Rodes-Cabau, 2010 | 9/85 (10.6%) | | | | | |
| ***Annulus rupture*** | | | | | | |
| Yamamoto, 2015 | 1/56 (1.8%) | | | | | |
| ***Bleeding*** | | | | | | |
| Zajarias, 2016 | 15/265 (43.4%) | | | | | |
| Kiani, 2020 | | 11.80% | | | 22.20% | |
| Shimura, 2020 | 71/288 (24.65%) | | | | | |
| ***Cardiac tamponade*** | | | | | | |
| Puls, 2014 | 3/144 (2%) | | | | | |
| Shimura, 2017 | 11/353 (3.11%) | | | | | |
| Yamamoto, 2015 | 3/56 (5.4%) | | | | | |
| Shimura, 2020 | 6/288 (2.08%) | | | | | |
| ***Conversion to open surgery*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| Shimura, 2017 | 8/353 (2.27%) | | | | | |
| Sami, 2020 | 284/2865 (9.9%) | | | | | |
| Voigtlander, 2020 | 7/956 (0.9%) | | | | | |
| Shimura, 2020 | 0/288 (0%) | | | | | |
| ***Coronary obstruction*** | | | | | | |
| Shimura, 2017 | 3/353 (0.85%) | | | | | |
| ***Delirium*** | | | | | | |
| Bagienski, 2017 | 105/127 (82.68%) | | | | | |
| Goudzwaard, 2020 | 24/97 (24.7%) | | | | | |
| ***Life-threatening bleeding*** | | | | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Alfredsson, 2016 | | 391/6100 (6.4%) | | | | |
| Green, 2012 | 7/76 (9.21%) | | | | | |
| Puls, 2014 | 15/144 (10%) | | | | | |
| Stortecky, 2012 | 4/24 (16.7%) | | | | | |
| Shimura, 2017 | 33/353 (9.35%) | | | | | |
| Yamamoto, 2015 | 4/56 (7.1%) | | | | | |
| Shimura, 2020 | 14/288 (4.86%) | | | | | |
| ***Major bleeding*** | | | | | | |
| Green, 2015 | | 10/110 (9%) | | | | |
| Koifman, 2015 | 6/238 (3%) | | | | | |
| Puls, 2014 | 12/144 (8%) | | | | | |
| Shimura, 2017 | 77/353 (21.81%) | | | | | |
| Yamamoto, 2015 | 9/56 (16.1%) | | | | | |
| Mach, 2020 | 1/39 (2.56%) | | | | | |
| Shimura, 2020 | 33/288 (11.46%) | | | | | |
| ***Minor bleeding*** | | | | | | |
| Koifman, 2015 | 31/238 (13%) | | | | | |
| Puls, 2014 | 50/144 (35%) | | | | | |
| Shimura, 2017 | 62/353 (17.56%) | | | | | |
| Mach, 2020 | 4/39 (10.26%) | | | | | |
| ***Major vascular complications*** | | | | | | |
| Bogdan, 2016 | | 0/79 (0%) | | | | |
| Green, 2015 | | 7/110 (6%) | | | | |
| Koifman, 2015 | 37/238 (16%) | | | | | |
| Shimura, 2017 | 18/353 (5.1%) | | | | | |
| Yamamoto, 2015 | 8/56 (14.3%) | | | | | |
| Shimura,2020 | 9/288 (3.13%) | | | | | |
| ***Minor vascular complications*** | | | | | | |
| Bogdan, 2016 | | 2/79 (2.5%) | | | | |
| Koifman, 2015 | 26/238 (11%) | | | | | |
| Shimura, 2017 | 17/353 (4.82%) | | | | | |
| ***Major access site complications*** | | | | | | |
| Rodes-Cabau, 2010 | 12/85 (14.1%) | | | | | |
| Stortecky, 2012 | 2/24 (8.3%) | | | | | |
| Mach, 2020 | 1/39 (2.56%) | | | | | |
| ***Major vascular injury*** | | | | | | |
| Huded, 2016 | | 4/64 (6%) | | | | |
| ***Myocardial infarction*** | | | | | | |
| Rodes-Cabau, 2010 | | 2/85 (2.4%) | | | | |
| Yamamoto, 2015 | 0/56 (0%) | | | | | |
| Mach, 2020 | 2/39 (5.13%) | | | | | |
| Kiani, 2020 | | 0.63% | | | 1.98% | |
| Voigtlander, 2020 | 3/956 (0.3%) | | | | | |
| ***Need for second valve*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| ***Permanent pacemaker*** | | | | | | |
| Green, 2015 | | 10/110 (9%) | | | | |
| Koifman, 2015 | 12/238 (5%) | | | | | |
| Okoh, 2017 | 2/30 (6.67%) | | | | | |
| Puls, 2014 | 3/144 (2%) | | | | | |
| Rodes-Cabau, 2010 | | 7/85 (8.2%) | | | | |
| Yamamoto, 2015 | 5/56 (8.9%) | | | | | |
| Mach, 2020 | 5/39 (12.82%) | | | | | |
| Kiani, 2020 | | 11.75% | | | | |
| Sami, 2020 | 309/2865 (10.8%) | | | | | |
| Shimura, 2020 | 29/288 (10.07%) | | | | | |
| ***Percutaneous closure device failure*** | | | | | | |
| Bogdan, 2016 | | 10/79 (12.7%) | | | | |
| ***Renal failure*** | | | | | | |
| Green, 2015 | | 9/110 (8%) | | | | |
| ***Respiratory failure*** | | | | | | |
| Huded, 2016 | | 1/64 (2%) | | | | |
| ***Readmission*** | | | | | | |
| Green, 2015 | | 4/110(4%) | | | | |
| Huded, 2016 | | 7/64 (11%) | | | | |
| Patel, 2020 | | 16/74 (21.6%) | | | | |
| Kiani, 2020 | | 3.69% | | | | |
| Kundi, 2020 | | | | | 6492/13593 (47.76%) | |
| Shimura, 2020 | | | | | 64/288 (22.22%) | |
| ***Stroke*** | | | | | | |
| Alfredsson, 2016 | | 129/6100 (2.1%) | | | | |
| Green, 2015 | | 1/110 (1%) | | | | |
| Huded, 2016 | | 4/64 (6%) | | | | |
| Koifman, 2015 | 11/238 (5%) | | | | | |
| Rodes-Cabau, 2010 | | 3/85 (3.5%) | | | | |
| Okoh, 2017 | 0/30 (0%) | | | | | |
| Shimura, 2017 | 6/353 (1.7%) | | | | | |
| Stortecky, 2012 | 2/24 (8.3%) | | | | | |
| Traynor, 2017 | 2/60 (3.3%) | | | | | |
| Yamamoto, 2015 | 3/56 (5.4%) | | | | | |

For peer review only

| | | | | | | |
|---|---|---|---|---|---|---|
| Mach, 2020 | 1/39 (2.56%) | | | | | |
| Kiani, 2020 | | 2.56% | | | 5.70% | |
| Voigtlander, 2020 | 14/956 (1.5%) | | | | | |
| Shimura, 2020 | 2/288 (0.07%) | | | | | |
| ***Sepsis*** | | | | | | |
| Rodes-Cabau, 2010 | | 4/85 (4.7%) | | | | |
| Voigtlander, 2020 | 22/956 (2.3%) | | | | | |
| ***Transfusion*** | | | | | | |
| Green, 2012 | 24/76 (32.9%) | | | | | |
| Koifman, 2015 | 100/238 (43%) | | | | | |
| Puls, 2014 | 67/144 (47%) | | | | | |
| ***Valve embolization*** | | | | | | |
| Rodes-Cabau, 2010 | 1/85 (1.2%) | | | | | |
| ***Vascular complications*** | | | | | | |
| Bogdan, 2016 | | 12/79 (15.18%) | | | | |
| Voigtlander, 2020 | 43/956 (4.5%) | | | | | |
| Shimura, 2020 | 22/288 (7.64%) | | | | | |
| ***2-valve implantation*** | | | | | | |
| Shimura, 2017 | 9/353 (2.55%) | | | | | |
| Yamamoto, 2015 | 1/56 (1.8%) | | | | | |
| ***Length of hospital stay for frail TAVI recipients, days*** | | | | | | |
| Ghatak, 2012 | 12.1±8.4 | | | | | |
| Green, 2012 | 9±6 | | | | | |
| Huded, 2016 | 6±3.5 | | | | | |
| Okoh, 2017 | 7±8 | | | | | |
| Patel, 2016 | 9** | | | | | |
| Traynor, 2017 | 6.8±5.2 | | | | | |
| Yamamoto, 2015 | 10.3±7.5 | | | | | |
| Kiani, 2020 | 5 (3-8) | | | | | |
| ***Intensive care unit (ICU) stay, days*** | | | | | | |
| Patel, 2016 | 2.7** | | | | | |
| Yamamoto, 2015 | 4.9±4 | | | | | |
| Voigtlander, 2020 | 3.58±5.0 | | | | | |

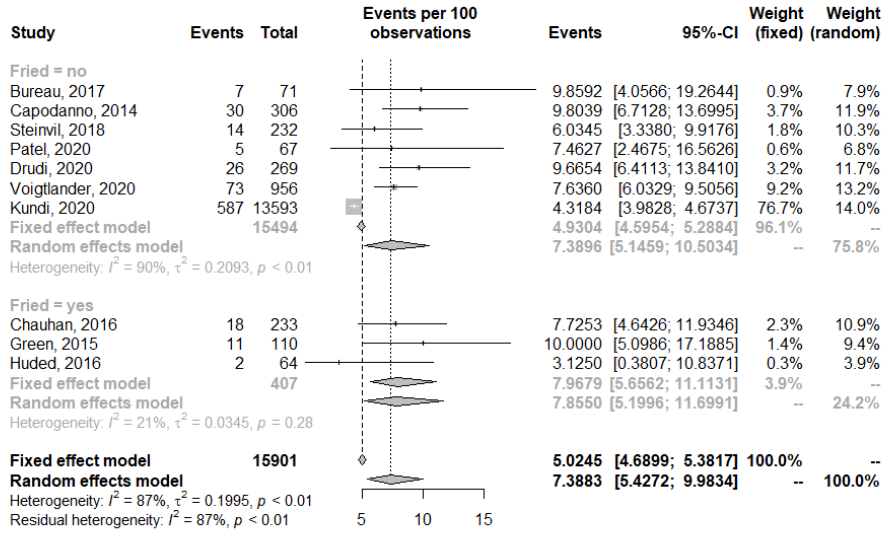
*= median + IQR

**=mean without standard deviation reported

For peer review only

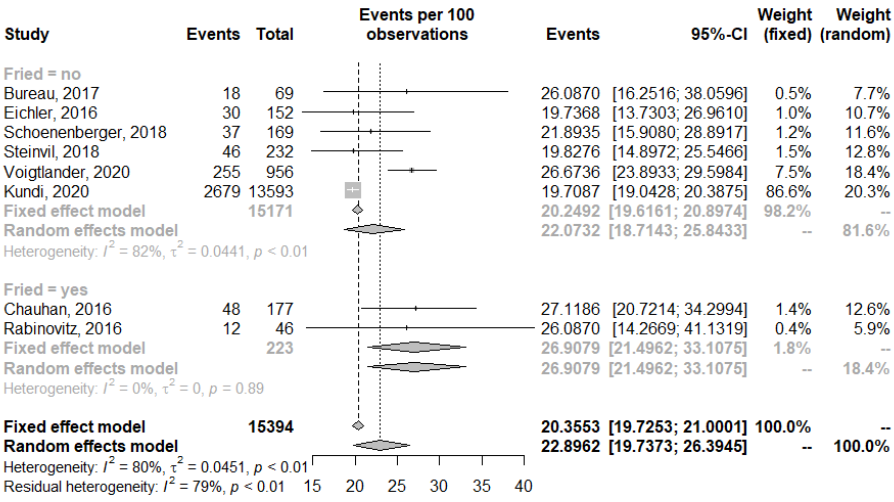
**Supplementary appendix E. Subgroup analyses of studies reporting frailty measurement using the Fried phenotype compared to non-Fried phenotype. The squares indicate the mortality reported by each study. The diamonds indicate the pooled estimates for mortality.**

## Subgroup analysis of 30-day mortality




Test for subgroup differences: Q=0.05 (p=0.8250)

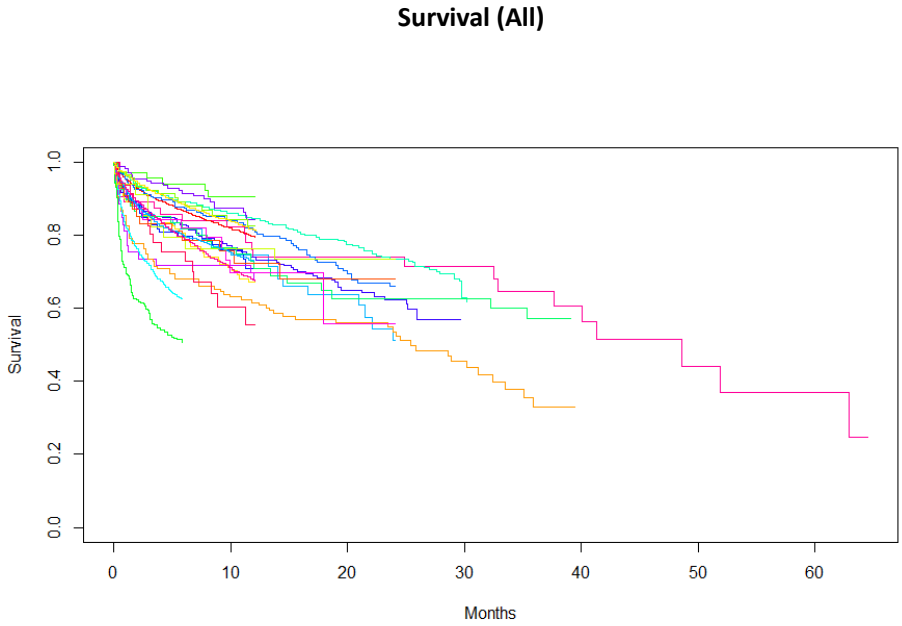
## Subgroup analysis of 1-year mortality

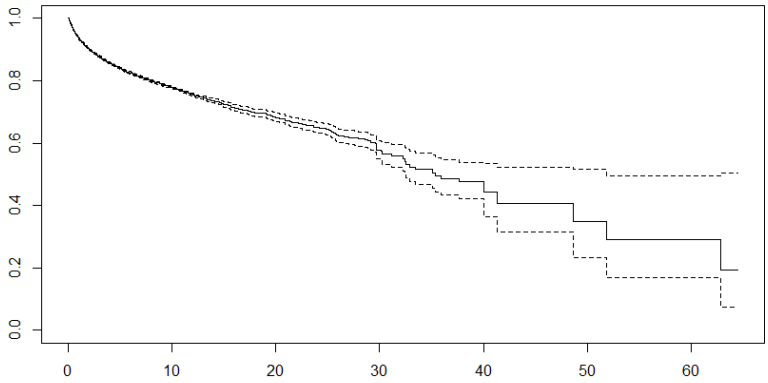



Test for subgroup differences: Q=2.02 (p=0.1551)

Appendix F. Survival of frail patients after TAVI

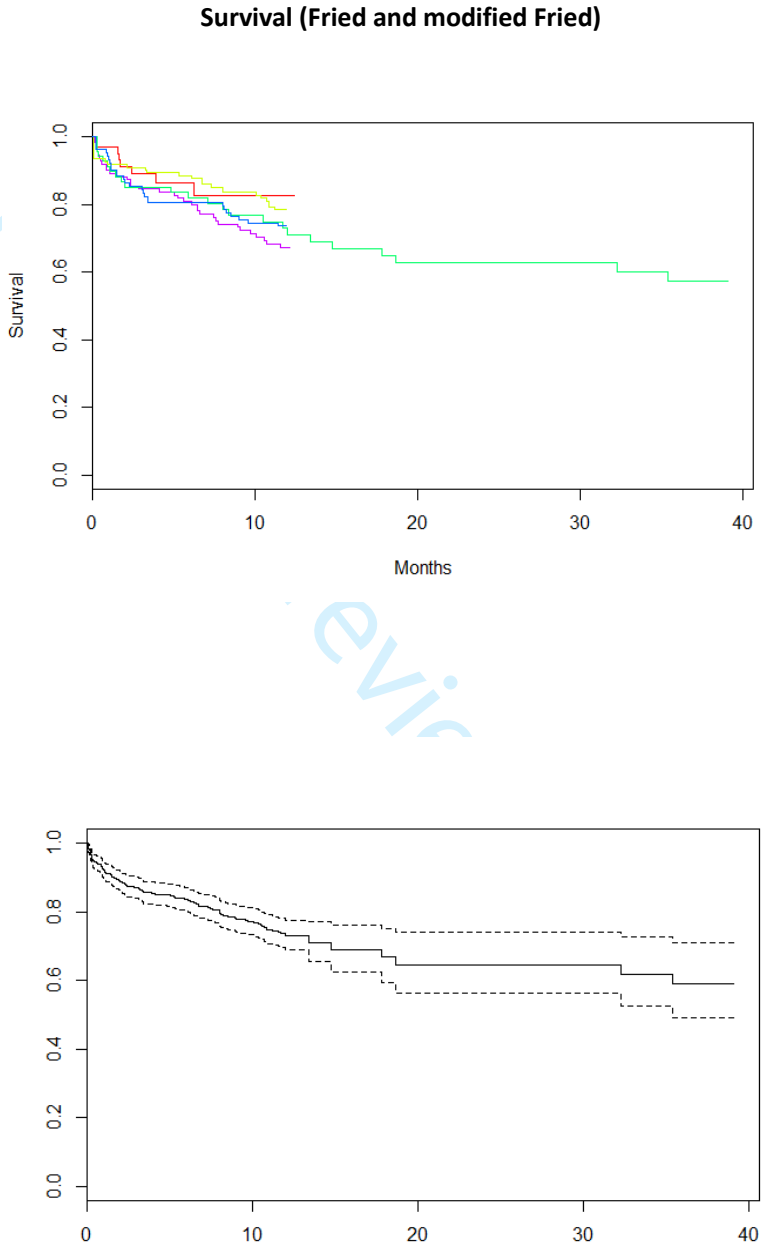
Survival after TAVI of frail patients.
Frailty was measured using single- and multi-dimensional measures.
Individual patient data estimated using algorithm developed by Guyot et al 2012.




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Survival after TAVI of frail patients.
Frailty was measured using the Fried and modified Fried frailty phenotype.
Individual patient data estimated using algorithm developed by Guyot et al 2012.



**Survival rate of frail patients after TAVI (combined all studies)**

| Time (Months) | n.risk | n.event | survival | Std.err | Lower 95%CI | Upper 95% CI |
|---|---|---|---|---|---|---|
| 1 | 44930 | 3200 | 0.934 | 0.00113 | 0.931 | 0.936 |
| 2 | 43260 | 4744 | 0.901 | 0.00136 | 0.899 | 0.904 |
| 3 | 42000 | 5911 | 0.877 | 0.0015 | 0.874 | 0.88 |
| 4 | 40226 | 6887 | 0.857 | 0.0016 | 0.853 | 0.86 |
| 5 | 38591 | 7643 | 0.84 | 0.00168 | 0.837 | 0.844 |
| 6 | 32073 | 8382 | 0.824 | 0.00175 | 0.82 | 0.827 |
| 7 | 30814 | 8835 | 0.812 | 0.00181 | 0.809 | 0.816 |
| 8 | 29512 | 9340 | 0.799 | 0.00188 | 0.795 | 0.802 |
| 9 | 27694 | 9738 | 0.788 | 0.00193 | 0.784 | 0.791 |
| 10 | 26214 | 10030 | 0.779 | 0.00197 | 0.775 | 0.783 |
| 11 | 24196 | 10438 | 0.767 | 0.00204 | 0.763 | 0.771 |
| 12 | 22329 | 10773 | 0.756 | 0.00209 | 0.752 | 0.76 |
| 13 | 1274 | 10857 | 0.746 | 0.00274 | 0.741 | 0.752 |
| 14 | 1194 | 10873 | 0.737 | 0.0036 | 0.73 | 0.744 |
| 15 | 1116 | 10893 | 0.724 | 0.00452 | 0.715 | 0.733 |
| 16 | 1043 | 10910 | 0.713 | 0.00523 | 0.702 | 0.723 |
| 17 | 974 | 10918 | 0.707 | 0.00555 | 0.696 | 0.718 |
| 18 | 915 | 10931 | 0.697 | 0.0061 | 0.685 | 0.709 |
| 19 | 835 | 10934 | 0.695 | 0.00624 | 0.683 | 0.707 |
| 20 | 775 | 10950 | 0.681 | 0.00699 | 0.668 | 0.695 |
| 21 | 720 | 10959 | 0.673 | 0.00743 | 0.658 | 0.687 |
| 22 | 662 | 10965 | 0.667 | 0.00774 | 0.652 | 0.682 |
| 23 | 606 | 10974 | 0.658 | 0.00823 | 0.642 | 0.674 |
| 24 | 554 | 10981 | 0.65 | 0.00867 | 0.633 | 0.667 |
| 25 | 363 | 10984 | 0.645 | 0.00908 | 0.627 | 0.663 |
| 26 | 327 | 10996 | 0.622 | 0.0108 | 0.602 | 0.644 |
| 27 | 284 | 10998 | 0.618 | 0.0111 | 0.597 | 0.64 |
| 28 | 244 | 11000 | 0.613 | 0.0116 | 0.591 | 0.636 |
| 29 | 198 | 11002 | 0.607 | 0.0122 | 0.584 | 0.632 |
| 30 | 134 | 11010 | 0.578 | 0.0154 | 0.549 | 0.609 |
| 31 | 71 | 11012 | 0.566 | 0.0174 | 0.533 | 0.601 |
| 32 | 65 | 11013 | 0.558 | 0.019 | 0.522 | 0.597 |
| 33 | 61 | 11017 | 0.524 | 0.0244 | 0.478 | 0.574 |
| 34 | 57 | 11018 | 0.515 | 0.0255 | 0.467 | 0.567 |
| 35 | 56 | 11018 | 0.515 | 0.0255 | 0.467 | 0.567 |
| 36 | 51 | 11021 | 0.487 | 0.0288 | 0.433 | 0.547 |
| 37 | 49 | 11021 | 0.487 | 0.0288 | 0.433 | 0.547 |
| 38 | 46 | 11022 | 0.476 | 0.03 | 0.421 | 0.539 |
| 39 | 45 | 11022 | 0.476 | 0.03 | 0.421 | 0.539 |

For peer review only

**Survival rate of frail patients after TAVI**
**(combined studies used Fried/modified Fried phenotype)**

| Time (Months) | n.risk | n.event | survival | Std.err | Lower 95%CI | Upper 95% CI |
|---|---|---|---|---|---|---|
| 1 | 422 | 36 | 0.922 | 0.0125 | 0.898 | 0.947 |
| 2 | 398 | 52 | 0.887 | 0.0148 | 0.858 | 0.916 |
| 3 | 382 | 59 | 0.871 | 0.0157 | 0.841 | 0.902 |
| 4 | 367 | 67 | 0.853 | 0.0166 | 0.821 | 0.886 |
| 5 | 359 | 69 | 0.848 | 0.0169 | 0.816 | 0.882 |
| 6 | 347 | 74 | 0.836 | 0.0175 | 0.803 | 0.871 |
| 7 | 333 | 82 | 0.817 | 0.0184 | 0.781 | 0.853 |
| 8 | 319 | 88 | 0.802 | 0.019 | 0.765 | 0.84 |
| 9 | 305 | 96 | 0.782 | 0.0198 | 0.744 | 0.821 |
| 10 | 297 | 100 | 0.771 | 0.0202 | 0.733 | 0.812 |
| 11 | 284 | 109 | 0.748 | 0.0211 | 0.708 | 0.79 |
| 12 | 108 | 114 | 0.73 | 0.0223 | 0.688 | 0.775 |
| 13 | 36 | 114 | 0.73 | 0.0223 | 0.688 | 0.775 |
| 14 | 35 | 115 | 0.71 | 0.0295 | 0.655 | 0.77 |
| 15 | 34 | 116 | 0.69 | 0.0349 | 0.625 | 0.762 |
| 16 | 32 | 116 | 0.69 | 0.0349 | 0.625 | 0.762 |
| 17 | 32 | 116 | 0.69 | 0.0349 | 0.625 | 0.762 |
| 18 | 30 | 117 | 0.668 | 0.0403 | 0.593 | 0.751 |
| 19 | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 20 | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 21 | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 22 | 28 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 23 | 27 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 24 | 27 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 25 | 26 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 26 | 26 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 27 | 26 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 28 | 25 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 29 | 25 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 30 | 24 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 31 | 24 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 32 | 23 | 118 | 0.645 | 0.0447 | 0.564 | 0.739 |
| 33 | 22 | 119 | 0.617 | 0.0508 | 0.525 | 0.725 |
| 34 | 22 | 119 | 0.617 | 0.0508 | 0.525 | 0.725 |
| 35 | 22 | 119 | 0.617 | 0.0508 | 0.525 | 0.725 |
| 36 | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |
| 37 | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |
| 38 | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |
| 39 | 21 | 120 | 0.589 | 0.0557 | 0.49 | 0.709 |

For peer review only

Supplementary appendix G. Quality of life among frail TAVI recipients

| Study | QoL measures | Baseline (SD) | Post-TAVI (SD) | Change in QoL post-TAVI | 30-day (SD) | Change in QoL at 30 days | 12-months (SD) |
|---|---|---|---|---|---|---|---|
| Okoh, 2017 | KCCQ overall score | | | | | -5.86 (p=.62) | |
| | KCCQ physical limitation | | | | | -8.34 (p=0.46) | |
| | KCCQ symptoms | | | | | -5.68 (p=0.65) | |
| | KCCQ quality of life | | | | | 6.36 (p=0.63) | |
| | KCCQ social limitation | | | | | -7.57 (p=0.58) | |
| Kobe, 2016 Subgroup of frail patients with a frailty score 4-6 | SF-36 physical functioning | 55.8 (15.3) | | | 49.6 (22.2) | | |
| | SF-36 role physical | 37.9 (32.8) | | | 33.8 (43.1) | | |
| | SF-36 bodily pain | 88.1 (23.2) | | | 74.4 (23.7) | | |
| | SF-36 role emotional | 88.2 (23.6) | | | 38.3 (42.3) | | |
| | SF-36 general health | 73.4 (18.7) | | | 62 (15.5) | | |
| | SF-36 vitality | 45.5 (24.6) | | | 43.4 (15.5) | | |
| | SF-36 social functioning | 91.5 (17.8) | | | 68.8 (26.4) | | |
| | SF-36 mental health | 80.4 (16.8) | | | 67.2 (23.2) | | |
| | SF-36 physical component summary | 42.9 (6.54) | | | 42.3 (8.02) | | |
| | SF-36 mental component summary | 55 (8.79) | | | 42.8 (9.86) | | |
| Kobe, 2016 Subgroup of frail patients with a frailty score 6-14 | SF-36 physical functioning | 40.9 (20.5) | | | 46.5 (30.2) | | |
| | SF-36 role physical | 25.6 (33.2) | | | 27.5 (39.7) | | |
| | SF-36 bodily pain | 69.1 (35.6) | | | 60.4 (34.8) | | |
| | SF-36 role emotional | 82.5 (32.9) | | | 45 (46.2) | | |
| | SF-36 general health | 63.1 (22.7) | | | 50.6 (20.9) | | |
| | SF-36 vitality | 35.4 (25.8) | | | 44.1 (24.9) | | |
| | SF-36 social functioning | 79.7 (28.7) | | | 67.9 (23.9) | | |
| | SF-36 mental health | 75 (16.3) | | | 66 (23.3) | | |
| | SF-36 physical component summary | 36.1 (9.57) | | | 38.6 (8.68) | | |
| | SF-36 mental component summary | 53 (9.23) | | | 45.2 (11.6) | | |
| Goudzwaard, 2020 | EQ-5D index | 0.55 (0.26) | | | | | 0.44 (0.33) |
| | EQ-VAS | 58.4 (16.8) | | | | | 63.4 (14.6) |
| Patel, 2020 | KCCQ 12 score | 36 (25) | 75 (70-80) | 37 (31-43) | | | |
| Morris, 2020 | KCCQ 12 score | 37 | 66 | | | | |

KCCQ, Kansas City Cardiomyopathy Questionnaire. SF-36, Short Form-36 questionnaire. EQ-5D, Euro-Qol-5-dimensions. VAS, visual analogue scale.

For peer review only

# MOOSE Checklist for Meta-analyses of Observational Studies

| Item No | Recommendation | Reported on Page No |
|---|---|---|
| Reporting of background should include | | |
| 1 | Problem definition | The second paragraph on page 4. |
| 2 | Hypothesis statement | NA |
| 3 | Description of study outcome(s) | The last paragraph on page 4. |
| 4 | Type of exposure or intervention used | The last paragraph on page 4. |
| 5 | Type of study designs used | The last paragraph on page 4. |
| 6 | Study population | The last paragraph on page 4. |
| Reporting of search strategy should include | | |
| 7 | Qualifications of searchers (eg, librarians and investigators) | Information specialist |
| 8 | Search strategy, including time period included in the synthesis and key words | Supplementary appendix A |
| 9 | Effort to include all available studies, including contact with authors | NA |
| 10 | Databases and registries searched | Supplementary appendix A |
| 11 | Search software used, name and version, including special features used (eg, explosion) | Supplementary appendix A |
| 12 | Use of hand searching (eg, reference lists of obtained articles) | Supplementary appendix A |
| 13 | List of citations located and those excluded, including justification | Supplementary appendix A and Figure 1 |
| 14 | Method of addressing articles published in languages other than English | NA |
| 15 | Method of handling abstracts and unpublished studies | The last paragraph on page 4. We included conference abstracts. |
| 16 | Description of any contact with authors | NA |
| Reporting of methods should include | | |
| 17 | Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested | Data synthesis section on page 5 |
| 18 | Rationale for the selection and coding of data (eg, sound clinical principles or convenience) | Data synthesis section on page 5 |
| 19 | Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability) | NA |
| 20 | Assessment of confounding (eg, comparability of cases and controls in studies where appropriate) | NA |
| 21 | Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results | Risk of bias assessment section on page 5 |
| 22 | Assessment of heterogeneity | Data synthesis section on page 5 |
| 23 | Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated | Data synthesis section on page 5 |
| 24 | Provision of appropriate tables and graphics | Tables 1-2, Figs 1 Supplementary appendix A-G |
| Reporting of results should include | | |
| 25 | Graphic summarizing individual study estimates and overall estimate | Table 1-2, Supplementary appendix D-G |
| 26 | Table giving descriptive information for each study included | Table 1, Supplementary appendix B |

For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

| | | |
|---|---|---|
| 27 | Results of sensitivity testing (eg, subgroup analysis) | Results section and Supplementary appendix E |
| 28 | Indication of statistical uncertainty of findings | Results section and Table 2 |

| Item No | Recommendation | Reported on Page No |
|---|---|---|
| Reporting of discussion should include | | |
| 29 | Quantitative assessment of bias (eg, publication bias) | Data synthesis section on page 5 and Principal findings section on page 9 |
| 30 | Justification for exclusion (eg, exclusion of non-English language citations) | NA |
| 31 | Assessment of quality of included studies | Principal findings section on page 9 |
| Reporting of conclusions should include | | |
| 32 | Consideration of alternative explanations for observed results | Principal findings section on page 9 |
| 33 | Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review) | Conclusion section on page 13 |
| 34 | Guidelines for future research | The last paragraph on page 12 |
| 35 | Disclosure of funding source | Page 13 |

*From*: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

For peer review only